

# EXHALE: physical equations and numerical algorithms

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## 1. Introduction and conventions

### 1.1 What the code solves

EXHALE integrates the one-dimensional, spherically symmetric equations of radiation hydrodynamics for the escaping upper atmosphere of an irradiated planet. The radial domain runs from a base near the lower atmosphere, which a run may place at the planet radius or at a stated pressure level (the  $1\ \mu\text{bar}$  level of the Koskinen et al. (2022) handoff is the one the molecular configurations use), out to the Roche-lobe (L1) radius of the planet–star system or, in the spherical domain mode, to a radius the user states. The unknowns advanced in time are the conserved hydrodynamic state  $u = (\rho, \rho v, \mathcal{E})$  on a finite-volume radial grid, together with the ionization and chemical state of the gas: hydrogen and helium always, the metastable He  $2^3S$  level, a set of trace metals, and a molecular network carrying  $\text{H}_2$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$ ,  $\text{HeH}^+$  and, optionally, the oxygen carriers  $\text{OH}$ ,  $\text{H}_2\text{O}$  and  $\text{CO}$ . The radiative source term couples the two: the stellar spectrum is attenuated along the radial column, photoionizes and heats the gas, and the gas cools through recombination, collisional and line processes whose rates depend on the composition the ionization solve returns.

The code is a heavily extended fork of ATES v2.0 (Caldiroli et al., 2021; Biassoni et al., 2024). The hydrodynamic core, the finite-volume discretization, the three-stage strong-stability-preserving Runge-Kutta integration and the choice of approximate Riemann solvers are recognizably those of ATES; the equation of state, the lower boundary condition, the reconstruction, the steady-state solvers, the transported species and essentially all of the chemistry are not. Every physics extension is selected at run time by the presence of a key in `input.inp` or of a companion file (`metals.inp`, `opacity.inp`, `base.inp`). Every option that adds a species, a transport process or an external field defaults to off, so that a configuration that states none of them solves the atomic H/He problem; the keys that default to on are parts of the physics of what is already selected, which a key can only turn off for comparison (among them the He/H charge exchange, secondary ionization, the molecular reaction heat, the resolved  $\text{H}_2$  photoabsorption channels, and the interdiffusion enthalpy flux of a run with `He_diffusion`; Table 2).

### 1.2 Units

The equations are solved in dimensionless form. The normalization is fixed once, in `input_read.f90`, from the planet radius  $R_0$ , the base temperature  $T_0$ , and the base number density  $n_0$ :

$$v_0 = \sqrt{\frac{k_B T_0}{m_H}}, \quad t_s = \frac{R_0}{v_0}, \quad p_0 = n_0 m_H v_0^2, \quad q_0 = \frac{n_0 m_H v_0^3}{R_0}, \quad (1)$$

where  $m_H = 1.67353284 \times 10^{-24}$  g is the mass of the hydrogen *atom*, not the atomic mass unit. Radii are in units of  $R_0$ , so the base level sits at  $r = 1$ , which is the lower face of the grid (Sec. 7.1); velocities in  $v_0$ ; mass densities in  $n_0 m_H$ ; pressures in  $p_0$ ; temperatures in  $T_0$ ; and volumetric energy rates in  $q_0$ . The mass unit being the hydrogen atom, the

density is the number of hydrogen-atom masses per unit volume, and a species contributes  $m_{\text{species}}/m_{\text{H}}$  to it; the species table therefore carries  $m(\text{HeI}) = 3.9715259$  and not 4.

The gravitational potential is measured in  $v_0^2$ , so it is fixed by the single dimensionless group

$$b_0 = \frac{GM_p m_{\text{H}}}{k_B T_0 R_0}, \quad (2)$$

the Jeans escape parameter at the base. Only three further dimensionless numbers enter the potential and the domain:  $\tilde{a} = a_{\text{orb}}/R_0$ , the mass ratio  $M_*/M_p$ , and the outer radius.

The base level itself may be stated in three ways, and a run that states two that disagree is stopped at startup: Log10 lower boundary number density gives  $n_0$  directly; Base BC: pressure states the base pressure in microbar and  $n_0$  follows from  $n_0 = p_{\text{base}}/(k_B T_0 \hat{n}_{\text{bc}})$ ; and a lower-atmosphere handoff (base.inp or a profile file) states  $p_{\text{base}}$  in bar and  $n_0$  follows the same way. Here  $\hat{n}_{\text{bc}}$  is the base particle count per H+He nucleus, which the composition module derives from the element ratios and, with a molecular base, from the H nuclei bound into  $\text{H}_2$  (Sec. 2.5).

### 1.3 Notation

One symbol stands for one quantity throughout this document. The table below fixes the symbols that appear in more than one section. Symbols introduced inside a published fit and defined where they appear (the  $x$  of a Verner-type cross section, the  $\lambda$  of a recombination-cooling fit, the  $\alpha$  of a spectral index) are local to that fit and are not in the table.

| symbol                                | quantity                                                                                                   | note                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $u$                                   | conserved state $(\rho, \rho v, \mathcal{E})$                                                              | the unknown of the hydrodynamics, Sec. 2.1                                     |
| $W$                                   | primitive state $(\rho, v, p)$                                                                             | what the reconstruction works on                                               |
| $\rho, v, p, T$                       | mass density, radial velocity, pressure, temperature                                                       |                                                                                |
| $\mathcal{E}, \rho e, e, \varepsilon$ | total energy density, internal energy density, internal energy per unit mass, internal energy per particle | $\mathcal{E} = \frac{1}{2} \rho v^2 + \rho e$                                  |
| $\Phi$                                | gravitational potential                                                                                    | $\Phi_i, \Phi_c$ at interfaces and cell centers                                |
| $\gamma_{\text{eff}}, c_s$            | effective adiabatic index, sound speed                                                                     | from the caloric equation of state (14)                                        |
| $m_{\text{H}}$                        | mass of the hydrogen atom                                                                                  | the code's mass unit, (1)                                                      |
| $\mu, \kappa$                         | dynamic viscosity, heat conductivity                                                                       | (21) only                                                                      |
| $n_X, n_{\text{tot}}, n_e$            | number density of species $X$ , of all heavy particles, of free electrons                                  |                                                                                |
| $x_{X^{q+}}, f_i$                     | ionization fraction of a stage, mixing ratio $n_i/n_{\text{tot}}$                                          | fractions of the element's own nuclei                                          |
| $E, h\nu$                             | photon energy                                                                                              | always subscripted where a particular energy is meant ( $E_k, E_{\text{th}}$ ) |
| $F_E, \mathcal{F}_k$                  | incident and attenuated flux per unit photon energy                                                        | (34)                                                                           |
| $\sigma, \tau$                        | photoionization cross section, optical depth                                                               |                                                                                |
| $\Gamma, \alpha, \beta$               | photoionization rate, recombination coefficient, electron-impact ionization coefficient                    | Sec. 4.2                                                                       |
| $\mathcal{H}, \Lambda$                | volumetric heating rate, volumetric cooling rate                                                           | $\text{erg cm}^{-3} \text{ s}^{-1}$ ; $Q = \mathcal{H} - \Lambda$              |
| $\xi_{\text{day}}$                    | dayside dilution of every stellar beam                                                                     | Table 1                                                                        |
| $J_i, J$                              | diffusive number flux of a carrier, diffusive mass flux of an element                                      | Secs. 6.5, 6.9                                                                 |
| $dF, S, R$                            | discrete flux difference, source, stationary residual                                                      | (6)–(11), (123)                                                                |
| $A_{\pm}, \Delta V_j, \Delta r_j$     | face areas, cell volume, cell width                                                                        | (5)                                                                            |

## 1.4 How to read this document

Every equation below is the discrete operator the code applies, taken from the source and not from a derivation. Where the code's expression differs from the textbook form (the spherical pressure term under PLM, the gravitational work term folded into the energy flux, the mean molecular weight that is never formed), the code's form is given and the difference is stated. Routines are named with the file they live in, for instance `RK_rhs.f90`, so that a statement here can be checked against one place in the tree.

Section 2. states the conservation laws, the equation of state and the source terms. The sections that follow it state the radiation field, the ionization and chemical network, the thermal source and the molecular and metal chemistry. Section 7. then gives the grid, the reconstruction, the Riemann solvers, the time integration and the operator splitting; Sec. 8. the boundary and initial conditions; and Sec. 9. the stationary residual, its acceptance test and the solvers that drive it to zero. A section on the post-processing tools closes the physics account, and Sec. 11. tabulates the input keys that select physics, with the default each of them takes when the key is absent.

## 2. Hydrodynamics

### 2.1 Conserved variables and the conservation laws

The conserved state of a cell is the triple

$$u = (\rho, \rho v, \mathcal{E})^\top, \quad \mathcal{E} = \frac{1}{2}\rho v^2 + \rho e, \quad (3)$$

with  $\rho e$  the internal energy density. The map between  $u$  and the primitive triple  $W = (\rho, v, p)$  is `UW_conversions.f90`: the density and the momentum are trivial, and the pressure follows from the caloric equation of state of Sec. 2.5,  $p = p(\rho, \rho e)$ , evaluated with the composition of the cell the state belongs to. For a reconstructed face state that cell is the cell the state was extrapolated from, because the composition ratios the equation of state needs are ratios of number densities and are therefore independent of the density.

The equations integrated are the Euler equations in spherical symmetry with gravity and a radiative source,

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v) &= 0, \\ \frac{\partial(\rho v)}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v^2) + \frac{\partial p}{\partial r} &= -\rho \frac{\partial \Phi}{\partial r} + S_\mu, \\ \frac{\partial \mathcal{E}}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 (\mathcal{E} + p) v) &= -\rho v \frac{\partial \Phi}{\partial r} + Q + S_\mathcal{E} - \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_d), \end{aligned} \quad (4)$$

where  $Q = \mathcal{H} - \Lambda$  is the net radiative heating minus cooling per unit volume (Sec. 5.) and  $S_\mu, S_\mathcal{E}$  are the viscous and conductive sources of Sec. 2.8, both off unless a run asks for them.  $q_d = \sum_s h_s J_s$  is the interdiffusion enthalpy flux of a mixture whose elements or transported carriers move relative to the mass-averaged velocity  $v$  (Cook, 2009, eqs. 11–13): it exists where `He_diffusion` moves helium against hydrogen and where a molecular carrier is transported, and is written out in Sec. 6.10. The factor  $4\pi$  that would appear in the face areas and the cell volume cancels between them and is simply not carried.

### 2.2 The discrete divergence

`RK_rhs.f90` assembles the right-hand side of (4) as a flux difference  $dF$  and a source  $S$ , so that one forward Euler stage of the marching loop is  $u \leftarrow u - \Delta t (dF - S)$  and the stationary residual is  $R = dF - S$  with the radiative source subtracted from its third row. With

$$A_\pm = r_{j\pm 1/2}^2, \quad \Delta V_j = \frac{r_{j+1/2}^3 - r_{j-1/2}^3}{3}, \quad \Delta r_j = r_{j+1/2} - r_{j-1/2}, \quad (5)$$

and  $F_\pm$  the numerical flux at the two faces of cell  $j$  (Sec. 7.3), the three rows are

$$dF_1(j) = \frac{A_+ F_+^\rho - A_- F_-^\rho}{\Delta V_j}, \quad (6)$$

$$dF_2(j) = \frac{A_+ F_+^{\rho v} - A_- F_-^{\rho v}}{\Delta V_j} + w_p \frac{p_R - p_L}{\Delta r_j}, \quad (7)$$

$$dF_3(j) = \frac{A_+ F_+^\mathcal{E} - A_- F_-^\mathcal{E} + \delta F_3}{\Delta V_j}, \quad (8)$$

with  $F^\rho = \rho v$ ,  $F^\mathcal{E} = v(\mathcal{E} + p)$ , and  $p_L, p_R$  the face pressures the Riemann solver returns at the lower and upper face. The weight  $w_p$  and the content of  $F^{\rho v}$  depend on the reconstruction, because the spherical pressure term is split differently in the two schemes (Sec. 2.3).

The term  $\delta F_3$  in (8) is the gravitational work,

$$\delta F_3 = A_+ F_+^\rho [\Phi_i(j) - \Phi_c(j)] - A_- F_-^\rho [\Phi_i(j-1) - \Phi_c(j)], \quad (9)$$

with  $\Phi_i$  the potential at the cell interfaces and  $\Phi_c$  at the cell centers (`set_gravity_grid.f90`). Expanding the bracket separates (8) into the energy flux of Caldiroli et al. (2021), their Eq. (14),  $F^\mathcal{E} + F^\rho \Phi$  taken at the interface potential, minus  $\Phi_c(j)$  times the mass-flux divergence, which is their energy source  $\Phi_j L_j^\rho$ . The two forms are algebraically the same operator; the code carries it as one term so that the mass flux the work term rides on is the very flux the continuity row differenced.

### 2.3 The spherical pressure term and gravity

The pressure gradient in spherical symmetry is

$$\frac{\partial p}{\partial r} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 p) - \frac{2p}{r}, \quad (10)$$

and the two reconstructions use the two sides of this identity. Under PLM, `Phys_flux` adds  $p$  to the momentum flux, so the first term of (10) is inside  $dF_2$  and the geometric correction is carried by the source;  $w_p = 0$ . Under WENO3 the pressure is left out of the flux and the whole gradient enters as the single face difference  $(p_R - p_L)/\Delta r_j$ ;  $w_p = 1$ . `Source.f90` accordingly returns

$$S_1 = 0, \quad S_2 = -\frac{\rho_L + \rho_R}{2} \frac{\Phi_i(j) - \Phi_i(j-1)}{\Delta r_j} + \underbrace{\frac{A_+ - A_-}{\Delta V_j} p_c}_{\text{PLM only}}, \quad S_3 = 0, \quad (11)$$

where  $\rho_L$  and  $\rho_R$  are the two reconstructed face densities that bound the cell and  $p_c$  is the cell's own pressure from the caloric equation of state. Because  $(A_+ - A_-)/\Delta V_j \rightarrow 2/r$  as the cell narrows, the PLM branch of (11) is the second term of (10).

The three physical terms of the momentum equation, the ram divergence, the spherical pressure gradient and the weight  $\rho \partial \Phi / \partial r$ , are gathered out of these pieces by `momentum_row_terms_of_cell` in the same file and stored. They are not  $dF_2$  and  $S_2$ : under PLM each of those holds a part of order  $2p/r$  that cancels against the other, so a magnitude built from them alone reads  $2p/r$  on a uniform pressure at rest, where the physical force is zero. The gathered terms are what the stationary residual scales the momentum row by (Sec. 9.1).

### 2.4 The gravitational potential

`grav_field.f90` carries two potentials, selected by `Domain` mode. The default is the full Roche potential of a synchronously rotating planet,

$$\Phi(r) = -\frac{b_0}{r} - \frac{b_0 \mathcal{M}}{\tilde{a} - r} - \frac{b_0(1 + \mathcal{M})}{2\tilde{a}^3} \left( \frac{\tilde{a} \mathcal{M}}{1 + \mathcal{M}} - r \right)^2, \quad \mathcal{M} = \frac{M_\star}{M_p}, \quad \tilde{a} = \frac{a_{\text{orb}}}{R_0}, \quad (12)$$

in code units, which is Eq. (2) of Caldiroli et al. (2021): the planetary term, the stellar tidal term and the centrifugal term about the center of mass. With `Domain` mode: `Spherical` the last two terms are dropped and  $\Phi = -b_0/r$ , which is what allows the domain to be carried past the L1 radius. The analytic derivative  $d\Phi/dr$  is provided beside it and is used

by the initial-condition sonic-point search; the flux assembly itself never differentiates  $\Phi$ , it differences the tabulated  $\Phi_i$  and  $\Phi_c$ , which are evaluated once at startup at the true face and center radii.

The default outer radius in Roche mode is the L1 (Hill) radius  $r_{\max} = (3\mathcal{M})^{-1/3}\tilde{a}$ ; Outer radius overrides it.

## 2.5 Equation of state

The thermal equation of state is that of an ideal gas of all particles including free electrons,

$$p = (n_{\text{tot}} + n_e)k_B T \quad \Longleftrightarrow \quad T = \frac{p}{n_{\text{tot}} + n_e} \text{ in code units,} \quad (13)$$

which is `comp_T_from_p` and `comp_p_from_T` of `composition.f90`. The particle and electron counts are built in one place, `get_species_densities` in the same file, from the density and the species mass fractions: `calc_ntot` adds one particle for every species the run carries (H I, H II, He I, He II, He III, every metal ion stage, and each of  $\text{H}_2$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$ ,  $\text{HeH}^+$ , OH,  $\text{H}_2\text{O}$ , CO), and `calc_ne` adds the free electrons weighted by the net charge of each species. The  $\text{He}2^3S$  column is not added by either: it is an excited level of He I and its particle and its nucleus are already counted there. Trace metals enter both sums only under the `eos_metals` policy of Sec. 6.8, on by default, so that the legacy H/He-only budget can be reproduced exactly.

The code never forms a mean molecular weight for the equation of state. The mean molecular weight of Caldiroli et al. (2021), their footnote 1, is a derived diagnostic here; (13) with the explicit particle count is what the hydrodynamics uses, so the temperature of a cell is a function of its composition and never of a fitted composition ratio.

The *caloric* half of the equation of state is `caloric_eos.f90`. For an atomic gas it is the monatomic relation  $\rho e = p/(\gamma_{\text{ad}} - 1)$  with  $\gamma_{\text{ad}} = 5/3$ , exactly as in ATES. This is not valid in a molecular layer: at the 400 to 1300 K of the  $\text{H}_2$  base the rotational ladder of  $\text{H}_2$  is populated (the  $J = 1$  level lies at 170 K), so an  $\text{H}_2$  molecule stores about  $\frac{5}{2}k_B T$  and the vibrational ladder adds to that above  $\sim 800$  K. Where the gas is 80%  $\text{H}_2$  the heat capacity is then wrong by a factor 1.67. The mixture relation used instead is

$$\rho e = \frac{3}{2}(n_{\text{tot}} + n_e)k_B T + n(\text{H}_2)k_B u_{\text{rv}}(T), \quad \frac{C_V}{k_B} = \frac{3}{2}(n_{\text{tot}} + n_e) + n(\text{H}_2)c_{\text{rv}}(T), \quad (14)$$

so that the exponent making  $\rho e = p/(\gamma_{\text{eff}} - 1)$  exact is  $\gamma_{\text{eff}} = 1 + (n_{\text{tot}} + n_e)k_B/C_V$ , a function of both the composition and the temperature. The mean rovibrational energy  $u_{\text{rv}}$  and the heat capacity  $c_{\text{rv}}$  are evaluated from the complete bound rovibrational ladder of  $\text{H}_2 X^1\Sigma_g^+$ , 302 levels to 51966 K, that is to the 4.478 eV dissociation limit (Roueff et al., 2019, Table 2 through VizieR); the same partition function is the internal partition function of the  $\text{H} + \text{H} \leftrightarrow \text{H}_2$  equilibrium constant (79), so the energy and the chemistry cannot disagree.  $\text{H}_2^+$ ,  $\text{H}_3^+$ ,  $\text{HeH}^+$  and the oxygen carriers are counted as monatomic, a stated approximation valid while they stay trace species (their largest ratio to  $n(\text{H}_2)$  in the molecular layer of the reference configuration is  $1.7 \times 10^{-7}$ ). Dissociation is not a heat capacity and is not in (14); it is a chemical source term. The branch between the two forms is structural, not a limit: a run with no molecules and a cell with  $n(\text{H}_2)$  exactly zero both take the constant- $\gamma_{\text{ad}}$  expression verbatim. The key `Caloric EOS: monatomic` forces the monatomic form everywhere.

The mapping  $\rho e \rightarrow p$  in the molecular branch is the inverse of (14) and costs one safeguarded Newton iteration per cell; the mapping  $p \rightarrow \rho e$  is direct. The sound speed used by the Riemann solvers and by the time step is  $c_s = \sqrt{\gamma_{\text{eff}} p / \rho}$  evaluated with the cell's own composition.

Two base composition scalars follow from the same module. The gas mass per hydrogen nucleus is  $m_{\text{H-nuc}} = m(\text{HI}) + m(\text{HeI})y + \sum_e A_e X_e$  with  $y = n_{\text{He}}/n_{\text{H}}$ ; the base particle count per H+He nucleus is

$$\hat{n}_{\text{bc}} = \frac{1+y+\sum_e X_e}{1+y} - \frac{1}{2} \frac{x_{\text{H}_2}}{1+y}, \quad x_{\text{H}_2} = \frac{2q_{\text{H}_2}(1+y)}{1+q_{\text{H}_2}}, \quad (15)$$

where  $x_{\text{H}_2}$  is the fraction of base hydrogen nuclei bound into  $\text{H}_2$  and the last term is present only with `Molecular base: True`. The mixing ratio  $q_{\text{H}_2}$  is the one a lower-atmosphere handoff states, or otherwise the chemical-equilibrium fit (84) at  $(p_{\text{base}}, T_0)$  (Sec. 6.7). The same  $x_{\text{H}_2}$  is imposed on the inflowing ghost composition, so the equation of state and the species state describe one gas.

## 2.6 The well-balanced option

`Well balanced: True` (default off) changes the coordinate the pressure is carried in. The local hydrostatic equilibrium of cell  $j$  is the constant-density one through its own  $(\rho_j, p_j)$ ,

$$p_{\text{eq},j}(r) = p_j - \rho_j [\Phi(r) - \Phi(r_j)], \quad (16)$$

(Eq. 16 of Käppeli & Mishra 2016), whose two face values are  $P_{\text{up}}(j) = p_j - \rho_j [\Phi_i(j) - \Phi_c(j)]$  and  $P_{\text{dn}}(j) = p_j + \rho_j [\Phi_c(j) - \Phi_i(j-1)]$ . The reconstruction then works on the departure from (16) rather than on the pressure itself, the Riemann pressure jump is formed as  $\Delta p = \Delta p_{\text{eq}} + q_R - q_L$  from quantities that are all of the size of the departure, and the momentum row is assembled from the face pressure measured against each side's own equilibrium. The two identities that make the substitution exact are

$$\begin{aligned} A_+ P_{\text{up}} - A_- P_{\text{dn}} - (A_+ - A_-) p_j &= -\rho_j \{A_+ [\Phi_i(j) - \Phi_c(j)] + A_- [\Phi_c(j) - \Phi_i(j-1)]\}, \\ P_{\text{up}} - P_{\text{dn}} &= -\rho_j [\Phi_i(j) - \Phi_i(j-1)], \end{aligned} \quad (17)$$

the first for the PLM assembly and the second for WENO3. Their left-hand sides are the equilibrium part of the pressure terms the row carries and their right-hand sides are the discrete gravitational source, so the two cancel in the algebra, `Source.f90` returns  $S_2 = 0$  and neither side is ever evaluated. This is the momentum source of Käppeli & Mishra (2014), their Eq. (2.26), in spherical geometry. What cancels is still the physics the row balances, so its magnitude,

$$\mathcal{W}_j = \frac{|\rho_j \{A_+ [\Phi_i(j) - \Phi_c(j)] + A_- [\Phi_c(j) - \Phi_i(j-1)]\}|}{\Delta V_j} \quad (\text{PLM}), \quad \mathcal{W}_j = \frac{|\rho_j [\Phi_i(j) - \Phi_i(j-1)]|}{\Delta r_j} \quad (\text{WENO3}), \quad (18)$$

is formed separately and is the term the momentum residual is read against under this option.

Under this option the lower boundary shares the same equilibrium: the interior pressure its outgoing characteristic reads is  $P_{\text{dn}}(1)$ , and the two ghost cells continue (16) below the level through the face state (Sec. 8.1), so a column at rest in the discrete equilibrium and meeting the reservoir's pressure at the level carries no flux through the base face either.

## 2.7 Low-Mach dissipation of the contact mode

The HLLC flux resolves the contact wave exactly, so the dissipation it applies to the entropy family is proportional to the contact speed  $S_* \simeq v$  and vanishes as the flow stalls; the acoustic families keep theirs at  $|v| \pm c_s$ . A stagnant layer can then carry a  $2\Delta r$  pattern in  $v$  and  $T$  that nothing in the scheme removes. `Low-Mach damping: (default off,`

`low_mach_dissipation.f90`) adds a gated fourth-difference stress (Jameson et al., 1981) to the numerical momentum flux,

$$D_p^{j+1/2} = \epsilon_4 g(M) \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1}), \quad D_{\mathcal{E}}^{j+1/2} = v_f D_p^{j+1/2}, \quad (19)$$

with  $\rho_f, v_f$  the face averages and  $\lambda_f = \frac{1}{2}[(|v| + c_s)_j + (|v| + c_s)_{j+1}]$  the face spectral radius. The gate  $g = [\max(0, 1 - M_f^2/M_{\text{th}}^2)]^2$ , written in  $M^2$  so that it is smooth where the velocity changes sign, is unity at rest and identically zero above  $M_{\text{th}}$  (default  $10^{-3}$ ). The pair  $(D_p, D_{\mathcal{E}})$  conserves total energy exactly and converts kinetic energy into heat and never the reverse; the mass flux is untouched. The term is added inside `RK_rhs.f90`, so the marching right-hand side and the stationary residual solve the same equation, which is what distinguishes it from the Shapiro filter of Sec. 7.6.

## 2.8 Viscosity and heat conduction

Viscosity: True and Conduction: True (both default off, `viscous_conduction.f90`) add the Navier-Stokes level of the same equations, the level the CETIMB code (the time-dependent upper-atmosphere model of Koskinen and collaborators, Koskinen et al. 2013, 2022) carries. For the Newtonian stress in spherical symmetry the only nonzero components are  $\tau_{rr} = \frac{4}{3}\mu(v' - v/r)$  and  $\tau_{\theta\theta} = \tau_{\varphi\varphi} = -\frac{1}{2}\tau_{rr}$ , so exactly

$$S_\mu = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{rr}) + \frac{\tau_{rr}}{r}, \quad q_\mu = \frac{4}{3}\mu \left( \frac{\partial v}{\partial r} - \frac{v}{r} \right)^2, \quad S_{\mathcal{E}} = v S_\mu + q_\mu + \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \kappa \frac{\partial T}{\partial r} \right). \quad (20)$$

These differ from the expressions printed as Eqs. (B5) and (B6) of Koskinen et al. (2022): those carry a different gradient term ( $\mu'v'$  against  $\frac{4}{3}\mu'v/r$ ), a geometric coefficient  $16/3$  against  $8/3$ , and a dissipation that is not sign definite, and the printed pair does not satisfy  $v S_\mu + q_\mu = \nabla \cdot (\tau \cdot v)$ . Physical correctness being the criterion, (20) is what is implemented and the discrepancy is recorded in the source.

Neither Koskinen paper prints the transport coefficients. The heat conduction coefficient is the number-weighted sum of the Banks & Kockarts (1973) coefficients of the constituents, as Salz et al. (2015) (their Eqs. 24–26, electron–ion and neutral hydrogen) and Sutton et al. (2015) (their Eq. 5, with helium) write it,

$$\kappa = \frac{1}{n} \left[ n_e 1.2 \times 10^{-6} T^{5/2} + n_{\text{HI}} 379 T^{0.69} + n_{\text{HeI}} 299 T^{0.69} + n_{\text{H}_2} k_{\text{H}_2}(T) \right] \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}, \quad (21)$$

$n$  the number density of all particles and  $T$  in K; ions contribute through  $n$  only. The hydrogen term is the atomic hydrogen conductivity of Watson et al. (1981),  $4.45 \times 10^4 (T/1000 \text{ K})^{0.7}$ , to within 2 per cent between 300 and 7000 K; at  $\text{He}/\text{H} = 9.7$  the helium term decides  $\kappa$  (0.81 of the hydrogen value on the LHS 1140 b states), and it agrees with the tabulated helium conductivity of Incropera et al. (2007) (Table A.4) to 1 per cent at 300 and 1000 K. None of these sources gives an  $\text{H}_2$  coefficient:  $k_{\text{H}_2}(T) = A T^s [1 + c E(\theta_v/T)]$ ,  $E(x) = x^2 e^x / (e^x - 1)^2$ ,  $A = 272.523$ ,  $s = 0.735652$ ,  $c = 0.370446$ ,  $\theta_v = 5987 \text{ K}$  (the  $v = 0 \rightarrow 1$  spacing of  $\text{H}_2$  in the Roueff et al. (2019) ladder), an Eucken form fitted to the tabulated conductivity of Incropera et al. (2007), Table A.4, over 200–2000 K (largest deviation 2.5 per cent, rms 1.05 per cent); it is valid there and an extrapolation outside. The molecular ions enter through  $n$  and  $n_e$  only. A continuation factor  $s$  on  $\kappa$  (`EXHALE_CONDUCTION_SCALE`) is part of the state identity (the `cond` token carries it). The viscosity keeps the hydrogen value tied to the Watson conductivity by the Chapman–Enskog/Eucken relation for a monatomic gas, Prandtl number  $2/3$ ,  $\mu(T) = \frac{4}{15} \frac{m_{\text{H}}}{k_B} 4.45 \times 10^4 (T/1000 \text{ K})^{0.7}$ . In code units  $\mu_{\text{code}} = \mu_{\text{cgs}} / (n_0 m_{\text{H}} v_0 R_0)$ , the inverse Reynolds number, and  $\kappa_{\text{code}} = \frac{15}{4} \mu_{\text{code}}$ .

Both operators are assembled once as tridiagonal coefficient triplets on the same faces and volumes as (5), and the same triplets serve as the source entering the stationary residual and as the matrix of a Crank-Nicolson update in the marching loop, so the marched fixed point and the zero of the residual are the same state. The inner boundary takes the flux through the base face from the anchored ghost, held fixed during the implicit step; the outer boundary carries zero diffusive flux, which is the condition an extrapolated outflow ghost admits. A step whose solve leaves a cell below the temperature floor of the equation of state fails with a named status rather than returning the floor, because returning it would add energy with no source behind it.

## 2.9 The two-dimensional approximation

A one-dimensional radial model of an irradiated sphere has to state how the stellar flux and the mass loss are shared over the planet. The mandatory key `2D approximate` method carries that statement, in the four forms of Table 1. The dilution factor  $\xi$  is applied to every stellar beam at the point where the beam is used, and in one place only (`dayside_dilution` in `parameters.f90`): the XUV photon grid, the stellar Ly- $\alpha$  beam and the FUV bands all read it. The stated fluxes therefore keep their meaning as the flux at the planet's orbit, and the correction to the mass-loss rate is applied once, at the end of the run (130).

Table 1: The `2D approximate` method options, as `input_read.f90` normalizes them, `dayside_dilution` applies  $\xi$  and `EXHALE_main.f90` applies the  $\dot{M}$  factor.

| Key word            | Stored as                    | $\xi$ | $\dot{M}$ factor |
|---------------------|------------------------------|-------|------------------|
| <code>Mdot/4</code> | <code>Mdot/4</code>          | 1     | 1/4              |
| <code>Rate/2</code> | <code>Rate/2 + Mdot/2</code> | 1/2   | 1/2              |
| <code>Rate/4</code> | <code>Rate/4 + Mdot</code>   | 1/4   | 1                |
| <code>alpha</code>  | <code>alpha</code>           | 1     | 1                |

The first row is the recipe of Salz et al. (2016) that the ATES paper adopts: the undiluted substellar rates with the resulting mass-loss rate divided by four. The second is the tidally locked reading, dayside averaged rates with the outflow leaving one hemisphere; the third the rapidly rotating one, rates reduced by four with the outflow leaving the whole surface. These are the three recipes of Caldiroli et al. (2021), their Sec. 2.4.

The fourth is different in kind. With `alpha` the run states a coefficient  $\alpha$  and the attenuated field carries the further factor  $1/(1 + \alpha\tau)$  inside the column integral (Sekiya et al., 1980; Odert et al., 2020), so that the cell-mean attenuation becomes

$$\langle e^{-\tau} \rangle_{\text{cell}} = \int_0^1 \frac{e^{-\tau_{\text{out}} - s \delta \tau}}{1 + \alpha(\tau_{\text{out}} + s \delta \tau)} ds, \quad (22)$$

which is the cell-mean attenuation (33) with the further factor inside the integral; it has no closed form and is integrated by composite three-point Gauss-Legendre on segments of optical depth at most 0.25. With  $\alpha = 0$ , the default and the only value for which the field is a pure exponential, it reduces to (33) itself.

The range of validity is stated at the routine:  $\xi = 1/2$  is the illuminated hemisphere seen by an *optically thin* absorber. Where a band is shielded the true shell average is lower, the slant columns away from the substellar point being longer;

it is 0.20 to 0.33 for the Lyman-Werner band over a molecular layer. No key asks for that refinement, so a shielded band is diluted by the optically thin factor.

### 3. The stellar radiation field and its absorption

#### 3.1 The incident spectrum

The star is represented by a plane beam that falls radially onto the substellar column. Its spectral shape is chosen once, by the `Spectrum` type: key of `input.inp`, and that single choice fills *every* band of the photon grid: the extreme ultraviolet and X-ray bands, and also the part below the hydrogen Lyman edge where the He  $2^3\text{S}$  metastable, the low-ionization-potential metals and hydrogen in  $n = 2$  absorb. The three types are implemented in `J_inc.f90` (`stellar_flux_ev`) and `sed_read.f90`:

**Power law** (`is_PL_sed`). Two power laws of one common index  $\alpha \equiv \text{PLind}$ , normalized separately on the EUV interval  $[E_{\text{low}}, E_{\text{mid}}]$  and on the X-ray interval  $[E_{\text{mid}}, E_{\text{top}}]$ . With  $L_X$  and  $L_{\text{EUV}}$  the two band luminosities,  $a$  the orbital distance and  $\mathcal{R} = 10^{L_X - L_{\text{EUV}}}$ ,

$$J_{\text{XUV}} = \frac{10^{L_X} + 10^{L_{\text{EUV}}}}{4\pi a^2}, \quad J_X = \frac{\mathcal{R}}{1 + \mathcal{R}} J_{\text{XUV}}, \quad J_{\text{EUV}} = \frac{1}{1 + \mathcal{R}} J_{\text{XUV}}, \quad (23)$$

and, for  $\alpha \neq -1$  and  $P_1 = \alpha + 1$ ,

$$J_{\text{inc}}(E) = \begin{cases} \frac{P_1 J_{\text{EUV}}}{E_{\text{mid}}^{P_1} - E_{\text{low}}^{P_1}} E^\alpha, & E < E_{\text{mid}}, \\ \frac{P_1 J_X}{E_{\text{top}}^{P_1} - E_{\text{mid}}^{P_1}} E^\alpha, & E \geq E_{\text{mid}}, \end{cases} \quad (24)$$

the logarithmic form replacing the ratio at  $\alpha = -1$ , and the X-ray normalization set to zero when `thereis_Xray` is false. Below  $E_{\text{low}}$  the same law is extrapolated, which is what the type means rather than an accident.

**Planck** (`spectrum_is_planck`). The photospheric blackbody seen at the orbit,

$$F_\nu = \pi B_\nu(T_{\text{eff}}) \left( \frac{R_\star}{a} \right)^2, \quad B_\nu = \frac{2h\nu^3/c^2}{e^{h\nu/kT_{\text{eff}}} - 1}, \quad (25)$$

converted to a flux per unit photon energy by  $F_E = F_\nu d\nu/dE$  with  $d\nu/dE = 2.417989242 \times 10^{14} \text{ Hz eV}^{-1}$ . For  $h\nu/kT_{\text{eff}} > 700$  the Wien form replaces  $1/(e^x - 1)$  so that no exponential overflows. The absolute scale comes from  $T_{\text{eff}}$  and  $R_\star$ , not from the two luminosities, so the grid-integrated flux generally differs from  $J_{\text{XUV}}$ ; `write_setup_report.f90` states both.

**Load** (`do_read_sed`). A measured spectral energy distribution read by `read_sed` (`sed_read.f90`): two columns, bin central wavelength in Å and flux at the planet in  $\text{erg cm}^{-2} \text{ s}^{-1} \text{ Å}^{-1}$ , ascending in wavelength. Between two rows the field is interpolated log-linearly in  $\log E$ , exact for the power-law segments a tabulated continuum is made of; outside the band it returns zero, and `spectrum_covers_ev` must be asked first so that the run stops rather than integrating a zero the input never stated.

A monochromatic option (`is_monochr`) puts the whole  $10^{L_{\text{EUV}}}/4\pi a^2$  at a single energy  $E_{\text{low}}$ .

**Bands of the photon grid.** `set_energy_vectors.f90` lays the grid out over at most six bands,

$$[E_{\text{floor}}, E_{\text{abs}}], [E_{\text{abs}}, 13.598 \text{ eV}], [13.598, 24.587], [24.587, 54.418], [54.418, E_{\text{mid}}], [E_{\text{mid}}, E_{\text{top}}],$$

with fixed bin budgets (20, 89, 50, 50, 50, 50).  $E_{\text{low}}$ ,  $E_{\text{mid}}$  and  $E_{\text{top}}$  are stated by the `[E_low, E_mid, E_high]` line of `input.inp` (the tutorial run uses 13.60, 123.98 and  $1.24 \times 10^3$  eV; with `Use only EUV? True` the top is held at  $1.24 \times 10^3$  eV and the X-ray normalization is zero).  $E_{\text{abs}}$  is `sub_lyman_absorber_threshold_eV()`, the lowest ionization threshold of an absorber of the bulk gas below the hydrogen edge: the He  $2^3\text{S}$  metastable at 4.768 eV when it is tracked, or the neutral stage of an active low-IP metal (K I 4.341, Na I 5.139, Ca I 6.113, Mg I 7.646, Fe I 7.902, Si I 8.152, S I 10.36 eV) when one is lower.  $E_{\text{floor}}$  is `photon_grid_floor_eV()`, which lowers that further to  $E_{\text{th}}(\text{H}, n=2) = E_{\text{th}}(\text{H})/4 = 3.3996$  eV, the head of the Balmer continuum, whenever the excited-hydrogen coupling is armed. The interval  $[E_{\text{floor}}, E_{\text{abs}}]$  carries no bulk-gas absorber: it exists so that the  $\text{H}(n=2)$  band is stated, that rate being a closed-form integral over it (Sec. 4.6) and not a sum over these bins.

**Every ionization threshold is a bin edge.** A photoionization rate is the rectangle sum  $\sum_k F_k \sigma_k \Delta E_k / E_k$ , so a bin that *contains* a threshold would charge the absorber its cross section, read at the bin center, over the part of the bin below the threshold where it cannot absorb at all. `ionization_thresholds_active` therefore collects every threshold the run carries (H I, He I, He II, He  $2^3\text{S}$ ,  $\text{H}_2$  and its two channel thresholds, and every photoionizable stage of an element with nonzero abundance together with its subshell turn-ons); each band is cut at those falling strictly inside it, and its budget is shared over the subintervals in proportion to their logarithmic widths (largest remainder, at least one bin each). The bins are geometric and  $E_k$  is the *geometric* mean of the edges, the more accurate abscissa of a rectangle rule for a power-law integrand. A grid whose points sat *on* the thresholds instead overstates the exact  $P_{\text{HI}}$ ,  $P_{\text{HeI}}$  and  $P_{\text{HeII}}$  by factors 1.095, 1.016 and 1.030.

**Dayside dilution.** Every stellar beam of the code is multiplied by the dilution  $\xi_{\text{day}}$  of Table 1: the XUV grid in `set_energy_vectors`,  $F_{\text{XUV}} \rightarrow \xi_{\text{day}} F_{\text{XUV}}$ ; the stellar Ly $\alpha$  beam in `lya_rt.f90` and `excited_hydrogen.f90`; and the four far-ultraviolet bands through `fuv_band_flux`.

**Bands stated by their integrated flux.** Four far-ultraviolet bands lie almost entirely below the photon grid and enter as band-integrated energy fluxes at the planet's orbit rather than through the grid, in the band order of `water_photolysis.f90`: LW (912–1201 Å), B2 (the H I Ly $\alpha$  line at 1215.67 Å, band 1202–1230 Å), B3 (1231–1450 Å) and B4 (1451–2304 Å). The edges are fixed by the physics of the absorbers and not chosen: 912 Å is the Lyman edge, 1201 Å the red end of the  $\text{H}_2$  Lyman and Werner line list, and the B2/B3/B4 edges follow the  $\text{H}_2\text{O}$  branching ratios (Sec. 6.6). Each band flux has one of two sources. A key in `input.inp` states it (Stellar LW flux, Stellar Ly $\alpha$  flux, Stellar FUV B3 flux, Stellar FUV B4 flux), and a stated value, zero included, always stands. Where no key is given and the spectrum is a loaded file (`Spectrum type: Load`), the three continuum bands are integrated from that file by `sed_band_integrated_flux` (`sed_read.f90`),

$$F_{\text{band}} = \int_{\lambda_{\text{lo}}}^{\lambda_{\text{hi}}} F_{\lambda} d\lambda, \quad (26)$$

the exact integral of the piecewise-linear table, the flux interpolated at each clipped band edge before the trapezoid (a constant or a linear spectrum integrates exactly whatever the nodes), the file being already at the planet and read once for the three bands: LW on any molecular run, B3 and B4 when the oxygen chemistry is on. A file that does not reach both edges of a band the run needs stops the run and asks for the key. The two analytic spectrum types cannot supply these bands (a power law is the XUV fit extrapolated downward, a Planck curve the photosphere, while the stellar far-ultraviolet is chromospheric), and the ionizing grid retained by `read_sed` ends at the lowest active threshold, so the file is re-read for them. Band B2 is never integrated: a  $\text{Ly}\alpha$  line flux reconstructed from observations is a better number than a trapezoid over whatever rows the file has across  $30\text{ \AA}$ , so it is always a key. `fuv_band_flux` returns  $\xi_{\text{day}} F_{\text{band}}$ , the dilution being applied where the beam is used so that the stated or integrated values keep their meaning, the band flux at the planet's orbit; the setup report states all four and names the source of each.

### 3.2 Photoionization cross sections

All cross sections are carried in units of  $10^{-18}\text{ cm}^2$  (`cross_sec.f90`) and are evaluated once, at the grid centers, into the vectors `s_hi`, `s_hei`, `s_heii`, `s_heiTR`, `s_h2` and the table `sigma_tab`.

**Hydrogenic ions** (H I, He II), function `sigma`. With  $\epsilon = \sqrt{E/E_{\text{th}} - 1}$ ,

$$\sigma(E) = \frac{6.3}{Z^2} \left( \frac{E_{\text{th}}}{E} \right)^4 \frac{\exp[4 - 4 \arctan(\epsilon)/\epsilon]}{1 - \exp(-2\pi/\epsilon)}, \quad E > E_{\text{th}}, \quad (27)$$

and  $6.3/Z^2$  exactly at threshold, zero below it.  $E_{\text{th}}$  is the *measured* potential of the ion, the same energy the grid uses as a band edge and the photoelectron is charged against.

**He I ground singlet** The Verner et al. (1996) single-shell fit,

$$x = \frac{E}{E_0} - y_0, \quad z = \sqrt{x^2 + y_1^2}, \quad Q = 5.5 - P/2, \quad \sigma = \sigma_0 [(x-1)^2 + y_w^2] z^{-Q} \left( 1 + \sqrt{z/y_a} \right)^{-P}, \quad (28)$$

with the published parameters  $(E_0, \sigma_0, y_a, P, y_w, y_0, y_1) = (13.61, 949.2, 1.469, 3.188, 2.039, 0.4434, 2.136)$  and  $E_{\text{th}}$  taken as the global He I threshold (it only gates the fit). The legacy ATES two-term fit  $0.6935/[(E/100)^{1.82} + (E/100)^{3.23}]$  is restored by `ates_photoion_rate` (default off).

**He  $2^3\text{S}$  metastable** Two forms of (28) joined by a log-linear bridge: wing A from the 4.768 eV threshold to the Cooper minimum at 34.70 eV, wing B from the resonance-averaged bump at 45.59 eV upward, and a power law joining them continuously in between. Each wing is a fit at the few-percent level to the Norcross (1971) and TOPbase background.

**Metal ions** `metal_photoion_sigma` returns the *shell sum* over the seven subshells of Verner et al. (1996) and Verner & Yakovlev (1995); the 1995 form is (28) with  $y_0 = y_1 = 0$  and  $Q = 5.5 + \ell - P/2$ . Above the  $E_{\text{max}}$  of the 1996 outer-shell table the handover to the 1995 subshell fit is made, as in Verner's own `phfit2`; extrapolating the outer-shell fit is not conservative (the Fe I row rises as  $E^{2-Q}$  and reaches 24.9 Mb at 1240 eV against a true 0.51 Mb). An inner-shell absorption is charged to one ionization of the absorbing stage with a photoelectron of  $h\nu - E_{\text{th}}(\text{outer})$ , which approximates the Auger cascade and undercounts the charge state by at least one stage; this is the approximation the published exoplanet photochemistry models make (Cecchi-Pestellini et al., 2009;

Locci et al., 2022). Fluorescence is neglected: with  $E_{\text{top}} = 1240$  eV only C, N, O and Na I can have a K hole, and their K yields (Krause, 1979) are 0.28, 0.52, 0.83 and 2.3 percent.

**H<sub>2</sub>** `sigma_H2` is a piecewise *measured* total photoabsorption:  $k_{\text{bx}} \times$  Backx et al. (1976) Table 1 over 15.4–18 eV with  $k_{\text{bx}} = 1.016415$ , Samson & Haddad (1994) Table 1 over 18–300 eV, and  $k_{\text{hi}} \times$  the Yan et al. (1998) eq. (19)  $E^{-7/2}$  sum-rule tail above 300 eV with  $k_{\text{hi}} = 0.885805$ , all interpolated in  $\log \sigma$  against  $\log E$ . No piece of the Yan fit survives below 300 eV: its three published pieces do not meet at their own join energies, stepping by a factor 1.719 at 18 eV and 0.619 at 85 eV.

**Opacity models** `opacity_models.f90` dispatches the cross sections through a model selected in the optional `opacity.inp`: A analytic (the curves above, the default), C constant above threshold, P constant with the Robinson & Catling (2012) pressure broadening  $f(p) = 1 + a(p/p_{\text{piv}})^n$ , and T a two-column table. In model P the multiplier  $f(p)$  of a cell is stored in `opa_pf(j)` and carried identically in the optical depth and in the local absorption, so that the photons a cell removes from the beam are the photons it absorbs.

### 3.3 Transfer along the radial ray

**The continuous form.** What the code evaluates is the frequency-resolved absorption of the incident beam along the substellar radial ray. With  $F_E(a)$  the incident flux per unit photon energy at the orbital distance  $a$  (for the power law,  $J_{\text{XUV}} = L_{\text{XUV}}/4\pi a^2$  of (23); for the Planck and table types, Sec. 3.1), the flux reaching radius  $r$  is

$$F_E(r) = F_E(a) e^{-\tau_E(r)}, \quad \tau_E(r) = \sum_a \sigma_{a,E} \int_r^{r_{\text{top}}} n_a(r') dr', \quad (29)$$

the sum running over every photoionizing absorber  $a$  and the column starting at zero at the outer edge of the grid,  $r_{\text{top}} = r(N + N_g)$ : no absorber is carried beyond the domain, which is where these expressions differ from Eq. (4) of Caldiroli et al. (2021), whose integral runs to  $a$ . The photoionization rate of one particle of absorber  $a$  and the photoheating rate of the gas are then

$$\Gamma_a(r) = \int_{E_{\text{th},a}}^{\infty} \frac{F_E(r)}{E} \sigma_{a,E} dE, \quad \mathcal{H}(r) = \sum_a n_a(r) \int_{E_{\text{th},a}}^{\infty} F_E(r) \left(1 - \frac{E_{\text{th},a}}{E}\right) \sigma_{a,E} dE, \quad (30)$$

which are Eq. (3) of Caldiroli et al. (2021) and its rate counterpart,  $1 - E_{\text{th}}/E$  being the fraction of the photon energy carried off by the photoelectron. Two factors the code adds to (29) are stated where they enter below: the pressure-dependent opacity weight  $f(p)$  of the optional model P, and the secondary-ionization heat fraction  $f_{\text{heat}}(E)$  that multiplies the photoelectron share in  $\mathcal{H}$  when secondary ionization is on (Sec. 3.5). The rest of this section is the discretization of (29) and (30) on the photon grid and on the radial cells, each line of which was read against `photoionization_field_at_cell_HHe` (`util_ion_eq.f90`), the one routine that forms the field and the rates of a cell.

The transfer is frequency-integrated over the grid of Sec. 3.1 and is pure absorption: there is no scattering term and no diffuse field in the XUV. For cell  $j$  the column of each absorber *outside* the cell is accumulated from the top down by `advance_starward_columns` (`util_ion_eq.f90`) on the same rectangle rule the grid uses,

$$N_a^{\text{out}}(j) = \sum_{j' > j} n_a(j') \Delta r_{j'} R_0 f(p_{j'}), \quad (31)$$

and the cell's *own* depth is formed from its own density rather than as a difference of two columns, which is the same number in exact arithmetic and loses its digits to cancellation for a thin cell. The two optical depths at photon energy  $E_k$  are

$$\tau_k^{\text{out}} = \sum_a \sigma_{a,k} N_a^{\text{out}}, \quad \Delta\tau_k = \sum_a \sigma_{a,k} n_a(j) \Delta r_j R_0 f(p_j), \quad (32)$$

the sum running over H I, the He I ground singlet, He II, He 2<sup>3</sup>S when tracked, H<sub>2</sub> when the molecular chemistry is on, and every photoionizable metal stage.

The attenuated flux the cell's *rate* sees is the *mean* of the beam over the cell, not its value at either face (cell\_mean\_attenuation in utilities.f90). Since the densities are uniform inside a cell,  $\tau(s) = \tau^{\text{out}} + s \Delta\tau$  and

$$\langle e^{-\tau} \rangle_{\text{cell}} = e^{-\tau^{\text{out}}} \frac{1 - e^{-\Delta\tau}}{\Delta\tau}, \quad (33)$$

the second factor being absorbed\_fraction\_per\_unit\_depth, which switches to its series  $1 - \Delta\tau/2 + \Delta\tau^2/6$  below  $\Delta\tau = 10^{-8}$ . Only this mean makes the photons the cell absorbs equal the photons the beam loses across it; a one-point rule at the inner face is low by 42 percent at  $\Delta\tau = 1$ , the depth a cell carries at the ionization front. The optional alpha form of the 2-D approximation replaces (33) by (22), which has no closed form.

The working flux of the cell is therefore

$$\mathcal{F}_k(j) = F_{\text{XUV},k} \langle e^{-\tau} \rangle_{k,\text{cell}} f(p_j), \quad (34)$$

the model-P factor being folded once into the shared flux weight rather than into each cross section, which is identical because every integrand below is linear in exactly one cross section.

### 3.4 Photoionization and photoheating rates

Every integral is the rectangle sum over the grid,  $\sum_k (\cdot) \Delta E_k$ , with  $\Delta E_k$  the exact bin width, so  $\sum_k \Delta E_k = E_{\text{top}} - E_{\text{floor}}$  exactly. For absorber  $a$  the photoionization rate per particle and the photoheating rate of *one* particle of that absorber are

$$P_a = \sum_k \mathcal{F}_k \frac{\sigma_{a,k}}{E_k} \Delta E_k \times 10^{-18} \text{ erg}^{-1} \text{ eV}, \quad (35)$$

$$h_a^{(1)} = \sum_k \mathcal{F}_k \left( 1 - \frac{E_{\text{th},a}}{E_k} \right) f_{\text{heat}}(E_k) \sigma_{a,k} \Delta E_k \times 10^{-18} \text{ [erg s}^{-1}\text{]}, \quad (36)$$

where the factor  $1 - E_{\text{th}}/E$  is the share of the photon left as photoelectron kinetic energy (photoelectron\_share; the key photoheat\_photon\_fraction, default off, replaces it by a constant fraction), and  $f_{\text{heat}}$  is the secondary-ionization heat fraction of Sec. 3.5. The heating of the cell is the contraction of these one-particle rates with the composition,

$$\mathcal{H} = \sum_a n_a h_a^{(1)}, \quad (37)$$

performed in exactly one place (photoheating\_of\_cell) and split by absorber into the six columns H I, He I, He II, He 2<sup>3</sup>S, H<sub>2</sub>, metals. The heating efficiency written to the output is

$$\eta_{\text{heat}} = \frac{\mathcal{H}}{\sum_k \mathcal{F}_k \left( \sum_a \sigma_{a,k} n_a \right) \Delta E_k \times 10^{-18}}, \quad (38)$$

the ratio of the energy deposited to the energy absorbed, set to zero in a transparent or unilluminated cell.

**Molecular channels.**  $H_2$  absorption is resolved into four mutually exclusive final states whose cross sections are *shares* of  $\sigma_{H_2}$ , so the opacity and the total  $H_2$  destruction rate are unchanged. Each channel is charged its own threshold in (36), and the two deposits that are not a photoelectron, the kinetic energy release of the double channel and the internal excitation the neutral channel radiates away, are stated with the channels themselves in Sec. 6.2.

**Self-consistency of the field with the composition.** Nothing in (32) depends on any cell inside  $j$ , which is what lets the equilibrium sweep march outside in and re-form the field of one cell against its own composition (Sec. 4.4); the cell mean (33) is formed once per pass so that no second attenuation is stacked on the first. The rates are constants of the cell's nonlinear solve, carrying the field dependence inside the residual meaning a re-integration of the photon grid at every residual evaluation.

### 3.5 Secondary ionization by photoelectrons

A photoelectron faster than  $E_{\text{sec}} = 30 \text{ eV}$  does not thermalize completely: part of its energy ionizes further neutrals. `electron_energy_degradation.f90` divides the labor between two published determinations. Shull & van Steenberg (1985) supply the energy budget of the ionization channels as a function of the ionized fraction  $x$ ,

$$\phi_{HI}(x) = 0.3908 (1 - x^{0.4092})^{1.7592}, \quad \phi_{HeI}(x) = 0.0554 (1 - x^{0.4614})^{1.6660}, \quad (39)$$

whose form vanishes at  $x = 1$  where no neutral is left. Dalgarno et al. (1999) supply the heating fraction and everything that depends on molecular hydrogen; their Table 4 mean energies per ion pair,  $W(x, E_0) = a(1 + cx^b)$ , give the energy dependence,

$$f_{HI}(x, E_0) = \phi_{HI}(x) \frac{W_{H^+}(x, 1 \text{ keV})}{W_{H^+}(x, E_0)}, \quad f_{HeI}(x, E_0) = \phi_{HeI}(x) \frac{W_{He^+}(x, 1 \text{ keV})}{W_{He^+}(x, E_0)}, \quad (40)$$

so that the high-energy limit is exactly Shull & van Steenberg's. The division of that energy among the targets is recomputed on the cell's own densities, using the collision weight 1.89 that Dalgarno et al.'s eqs. (9) and (10) imply (their coefficients 1.89 and 0.53 are reciprocal to 0.17 percent), with  $r_{\text{ref}} = n(\text{He I})/n(\text{H I}) = 0.1$  the reference composition of the fits:

$$w = (n_{HI} + 1.89 n_{H_2}) r_{\text{ref}} f_{HI} + n_{HeI} f_{HeI}, \quad c_{HI} = \frac{f_{\text{ion}} r_{\text{ref}} f_{HI}}{w}, \quad c_{H_2} = 1.89 c_{HI}, \quad c_{HeI} = \frac{f_{\text{ion}} f_{HeI}}{w}, \quad (41)$$

with  $f_{\text{ion}} = f_{HI} + f_{HeI}$ . These coefficients are *per target particle* (units  $\text{cm}^3$ ), so  $n_{HI} c_{HI} + n_{H_2} c_{H_2} + n_{HeI} c_{HeI} = f_{\text{ion}}$  identically and nothing is divided by a vanishing neutral density. The heat fraction is

$$f_{\text{heat}}(E_0) = 1 - [1 - \eta_{\text{mix}}(x, E_0)](1 - x) + f_{\text{diss}} + f_{\text{vib}} + f_{\text{fluor}}, \quad (42)$$

with  $\eta_D = 1 + (\eta_{D,0} - 1)/(1 + cx^b)$  their eq. (14) and  $\eta_{\text{mix}}$  their eq. (12) mixing of the H/He and  $H_2$ /He efficiencies. The factor  $(1 - x)$  is this code's closure and not either source's: the channels that compete with heat need a neutral to act on, and Dalgarno et al.'s form leaves 10 percent of the primary energy unthermalized even at  $x = 1$ . The three molecular terms (fragment kinetic energy, direct vibrational excitation of  $v = 1, 2$ , and the vibrational energy the B and C states return on fluorescing) vanish identically with  $n(H_2)$ ; the last two carry the collisional quench fraction of `h2_vibrational_relaxation.f90`, since a radiated excitation is not heat.

The secondary rates are added to the primary ones of (35): for an absorber  $a$  whose photoelectron energy exceeds  $E_{\text{sec}}$ ,

$$P_{\text{HI}}^{\text{sec}} = \sum_k \mathcal{F}_k \frac{\sigma_{a,k} n_a}{E_k} c_{\text{HI}}(E_k) \frac{E_k - E_{\text{th},a}}{E_{\text{th}}(\text{HI})} \Delta E_k \times 10^{-18} \text{ erg}^{-1} \text{ eV}, \quad (43)$$

and likewise for He I and H<sub>2</sub>; the H<sub>2</sub> secondaries additionally make one proton per 22 H<sub>2</sub><sup>+</sup> ions (Dalgarno et al., 1999, after their eq. 10), which is an added yield and not a share.

**Staged activation.** The key `Secondary_ionization` is *on* by default (`use_sec_ion = .true.`) but *staged*: the runtime flag `sec_ion_active` stays false until the wind has first converged without the coupling, because from a cold initial condition the base feedback amplifies the startup transient into a runaway (`EXHALE_main.f90`, the block that sets `sec_ion_active = .true.` once any of the mass-flux, step-change, plateau or steady gates is met, and again before the JFNK handover so that Newton is never asked to solve a system the marched state does not approximately satisfy). `Secondary_ionization: Immediate` applies it from step 0 and `False` disables it entirely. Photoelectrons between 13.6 and 30 eV are still treated as depositing all their energy as heat, which is a limit of the published tables and not a physical threshold.

### 3.6 Reprocessed and non-stellar fields

Three fields in the code are not the direct stellar beam.

**Recombination radiation.** The ionizing photons emitted when H II, He II and He III recombine are absorbed on the spot (`recombination_radiation_absorbed` in `util_ion_eq.f90`): each cell keeps the fraction  $1 - e^{-\tau}$  of every channel at its own optical depth and shares it among H I, He I, He II, H<sub>2</sub> and the metals in the ratio of their absorption coefficients; the part re-absorbed by the recombined species itself is the on-the-spot cancellation of the net coefficients (58). The H II ground captures (`H_rec_escape`, default on) can be taken only by H I and the low-potential metals. With `He_rec_coupling` (default on) each He II cascade exit is represented at one photon energy: the ground-capture continuum starting at  $E_{\text{th}}(\text{HeI})$ , the  $2^1P \rightarrow 1^1S$  resonance line at 21.2 eV, the  $2^3S \rightarrow 1^1S$  line at 19.8 eV, and the part of the  $2^1S$  two-photon continuum above the hydrogen edge. The last is not a line: integrating the Drake et al. (1969) shape over 13.598–20.62 eV gives 0.5564 ionizing photons per decay carrying 8.9646 eV, a mean in-band photon energy of 16.110 eV and hence 2.512 eV of photoelectron energy per photon, against 3.511 eV for a flat distribution over the same window. The He III channels are the ground capture at 54.4 eV, the direct capture into He II  $n = 2$  (Seaton, 1959), whose continuum starts at 13.605 eV, He II Ly $\alpha$  at 40.81 eV, and the He II  $2s$  two-photon continuum, whose spectrum is the fit of Nussbaumer & Schmutz (1984) carried in three bands (13.6–15.4, 15.4–24.6 and above 24.6 eV: 0.112, 0.577 and 0.736 photons per decay). Every case-B capture of He III ends in  $2s$  (the fraction of Pengelly 1964) or  $2p$ , and  $2s$  decays by two photons at  $526.823 \text{ s}^{-1}$  (Drake 1986 eq. 27 at  $Z = 2$ , with the  $\alpha$ -particle mass) unless a proton or He<sup>2+</sup> collision moves it to  $2p$  first (Pengelly & Seaton, 1964).

**Lyman-Werner band.** The 912–1201 Å beam is solved once for all of its absorbers together (`fuv_lw_photon_field`), because an H<sub>2</sub>O molecule cannot absorb a photon an H<sub>2</sub> line has already taken. Its transmission down to a face is  $T = (1 - A)e^{-\tau_c}$  with  $A = \int \sigma_{\text{pump}} f_{\text{shield}}(N_{\text{H}_2}) n_{\text{H}_2} dr$  summed along the path, each cell

contributing at its own  $(T, n)$  the same integral its own line rate takes (the table is built for a uniform slab, and reading  $A$  of the whole column at the temperature of the cell the beam has reached would count as taken photons the lines of the cells above have not taken, or the reverse), and across one cell the identity

$$T^{\text{out}} - T^{\text{in}} = e^{-\tau^{\text{out}}} \left[ (1 - A^{\text{out}}) (1 - e^{-\Delta\tau}) + \Delta A e^{-\Delta\tau} \right] \quad (44)$$

is exact, so the photons the cell removes split into a continuum share and a line share with nothing left over. The continuum factor  $e^{-\tau_c}$  is the Draine & Bertoldi (1996) eq. (40) term, identically 1 where the oxygen option is off. Line self-shielding is the tabulated function of  $(T, n, N_{\text{H}_2})$  of Sec. 6.3. CO is shielded on the same beam by the Visser et al. (2009) function  $\Theta(N_{\text{CO}}, N_{\text{H}_2})$  (`co_self_shielding_table.f90`).

**Ly $\alpha$  and the Balmer continuum** are described with the excited-hydrogen model in Sec. 4.6.

## 4. Ionization and chemical state

### 4.1 The species carried

The composition of a cell is the vector  $f_{\text{sp}}(j, \cdot)$  of species fractions, laid out once in `species_table.f90`:

| column(s) | species                                                                                       | note                                                                     |
|-----------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1–2       | H I, H II                                                                                     | fractions of the hydrogen nuclei                                         |
| 3–5       | He I, He II, He III                                                                           | fractions of the helium nuclei; column 3 <i>contains</i> the metastable  |
| 6         | He 2 <sup>3</sup> S                                                                           | the metastable triplet, tracked when Include He23S? is true (default on) |
| 7–33      | the 27 metal ion stages                                                                       | C, O, N, Mg, Si, Ca, Fe in three stages; Na, K, S in two                 |
| 34–37     | H <sub>2</sub> , H <sub>2</sub> <sup>+</sup> , H <sub>3</sub> <sup>+</sup> , HeH <sup>+</sup> | molecular option ( <code>thereis_mol</code> , default off)               |
| 38–40     | OH, H <sub>2</sub> O, CO                                                                      | oxygen option ( <code>thereis_oxychem</code> , default off)              |

Ten elements carry a nuclear abundance  $n_X/n_{\text{H}}$ , set by `metals.inp` (`metals_input_read.f90`) or by a lower-atmosphere handoff. Masses are in units of the hydrogen *atom*: helium weighs  $m_{\text{He}}/m_{\text{H}} = 3.9715259$  and the metal weights are the CIAAW values converted from the unified mass unit by  $1 \text{ u} = 0.99223573 m_{\text{H}}$ .

### 4.2 The cell system

For each cell the code solves a system of algebraic balance equations at fixed temperature, density and radiation field. The unknowns are stage *fractions*, and one stage of each element is closed as the remainder rather than being an unknown:  $n(\text{HI}) = (1 - x_{\text{H}^+})n_{\text{H}}$ ,  $n(\text{HeI}) = (1 - x_{\text{He}^+} - x_{\text{He}^{2+}})n_{\text{He}}$ , and for a metal element  $n(X^0) = (1 - x_{X^+} - x_{X^{2+}})n_X$ . The layout is

| rows                    | contents                                                                               |
|-------------------------|----------------------------------------------------------------------------------------|
| 1–3                     | $x_{\text{H}^+}, x_{\text{He}^+}, x_{\text{He}^{2+}}$                                  |
| 4                       | He $2^3\text{S}$ (triplet systems) <i>or</i> $\text{H}_2$ (molecular systems)          |
| 5–8                     | $\text{H}_2^+, \text{H}_3^+, \text{HeH}^+, \text{He } 2^3\text{S}$ (molecular systems) |
| $m_{\text{base}} \dots$ | two rows per metal element ( <code>metal_row_base()</code> )                           |
| $i_{\text{ox}} \dots$   | OH and $\text{H}_2\text{O}$ (oxygen option)                                            |

with  $m_{\text{base}} = 4$ , or 5 with the triplet. In the molecular layout  $i_{\text{ox}} = 8$ , or 9 with the triplet, and  $m_{\text{base}} = i_{\text{ox}}$ , raised by two when the oxygen chemistry is on (`metal_row_base`, `oxygen_row_base`: one definition, read by the residual and by the driver alike). Seven residual routines cover the configurations: `System_H`, `System_HeH`, `System_HeH_metals`, `System_HeH_TR`, `System_HeH_TR_metals`, `System_HeH_mol` and `System_HeH_mol_metals`. The shared rows live once, in `ion_residual_core.f90`.

**The H/He rows.** With photoionization rates  $\Gamma$ , recombination coefficients  $\alpha$  (case B plus the ground captures whose photons are not re-absorbed on the spot, (58)) and electron-impact ionization coefficients  $\beta$ , `heh_rows` writes

$$0 = n_{\text{HI}}\Gamma_{\text{HI}} + (n_{\text{HI}}\beta_{\text{HI}} - \alpha_{\text{HII}}n_{\text{HII}})n_e, \quad (45)$$

$$0 = n_{\text{HeI}}\Gamma_{\text{HeI}} + (n_{\text{HeI}}\beta_{\text{HeI}} - \alpha_{\text{HeII}}n_{\text{HeII}})n_e, \quad (46)$$

$$0 = n_{\text{HeII}}\Gamma_{\text{HeII}} + (n_{\text{HeII}}\beta_{\text{HeII}} - \alpha_{\text{HeIII}}n_{\text{HeIII}})n_e, \quad (47)$$

with  $n_e$  the charge sum of every tracked ion,  $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + \sum_i Z_i n_{\text{m},i} + n_{\text{H}_2^+} + n_{\text{H}_3^+} + n_{\text{HeH}^+}$  (`calc_ne`), so charge neutrality is imposed identically rather than solved for. Each metal element contributes

$$0 = n_{X^0}\Gamma_{X^0} + (n_{X^0}\beta_{X^0} - \alpha_{X^+}n_{X^+})n_e, \quad 0 = n_{X^+}\Gamma_{X^+} + (n_{X^+}\beta_{X^+} - \alpha_{X^{2+}}n_{X^{2+}})n_e, \quad (48)$$

the second row being replaced by  $x_{X^{2+}} = 0$  for a two-stage element, and the whole pair by  $x_{X^+} = x_{X^{2+}} = 0$  for an element with zero abundance.

**The triplet row.** With the metastable tracked, the He rows are rewritten in the form of Oklopčić & Hirata (2018) (rows 2 and 3 summed, the triplet and the singlet ground state distinguished) and the fourth row is the  $2^3\text{S}$  balance,

$$0 = -n_{2^3\text{S}}\Gamma_{2^3\text{S}} + n_e [n_{\text{HeII}}\alpha_{2^3\text{S}} + n_{1^1\text{S}}q_{13} - n_{2^3\text{S}}(q_{31g} + q_{31a} + q_{31b}) - n_{2^3\text{S}}\beta_{2^3\text{S}}] - n_{2^3\text{S}}(A_{31} + n_{\text{HI}}Q_{31}), \quad (49)$$

where  $q_{13}$  is the collisional feed of  $2^3\text{S}$  out of the singlet ground state: the direct excitation  $1^1\text{S} \rightarrow 2^3\text{S}$  plus the excitation of every higher triplet term  $n \leq 5$  of Bray et al. (2000, Table 2), each of which cascades into  $2^3\text{S}$  within the triplets (0.019 of the direct rate at 5000 K, 0.107 at  $10^4$  K, 0.331 at  $2 \times 10^4$  K);  $q_{31g} = q_{13}^{\text{direct}}(g_1/g_3)e^{E_{13}/kT}$  is the detailed-balance reverse of the direct excitation (written with the Boltzmann factor cancelled, so it stays finite where  $q_{13}$  underflows, and not negligible in cool gas:  $q_{31g}/(q_{31a} + q_{31b}) = 0.055$  at  $10^4$  K, 0.15 at 5000 K, 0.56 at 3000 K and 2.7 at 2000 K);  $q_{31a}$  and  $q_{31b}$  are the collisional conversions to  $2^1\text{S}$  and to  $2^1\text{P}$ , the latter together with every singlet level above it (0.019 of  $q_{31a} + q_{31b}$  at  $10^4$  K), all of the form  $C \Upsilon e^{-E/kT}/(g_i\sqrt{T})$  with the Maxwellian constant  $C = h^2/[(2\pi m_e)^{3/2}k^{1/2}] = 8.629132 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1} \text{ K}^{1/2}$ , the effective collision strengths of Bray et al. (2000, Tables 2 and 3) for the network's own transitions

and for the higher levels of the sums alike (interpolated inside the tabulated  $10^{3.75}$ – $10^{5.75}$  K and continued outside it with the temperature dependence of the CHIANTI v11.0.2 strength of the same transition, scaled to the table end) and the NIST level energies,  $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$  the forbidden radiative decay,  $\beta_{2^3S}$  the electron-impact ionization of the metastable at its own 4.8 eV threshold, and  $Q_{31}$  the *total* He( $2^3S$ )+H ionization rate coefficient. Both channels of  $Q_{31}$  quench the metastable, so the sink takes the sum; only the Penning branch  $f_P = 0.9$  leaves a lasting proton, and it appears as the source  $f_P Q_{31} n_{2^3S} n_{\text{HI}}$  in the  $\text{H}^+$  row. The remaining 10 percent is associative ionization to  $\text{HeH}^+$ , which the atomic systems drop (in an atomic gas  $\text{HeH}^+$  dissociatively recombines straight back) and the molecular system carries explicitly.

**The molecular rows.** `mol_heh_rows` adds four species balances on the Koskinen et al. (2022) Table 1 network (Sec. 6.1). Schematically, writing  $g$  for photorates and  $k_n$  for the Table-1 coefficients,

$$\begin{aligned} \text{H}_2 : \quad 0 = & k_6 n_e n_{\text{H}_3^+} + k_9 n_{\text{H}_2^+} n_{\text{HI}} + k_{11} n_{\text{H}_3^+} n_{\text{HI}} + k_{15} n_{\text{HI}}^2 \\ & - [g_{\text{H}_2} + g_{\text{LW}} + (k_{10} + k_{13}) n_{\text{HII}} + k_{12} n_{\text{tot}} + k_{14} n_e + k_8 n_{\text{H}_2^+} + (k_{17} + k_{20} + k_{23}) n_{\text{HeII}} + k_{18} n_{\text{HeH}^+} + k_{\text{H}_2}^{\text{ion}} n_{2^3S}] n_{\text{H}_2}, \end{aligned} \quad (50)$$

$$\begin{aligned} \text{H}_2^+ : \quad 0 = & (g_{\text{H}_2} - g^{\text{di}} - g^{\text{dd}} - g^{\text{nd}}) n_{\text{H}_2} + k_{10} n_{\text{HII}} n_{\text{H}_2} + k_{11} n_{\text{H}_3^+} n_{\text{HI}} + k_{19} n_{\text{HeH}^+} n_{\text{HI}} + k_{23} n_{\text{HeII}} n_{\text{H}_2} \\ & + f_P k_{\text{H}_2}^{\text{ion}} n_{2^3S} n_{\text{H}_2} - (k_5 n_e + k_8 n_{\text{H}_2} + k_9 n_{\text{HI}}) n_{\text{H}_2^+}, \end{aligned} \quad (51)$$

$$\text{H}_3^+ : \quad 0 = k_8 n_{\text{H}_2^+} n_{\text{H}_2} + k_{13} n_{\text{HII}} n_{\text{H}_2} + k_{18} n_{\text{HeH}^+} n_{\text{H}_2} - [(k_6 + k_7) n_e + k_{11} n_{\text{HI}}] n_{\text{H}_3^+}, \quad (52)$$

$$\text{HeH}^+ : \quad 0 = k_{20} n_{\text{HeII}} n_{\text{H}_2} + (1 - f_P) n_{2^3S} (Q_{31} n_{\text{HI}} + k_{\text{H}_2}^{\text{ion}} n_{\text{H}_2}) - (k_{16} n_e + k_{18} n_{\text{H}_2} + k_{19} n_{\text{HI}}) n_{\text{HeH}^+}, \quad (53)$$

and the  $\text{H}^+$  row gains  $(g^{\text{di}} + 2g^{\text{dd}}) n_{\text{H}_2} + k_9 n_{\text{H}_2^+} n_{\text{HI}} + k_{17} n_{\text{HeII}} n_{\text{H}_2} - (k_{10} + k_{13}) n_{\text{HII}} n_{\text{H}_2}$ . The factor 2 on the double photoionization branch is the stoichiometry, two protons out of one event, while the  $\text{H}_2$  destroyed is still one per event, which is why (50) is not doubled with it.

**The oxygen rows.** With the oxygen option (Sec. 6.6; `thereis_oxochem`, solved only by `System_HeH_mol_metals` since oxygen is a metal element) two more unknowns join the cell system,  $x_{\text{OH}} = n_{\text{OH}}/n_{\text{O}}$  and  $x_{\text{H}_2\text{O}} = n_{\text{H}_2\text{O}}/n_{\text{O}}$  with  $n_{\text{O}}$  the oxygen family of the cell, and `oxygen_carrier_rows` adds their balances on the reactions O1 ( $\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$ ), O2 ( $\text{O} + \text{H}_2 \rightarrow \text{OH} + \text{H}$ ), their reverses O1r and O2r, the three  $\text{H}_2\text{O}$  photolysis channels O3 ( $\rightarrow \text{OH} + \text{H}$ ), O4 ( $\rightarrow \text{H}_2 + \text{O}(^1\text{D})$ ) and O5 ( $\rightarrow \text{O} + \text{H} + \text{H}$ ), and the OH photolysis O7 ( $\rightarrow \text{O} + \text{H}$ ):

$$\text{OH} : \quad 0 = k_{1r} n_{\text{H}_2\text{O}} n_{\text{HI}} + k_2 n_{\text{O}^0} n_{\text{H}_2} + (J_3 + J_4) n_{\text{H}_2\text{O}} - k_{1r} n_{\text{OH}} n_{\text{H}_2} - k_{2r} n_{\text{OH}} n_{\text{HI}} - J_7 n_{\text{OH}}, \quad (54)$$

$$\text{H}_2\text{O} : \quad 0 = k_1 n_{\text{OH}} n_{\text{H}_2} - k_{1r} n_{\text{H}_2\text{O}} n_{\text{HI}} - (J_3 + J_4 + J_5) n_{\text{H}_2\text{O}}, \quad (55)$$

and the  $\text{H}_2$  row (50) gains the exchange with the water cycle,  $-(k_1 n_{\text{OH}} + k_2 n_{\text{O}^0}) n_{\text{H}_2} + (k_{1r} n_{\text{H}_2\text{O}} + k_{2r} n_{\text{OH}}) n_{\text{HI}}$ . Here  $n_{\text{O}^0}$  is the *free* atomic oxygen, the remainder of the family once the nuclei bound in OH,  $\text{H}_2\text{O}$  and CO are removed,  $n_{\text{O}^0} = n_{\text{O}} - n_{\text{OH}} - n_{\text{H}_2\text{O}} - n_{\text{CO}}$ , which the metal block then splits over O I/O II/O III from its own balance: the O I column of the output means free atomic oxygen when the option is on. The hydrogen closure gains the nuclei the carriers hold,  $n_{\text{OH}} + 2n_{\text{H}_2\text{O}}$ . The forward coefficients  $k_1$  and  $k_2$  are the Baulch et al. (2005) evaluations ( $k_1 = 3.6 \times 10^{-16} T^{1.52} e^{-1740/T}$ ,  $k_2 = 6.34 \times 10^{-12} e^{-4000/T} + 1.46 \times 10^{-9} e^{-9650/T} \text{ cm}^3 \text{ s}^{-1}$ ); the reverses are not transcribed but generated by detailed

balance,  $k_r = k_f/K_c(T)$  with  $K_c$  from the NIST-JANAF Shomate Gibbs energies (Chase, 1998) of H, H<sub>2</sub>, O, OH, H<sub>2</sub>O, CO and C (oxygen\_rates.f90), so that the hot limit of the network is the chemical equilibrium of the same thermochemistry.  $J_3, J_4, J_5$  and  $J_7$  are the band-resolved photolysis rates of Sec. 6.6 summed over the four far-ultraviolet bands with their quantum yields. The O4 branch appears in the OH row as if it made OH directly: its product O(<sup>1</sup>D) has exactly one sink in the network, O6 (O(<sup>1</sup>D)+H<sub>2</sub> → OH+H), with a lifetime of  $1.6 \times 10^{-3}$  s at the HD 189733 b base, so its local steady state is exact, the flux through O6 equals the O(<sup>1</sup>D) production  $J_4 n_{\text{H}_2\text{O}}$ , and the pair O4+O6 is one channel whose net effect on H<sub>2</sub> cancels (O4 makes one, O6 consumes one), which is why only O1 and O2 enter the H<sub>2</sub> row; the O(<sup>1</sup>D) density is evaluated for the output only. CO is *not* a row of this system: it is a transported carrier (Sec. 6.5) with no formation channel in the network, destroyed one-sidedly by He<sup>+</sup>+CO → C<sup>+</sup>+O+He at the Langevin value  $1.6 \times 10^{-9}$  cm<sup>3</sup> s<sup>-1</sup> (UMIST RATE22 entry 4068; Millar et al., 2024) and by photodissociation on the Lyman-Werner beam with the Visser et al. (2009) shielding of the star-ward CO and H<sub>2</sub> columns, both inside the carrier step; the cell system sees the CO of the cell as a frozen background and takes the He<sup>+</sup> it consumes,  $-k_{\text{D1}} n_{\text{CO}} n_{\text{HeII}}$ , in the He<sup>+</sup> row, so the one reaction is complete over the pair of operators and counted once in each. Each oxygen row is divided by its own turnover rate  $n_{\text{O}}[(k_1 + k_{1r} + k_2 + k_{2r})n_{\text{H}} + J_3 + J_4 + J_7]$  (OH) and  $n_{\text{O}}[(k_1 + k_{1r})n_{\text{H}} + J_3 + J_4 + J_5]$  (H<sub>2</sub>O) before the Newton solve, as the molecular rows are, because the oxygen family is  $\sim 10^{-4}$  of the hydrogen and the block would otherwise sit far below the helium one.

### 4.3 The rate coefficients

All of the following live in Cool\_coeff.f90 unless stated. The default set is selected by Legacy\_HHe\_rates: False (default) and Atomic\_rate\_set: default; the two alternatives are the Hui & Gnedin (1997) fits and the Koskinen et al. (2022) Table 1 fits, and the three meet at the accessor functions at the end of that module.

**Recombination.** Radiative recombination follows the Badnell (2006, 2023 update) total ground-level fit

$$\alpha_{\text{RR}}(T) = \frac{A}{\sqrt{T/T_0} \left(1 + \sqrt{T/T_0}\right)^{1-B'} \left(1 + \sqrt{T/T_1}\right)^{1+B'}}, \quad B' = B + C e^{-T_2/T}, \quad (56)$$

implemented as rr\_badnell (its dummy T0 keeps the published notation and is not the code's temperature scale). Dielectronic recombination of He II adds the adf09 three-term sum  $T^{-3/2} \sum_i c_i e^{-E_i/T}$ . Case B is formed by subtracting the capture into the ground level,  $\alpha_1$ , from the Milne relation on the ground-state photoionization cross section the radiative transfer itself absorbs with (hydrogenic for H I and He II, the Verner et al. (1996) fit for He I),

$$\alpha_1(T) = \frac{g_n}{g_+} \sqrt{\frac{2}{\pi}} \frac{1}{c^2 (m_e k T)^{3/2}} \int_0^\infty (I+E)^2 \sigma(I+E) e^{-E/kT} dE, \quad (57)$$

( $g_n/g_+ = 2, 1/2, 2$  for H I, He I, He II; ground\_capture\_milne, a 24-point Gauss–Legendre rule in  $\ln(1+E/I)$  accurate to  $10^{-10}$ ), so that the ground captures and the absorption of the photons they emit are one process and  $\alpha_B = \alpha_A - \alpha_1$  for H II, He II and He III.  $\alpha_1$  is 0.998 of Hummer (1994) for H I, 0.999 of his hydrogenic scaling for He II and 1.006–1.011 of Hummer & Storey (1998) for He I. With the metastable tracked, the case-B captures of He II are split between the triplets (all ending in 2<sup>3</sup>S) and the excited singlets (ending in 1<sup>1</sup>S) in the ratio of the two sums of Hummer & Storey (1998, Table 5), a triplet share of 0.755 at 10 K, 0.771 at 10<sup>4</sup> K and 0.781 at  $2.5 \times 10^4$  K.

**The coefficients the balance uses.** A ground capture emits a photon above the edge of the recombined species; the part of those photons the same species re-absorbs in the cell is a capture and a re-ionization that cancel. The coefficients of the rows are therefore

$$\alpha_{\text{HII}} = \alpha_{\text{B}} + y_{\text{HI}} \alpha_{\text{I}}^{\text{H}}, \quad \alpha_{\text{HeII}} = \alpha_{\text{B}} + y_{\text{gnd}} \alpha_{\text{I}}^{\text{He}}, \quad \alpha_{\text{HeIII}} = \alpha_{\text{B}} + y_{\text{HeII}} \alpha_{\text{I}}^{\text{He}^+}, \quad (58)$$

$y_r = 1 - n_r \sigma_r \Delta r (1 - e^{-\tau}) / \tau$  the fraction of the photons not re-absorbed by  $r$  in the cell,  $\tau$  the cell optical depth at the edge of every absorber (ground\_capture\_escape\_weights).  $y \rightarrow 0$  is case B, where the gas is thick at the edge;  $y \rightarrow 1$  case A, in the thin outer wind. H\_rec\_escape (default on) governs  $y_{\text{HI}}$  and He\_rec\_coupling (default on) the two helium weights; a switch that is off sets its weight to zero. The photons taken by other absorbers ionize them (Sec. 3.6). The metal stages take Badnell RR + adf09 DR through the same two forms, with the one gap (Ca II, K-like) handled separately and noted at the code site.

**Collisional ionization.** Voronov (1997) Table I,

$$\beta(T) = A \frac{(1 + P\sqrt{U}) U^K e^{-U}}{X + U}, \quad U = \frac{\Delta E}{k_B T}, \quad (59)$$

with  $(\Delta E, P, A, X, K)$  equal to  $(13.6, 0, 2.91 \times 10^{-8}, 0.232, 0.39)$  for H I,  $(24.6, 0, 1.75 \times 10^{-8}, 0.180, 0.35)$  for He I and  $(54.4, 1, 2.05 \times 10^{-9}, 0.265, 0.25)$  for He II, and the metal stages take their own rows of the same table. The  $\Delta E$  here is Voronov's own fit coefficient, not the measured potential (Mg II 15.2 against 15.035 eV), for the metals as for H and He; the energy *removed* per ionization of H and He is the measured one,  $E_{\text{th}}$  in erg. Voronov's fits cover 1 eV–20 keV (3 eV for He II, 2 eV for N II and O II); below that, in the 2000–10<sup>4</sup> K of a photoionized wind, the same expression is evaluated, which keeps the exact threshold factor  $e^{-U}$  but continues the fitted prefactor where no data constrain it, and no accuracy is claimed there.

**Charge exchange.** charge\_exchange.f90 carries the reaction list of Huang et al. (2023) Table 4, 63 rows, plus two rows that are not in it, but not all of that table's rate coefficients. A rate that a source calculated or measured is carried as the source gives it, from the original where it could be read (among others Kingdon & Ferland 1996 with Butler & Dalgarno 1979 for N+H<sup>+</sup>, Dutta et al. 2001 and Watanabe et al. 2002 for Na+H<sup>+</sup>, Watanabe et al. for K+H<sup>+</sup>, Rutherford & Vroom 1972 for Fe+X<sup>+</sup>, Zhao et al. 2005 for S+H<sup>+</sup>, Satta et al. 2013 for Si+He<sup>+</sup>, Chenel et al. 2010 for S+C<sup>+</sup>, Kwolek et al. 2019 for Na+Ca<sup>+</sup>); where the source gives one direction only, the other is computed from it by detailed balance,  $k_r = k_f [Q(X^s)Q(Y^{t+1})] / [Q(X^{s+1})Q(Y^t)] e^{[I(X^s) - I(Y^t)] / kT}$  with NIST internal partition functions and ionization energies, in place of the fitted reverses Table 4 prints. Ten reverses of Table 4 are not carried (A2, A6, A10, A22, C2, C4, C6, D18, D28, D30), because their forward channel ends in an excited or radiative product and so has no collisional reverse from the populated states. One pair, D1/D2 (C+Si<sup>+</sup>), stays as printed: the paper Table 4 cites for it contains no such rate. src/tests/charge\_exchange\_detailed\_balance checks every carried pair against detailed balance and every carried rate against the threshold law. The default active set is Group A, the 23 metal+H/H<sup>+</sup> reactions; cx\_full adds the metal+He (C) and metal+metal (D) groups. The He↔H pair (Group B), He<sup>0</sup> + H<sup>+</sup> ⇌ He<sup>+</sup> + H<sup>0</sup>, is applied separately by he\_h\_cx\_fvec so that it is available in *every* system containing helium and is never counted twice; it is gated by He\_H\_charge\_exchange (default on). Group E is the two reactions outside Table 4, O<sup>2+</sup> + H<sup>0</sup> → O<sup>+</sup> + H<sup>+</sup> and

$\text{N}^{2+} + \text{H}^0 \rightarrow \text{N}^+ + \text{H}^+$  (Barragán et al. 2006, Tables 1 and 2, held at the ends of  $10^2$ – $10^5$  K), on by default at the published rates because the absence of the first leaves the O III profile wrong by decades in the region a transit probes; `cx_02p_H 0` and `cx_N2p_H 0` in `metals.inp` restore the Table-4-only set exactly. Each reaction enters as  $R = k n_{\text{donor}} n_{\text{acceptor}}$ , so the zero-abundance guard is automatic. The energy defect of each applied reaction, less the excitation of a product state that leaves as radiation, is heating channel 20 (Sec. 5.1).

**The helium-hydrogen pair.**  $\text{He}^+ + \text{H} \rightarrow \text{He} + \text{H}^+$  has a radiative and a non-radiative channel, and both are carried (`charge_exchange.f90`). The radiative one is row (19) of Stancil et al. (1998) Table 1,  $k = 1.20 \times 10^{-15} (T/300)^{0.25} \text{ cm}^3 \text{ s}^{-1}$ , the direct radiative charge transfer of Zygelman et al. (1989) weighted by the 1/4 of the singlet approach. The non-radiative one is the Maxwellian average of the  $\text{H}^+ + \text{He}(1s^2)$  cross section of Loreau et al. (2014) turned into the exothermic direction by microscopic reversibility,  $E_r = E_f + 10.989 \text{ eV}$  in the center-of-mass frame with statistical weights 4 and 1, so that the reverse threshold lies at  $x_{\text{th}} = 13.655 \text{ eV/u}$ . Their calculation starts at  $15 \text{ eV/u}$  and their fit stays finite below the threshold, where the channel is closed, so since 2026-09-26 the reverse cross section between the threshold and  $15 \text{ eV/u}$  is

$$\sigma_r(x) = \sigma_r(15) \sqrt{\frac{x - x_{\text{th}}}{15 - x_{\text{th}}}}, \quad x_{\text{th}} \leq x \leq 15 \text{ eV/u}, \quad (60)$$

and zero below  $x_{\text{th}}$ : by reciprocity  $\sigma_f v$  is constant at low forward energy, capture on the ion-induced-dipole potential with a transfer probability independent of energy, an assumption over forward energies of 0– $1.08 \text{ eV}$  that no calculation tests. Before,  $\sigma_r$  was zero below  $15 \text{ eV/u}$ , which closed a channel that is open: the forward rate then fell as  $e^{-E/kT}$  and was  $1.5 \times 10^{-21} \text{ cm}^3 \text{ s}^{-1}$  at  $10^3 \text{ K}$ , where it is now  $9.56 \times 10^{-17}$  ( $9.64 \times 10^{-17}$  at  $10^4 \text{ K}$ , against  $4.41 \times 10^{-17}$ ). The rate is tabulated from  $300 \text{ K}$  to  $10^5 \text{ K}$  (`generator_cooling_data/heii_h_charge_exchange_rate.py`), interpolated linearly in  $\ln k$  against  $1/T$ , the first chord continued below  $300 \text{ K}$  and the  $10^5 \text{ K}$  value held above. The non-radiative channel is 3.3 per cent of the radiative one at  $10^4 \text{ K}$  and 5.9 per cent at  $10^3 \text{ K}$ , so  $\text{He}^+ + \text{H}$  is removed almost entirely by radiative charge transfer; the reverse  $\text{He} + \text{H}^+ \rightarrow \text{He}^+ + \text{H}$  is the detailed-balance reverse of the non-radiative channel alone.

**Penning ionization of the metastable.** The total  $\text{He}(2^3\text{S}) + \text{H}$  ionization rate coefficient is the Maxwell-Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections as collected by García Muñoz (2025),

$$Q_{31}(T) = 10^{-9} \exp \left[ -\frac{86.4804}{T} - 0.286766 \ln T + 0.0868445 (\ln T)^2 - 5.73001 \times 10^{-3} (\ln T)^3 \right] \text{ cm}^3 \text{ s}^{-1}, \quad (61)$$

smooth, positive and single-peaked at  $3.8 \times 10^3 \text{ K}$ . The code notes at that site why the two-branch power law of Taylor et al. (2025) Table 2 is *not* used: it jumps by a factor 1.5724 at  $4000 \text{ K}$ , which a rate coefficient cannot do, and it disagrees with that paper's own Figure 19; the expression above agrees with Figure 19 to 10–20 percent. The corresponding  $\text{He}(2^3\text{S}) + \text{H}_2$  coefficient is  $k = a T^b e^{c/T}$  with the Cohen & Lane (1977) amplitudes of the García Muñoz (2025) network file, reproducing its Table A.5 to 0.13 percent. Both split 0.9 : 0.1 between the Penning and the associative branch.

## 4.4 Solving one cell

**The solver.** `solve_ieq` (`newton_solver.f90`) tries a damped Newton iteration with an *analytic* Jacobian first (`newton_dense`: backtracking Armijo line search on  $\|f\|_2$ , at most 60 iterations and 20 line-search steps), and on

any failure restores the entry guess and calls the in-tree MINPACK hybrd1 from the same guess, so the result can never be worse than the forward-difference solve. Each residual routine ships its own exact Jacobian; the photoionization rates are constants of the solve, so those Jacobians are exact for the system actually solved (Sec. 3.4). Run-wide counters record how many cells each route handled, and EXHALE\_FORCE\_HYBRD1 forces the MINPACK path for an A/B comparison. `use_newton_ieq` is on by default.

**The sweep.** `ioniz_eq` (`ionization_equilibrium.f90`) takes the temperature and the total density as *inputs* and overwrites the species fractions in place; chemistry rearranges nucleons and electrons and creates no mass, so the mass density is never corrected by this step and the mass sum of the returned composition is a check against it (`rho_recon`), never a correction. The grid is traversed from the top down in blocks; the columns of (31) are advanced with the composition already solved, so the field each cell is solved at is self-consistent with the states above it in one traversal. Inside a cell the pair (field, composition) is iterated at most `xuv_self_field_passes` times, stopping when the accepted stage fractions move by less than `xuv_self_field_tol`. Cells within a block are independent and the sweep is OpenMP-parallel with the per-cell scratch declared `threadprivate`; blocks are of fixed even length so that the vectorized transcendental calls pair the same lanes at any thread count.

**Acceptance classes.** Every cell's returned state carries a class, and the contract is that classes 1, 2, 3 and 5 are *roots* of the requested equations (each passed the same two tests, a state inside the element simplex and a normalized reaction residual at or below `ieq_res_tol`) and differ only in how the root was found:

| class | meaning                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------|
| 1     | root of the last attempt made                                                                                                 |
| 2     | root of the best earlier attempt                                                                                              |
| 3     | root after projecting the closest attempt onto the element budget                                                             |
| 4     | not a root; accepted under the relaxation amnesty and reported                                                                |
| 5     | root of the constrained element-conserving continuation solve                                                                 |
| 6     | not a root, residual above <code>ieq_nonroot_res_cap</code> : the cell <i>keeps the composition it entered the sweep with</i> |

A residual that is not a finite number is a class-4 non-root by construction, since NaN passes every tolerance test. A consumer counting states its chemistry does not describe counts classes 4 and 6 and nothing else.

**The constrained continuation.** `constrained_chemical_equilibrium.f90` is the second formulation of the molecular network, and it removes the possibility of an unphysical iterate instead of repairing one. The unknowns are  $u_k = \ln n_k$ , so  $n_k > 0$  for every iterate, converged or not (the logarithmic composition variables of the CEA method, Gordon & McBride 1994); element conservation is an explicit residual row per element rather than a closure, so a species carrying nuclei of two elements ( $\text{HeH}^+$ , OH,  $\text{H}_2\text{O}$ ) satisfies both rows at the accepted root; and  $n_e$  is the charge-neutrality sum, never an unknown. Because the network is bistable in its starting point (a molecular basin in the shielded base, an atomic basin above the  $\text{H}_2 \rightarrow \text{H}$  front), the solve is a natural-parameter continuation (Allgower & Georg, 2003) in the radiation field:  $\lambda$  scales the field from six decades below the cell's own, where the composition is

known in closed form from the  $\text{H}_2$  dissociation equilibrium of the local  $p, T$ , up to  $\lambda = 1$ , each rung starting from the previous rung's root and a failed rung being bisected geometrically. The rungs use the general MINPACK `hybrd` and not `hybrd1`, whose fixed initial trust radius  $100\|x\|$  would open a first step of  $e^{10^4}$  in every density when  $\|u\| \sim 130$ . A transported fraction, when one is imposed, enters as a further constraint row.

#### 4.5 Ionization carried by the flow

Where the recombination time is not short compared with the flow time, the local balance of Sec. 4.2 is not the answer. Two places in the code carry the ionization state along the flow instead.

**The implicit advection systems.** `System_implicit_adv_H`, `_HeH` and `_HeH_TR` each take one step of  $v dx/dr = R(x)$  into cell  $j$ , per hydrogen nucleus, with the variable-step second-order backward differentiation formula (BDF2):  $x^{\text{up}} \equiv a_1 x_{j-1} - a_2 x_{j-2}$  is the history of the step and  $c_1 = g h_j / v_j$  its rate weight,  $h_j = r_j - r_{j-1}$ ,  $w = h_j / h_{j-1}$ ,  $a_1 = (1+w)^2 / (1+2w)$ ,  $a_2 = w^2 / (1+2w)$ ,  $g = (1+w) / (1+2w)$ , so that the rates and the velocity are both those of cell  $j$ . The first step of the recursion, and every step after a cell it does not integrate, is backward Euler ( $a_1 = g = 1$ ,  $a_2 = 0$ ;  $x^{\text{up}} = x_{j-1}$ ), as is a BDF2 step whose populations come out negative. BDF2 is L-stable, so a population that relaxes within the cell settles on its local equilibrium (the trapezoid rule, A- but not L-stable, would alternate about it); `post_process_adv` states the formula, its derivation and its zero-stability bound  $w \leq 1 + \sqrt{2}$ . The systems solve

$$0 = x_{\text{HI}}^{\text{up}} - x_{\text{HI}} + c_1 [-(\Gamma_{\text{HI}} + \beta_{\text{HI}} x_e n_{\text{H}}) x_{\text{HI}} + \alpha_{\text{HIII}} x_{\text{HII}} x_e n_{\text{H}}], \quad (62)$$

$$0 = x_{\text{HeI}}^{\text{up}} - x_{\text{HeI}} + c_1 [-(\Gamma_{\text{HeI}} + \beta_{\text{HeI}} x_e n_{\text{H}}) x_{\text{HeI}} + \alpha_{\text{HeII}} x_{\text{HeII}} x_e n_{\text{H}}], \quad (63)$$

$$0 = x_{\text{HeIII}}^{\text{up}} - x_{\text{HeIII}} + c_1 [(\Gamma_{\text{HeII}} + \beta_{\text{HeII}} x_e n_{\text{H}}) x_{\text{HeII}} - \alpha_{\text{HeIII}} x_{\text{HeIII}} x_e n_{\text{H}}], \quad (64)$$

with  $x_e = x_{\text{HIII}} + (\text{He}/\text{H})_{\text{local}}(x_{\text{HeII}} + 2x_{\text{HeIII}}) + x_e^{\text{metal}}$ , and the same  $\text{He} \leftrightarrow \text{H}$  charge-exchange terms on rows 1 and 2. The helium ratio is the global one on the legacy path and the *local*, radius-dependent one when the element diffusion is on, since the global value would misstate  $n_e$  by the separation factor. These are the systems of the `_adv` post-process (Sec. 10.1).

**Transported protons.** With `Ionization transport` (default off)  $\text{H}^+$  is carried as a fifth transported species on the hydrodynamic face mass flux inside the Runge-Kutta stages (107), in the same way the molecular carriers are (Sec. 6.5), and its local balance is then a constraint rather than a closure.

#### 4.6 Excited hydrogen and the $\text{Ly}\alpha$ field

`excited_hydrogen.f90` solves the  $2s/2p$  statistical equilibrium of Christie et al. (2013, their eqs. (12)–(13)) as a  $2 \times 2$  linear system,

$$\begin{pmatrix} L_{2p} & -M_{12} \\ -M_{21} & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix}, \quad n_{2p} = \frac{S_{2p} L_{2s} + M_{12} S_{2s}}{L_{2p} L_{2s} - M_{12} M_{21}}, \quad (65)$$

whose sources are the recombination cascade, collisional excitation from  $1s$  and radiative pumping by the  $\text{Ly}\alpha$  mean intensity, and whose sinks are radiative decay, the two-photon continuum, collisional de-excitation,  $\ell$ -mixing and photoionization by the stellar Balmer continuum. The atomic data are in one place (`hydrogen_n2_rates.f90`):  $A_{2p1s} =$

$6.2649 \times 10^8 \text{ s}^{-1}$  (NIST ASD, the nonrelativistic dipole value carrying the reduced-mass factor),  $A_{2s1s} = 8.22461 \text{ s}^{-1}$  (Drake 1986 eq. 27, with the finite nuclear mass and his relativistic factor),  $g_{1s} = g_{2s} = 2$ ,  $g_{2p} = 6$ ,  $f_{\text{Ly}\alpha} = 0.4162$ , the  $1s \rightarrow 2s$  and  $1s \rightarrow 2p$  collisional rates  $C\Upsilon e^{-E/kT}/(2\sqrt{T})$  with the CHIANTI v11 effective collision strengths (the rates the H I cooling uses too), their detailed-balance inverses, and the case-B coefficient of the ionization balance (the captures into  $n \geq 2$ ) split into  $2s$  and  $2p$  with the case-B fraction of Pengelly (1964, Table I),  $\alpha_{2s} = f_{2s}(T)\alpha_B$ ,  $\alpha_{2p} = \alpha_B - \alpha_{2s}$  ( $f_{2s} = 0.322$  at  $10^4 \text{ K}$ ). The  $2s \rightarrow 2p$   $\ell$ -mixing rate is

$$M_{2s2p} = n_e C_{2s2p}^e + n_{\text{HII}} q_{2s2p}^p + n_{\text{HeII}} q_{2s2p}^{\text{He}^+} + n_{\text{HeIII}} q_{2s2p}^{\text{He}^{2+}}, \quad (66)$$

$C^e$  the electron rate of Seaton (1955) eq. (55), his approximation V, Maxwell-averaged in closed form with the same two final states and separations ( $5.78 \times 10^{-5} \text{ cm}^3 \text{ s}^{-1}$  at  $10^4 \text{ K}$ ; the Christie et al. Table 2 rate,  $5.35 \times 10^{-6}$ , averages a cross section whose constant is one tenth of Seaton's first-order dipole constant, and is not used), and the ion rates the impact-parameter cross section of Pengelly & Seaton (1964), eqs. (34)–(41), Maxwell-averaged in closed form with their cut-off  $R_c = \min(1.12\hbar v/\Delta E, 0.72v\tau, R_D)$ ; the Lamb shift and the fine structure set  $\Delta E$  ( $2s_{1/2}$ – $2p_{1/2}$   $0.035 \text{ cm}^{-1}$ ,  $-2p_{3/2}$   $0.331 \text{ cm}^{-1}$ ), which is the smallest cut-off below  $n_e \sim 4 \times 10^{12} \text{ cm}^{-3}$ . The proton coefficient is  $4.8 \times 10^{-4} \text{ cm}^3 \text{ s}^{-1}$  at  $10^4 \text{ K}$ , 8.3 times the electron one, so  $2s$  is mixed into  $2p$  faster than it decays by two photons wherever  $n_e = n_p \gtrsim 1.5 \times 10^4 \text{ cm}^{-3}$ . The removal of a  $2p$  atom without emission of a line photon is

$$D_{2p} = n_e q_{2p1s} + \Gamma_{2p} + M_{2p2s} \frac{A_{2s1s}}{A_{2s1s} + n_e q_{2s1s} + M_{2s2p} + \Gamma_{2s}}, \quad (67)$$

$$M_{2p2s} = (g_{2s}/g_{2p})M_{2s2p}.$$

**Balmer continuum.** The  $n = 2$  photoionization rate and the photoelectric heating it delivers are integrals over  $[E_{\text{th}}(\text{H}, n=2), E_{\text{th}}(\text{H})]$  of the run's own spectrum with the hydrogenic cross section  $\sigma_2(E) = \sigma_2^{\text{thr}}(E_2/E)^3$ ,  $\sigma_2^{\text{thr}} = 1.4 \times 10^{-17} \text{ cm}^2$  for  $2s$  and  $2p$  alike. The code states no source for that value; it is, to its two figures, the  $n = 2$  threshold value of the Kramers cross section with the bound-free Gaunt factor of Seaton (1959, eqs. 3 and 10),  $1.378 \times 10^{-17} \text{ cm}^2$ , and the exact hydrogenic threshold values are  $1.478 \times 10^{-17}$  ( $2s$ ) and  $1.355 \times 10^{-17} \text{ cm}^2$  ( $2p$ ), statistical mean  $1.386 \times 10^{-17} \text{ cm}^2$  (derived for this document); the  $(E_2/E)^3$  form with a constant threshold value leaves out the rise of the Gaunt factor above the edge:

$$\Gamma_2 = \frac{\sigma_2^{\text{thr}} E_2^3}{h\nu_{\text{eV}}} \int F_E E^{-4} dE, \quad h_2^{(1)} = \sigma_2^{\text{thr}} E_2^3 \int F_E (E^{-3} - E_2 E^{-4}) dE. \quad (68)$$

The quadrature resolves the field it integrates, which a fixed rule cannot: for a loaded table the nodes are the table's *own* rows, and each segment is integrated in closed form because the interpolation is a power law there; for the power law the whole band is closed form; only for the smooth Planck field is a composite five-point Gauss-Legendre rule on a geometric subdivision used. The feedback is lagged-explicit: it is computed once per timestep before the ionization and energy solve, and the frozen arrays are read inside `ioniz_eq` and never inside its Newton iteration. The whole module is gated by `use_excited_H`, which is armed by supplying `Stellar Teff [K]`: and `Stellar radius [R_sun]`:; it is *off* by default.

**The Ly $\alpha$  field.** `lya_rt.f90` closes the line with an escape probability rather than a Monte Carlo, because the columns involved reach  $\tau_0 \sim 10^8$  at line center. With  $\Delta\nu_D = \nu_{\text{Ly}\alpha} \sqrt{2kT/m_{\text{H}}}/c$ , the Voigt parameter  $a = (A_{2p1s}/4\pi)/\Delta\nu_D$  and the

line-center depth accumulated cell by cell from  $n_{\text{HI}}C_{\text{Ly}\alpha}/\Delta\nu_{\text{D}}$ , the escape probability of a photon emitted at the cell center is

$$\beta = \frac{1}{1 + \langle N \rangle}, \quad \langle N \rangle = \frac{4\sqrt{6}}{\pi^2} \frac{\tau_{\text{up}} + \tau_{\text{dn}}}{2} \Psi(\xi), \quad \xi = \frac{\tau_{\text{dn}}}{\tau_{\text{up}} + \tau_{\text{dn}}}, \quad \Psi(\xi) = \sum_{n \text{ odd}} \frac{\sin(n\pi\xi)}{n^2}, \quad (69)$$

which is Neufeld (1990) eq. (3.27) at zero continuum destruction, reducing for a mid-plane source to Harrington (1973) eq. (40),  $\langle N \rangle = 0.909316B$ . The form  $1/(1 + \langle N \rangle)$  and not  $1/\langle N \rangle$  is used because  $\langle N \rangle$  counts absorptions per photon generated, so a photon is emitted  $1 + \langle N \rangle$  times before it leaves; this is exact at both ends and needs no join between the thin and thick limits. Under  $R_{\text{II}}\text{-A}$  redistribution  $\langle N \rangle$  is independent of  $a$ , so the Voigt parameter survives only in the one-flight penetration of the stellar beam. The trapped internal field and the stellar term are

$$J_{\text{int}} = \frac{2h\nu^3}{c^2} \frac{g_{1s}}{g_{2p}} \frac{P(1 - \beta)}{(A_{2p1s}\beta + D_{2p}) n_{\text{HI}}}, \quad J_{\star} = \frac{\xi_{\text{day}} F_{\text{Ly}\alpha}}{4\pi \Delta\nu_{\star}} \langle T_s E \rangle, \quad J_{\text{Ly}\alpha} = J_{\text{int}} + J_{\star}, \quad (70)$$

with  $P = \alpha_{\text{B}} n_e n_{\text{HII}} + C_{1s2p} n_e n_{\text{HI}}$  the internal production,  $T_s = \text{erfc}(x_1/\sqrt{2}X_s)$  the fraction of the broad stellar profile whose wings reach depth  $\tau$ , and  $\langle \cdot \rangle$  the mean of the beam over the cell's own optical depth. The published solution requires  $(a\tau)^{1/3} > 10$  (Neufeld section Va); cells outside that are counted by `lya_wing_domain_record` and are carried there by the thin limit  $\beta \rightarrow 1$ , which (69) reaches smoothly. The alternative `jlya_mode = 1` reads an externally computed  $J_{\text{Ly}\alpha}(r)$ , for instance from the LaRT Monte Carlo code, and interpolates it onto the grid.

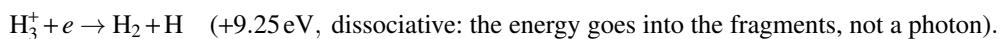
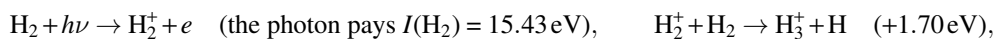
## 5. The thermal source: heating and cooling

### 5.1 The heating channels

The volumetric heating rate of a cell is a sum of physically distinct deposits, and there is exactly one list of them (`util_ion_eq.f90`): `heating_of_composition` fills every column of `heat_chan`, the total it returns is the sum of those columns in that order, and the header of `output/Heating_breakdown.txt` is built from the same names, so a deposit cannot exist in the energy equation and be missing from the breakdown.

| #     | name                                                                                             | what it is                                                                                                                                                                                                                             |
|-------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–6   | heat_HI . . . heat_metals                                                                        | photoionization of each absorber, $n_a h_a^{(1)}$ of (36)–(37)                                                                                                                                                                         |
| 7     | heat_Hpe[excitedH]                                                                               | photoelectric heating of H( $n=2$ ) by the Balmer continuum, (68)                                                                                                                                                                      |
| 8     | heat_Hdx[Lya-deexc]                                                                              | collisional de-excitation of the Ly $\alpha$ -pumped $n=2$ atoms (incl_deexc_heat, default on)                                                                                                                                         |
| 9     | heat_He_recomb                                                                                   | photoelectrons of the H I, He I, H <sub>2</sub> and metal ionizations driven by recombination radiation absorbed on the spot (He II and He III, and the H II ground captures taken by metals)                                          |
| 10–12 | heat_He23S_*                                                                                     | He(2 <sup>3</sup> S)+H Penning and associative branches, and He(2 <sup>3</sup> S)+H <sub>2</sub> Penning                                                                                                                               |
| 13–14 | heat_H2_LW_*                                                                                     | kinetic energy of the Lyman-Werner dissociation fragments, and collisional de-excitation of the H <sub>2</sub> fluorescence                                                                                                            |
| 15    | heat_mol_chem                                                                                    | net chemical heat of the collisional reactions of the H <sub>2</sub> /He network                                                                                                                                                       |
| 16–19 | heat_FUV_photolysis,<br>heat_oxygen_collisional,<br>heat_CO_Hep_transfer,<br>heat_CO_photodissoc | the oxygen option's deposits                                                                                                                                                                                                           |
| 20    | heat_charge_exchange                                                                             | energy defects of the applied charge-exchange reactions (metal+H/He/metal and He $\leftrightarrow$ H), less the excitation of a product state that leaves as radiation; negative for an endothermic reaction (charge_exchange_heating) |

Channels 1–6 are the photoheating integrals of Sec. 3.4, already carrying the heating efficiency  $f_{\text{heat}}$  of (42). Channel 15 exists because the return path that makes ionization energy a non-channel in an atomic gas is different in a molecular one (molecular\_reaction\_heat.f90). Follow one closed cycle:



The cycle returns to H<sub>2</sub> having turned one H<sub>2</sub> into H+H and having left  $I(\text{H}_2) - D_0(\text{H}_2) = 15.43 - 4.48 = 10.95 \text{ eV}$  in the gas, with no photon anywhere in it to carry that away. Measured on the converged He/H = 0.0793 hot Uranus, that is 81 percent of the total heating rate at  $1.02 r_{\text{base}}$ . Molecular reaction heat is therefore *on* by default; it is unreachable in an atomic run, so no non-molecular result moves.

**Where each channel's energy goes (corrected 2026-09-17).** That the energy stays in the gas does not make all of it translational, and until item L7g the ledger deposited each dissociative recombination's whole enthalpy as heat. Each channel now deposits what its measured product state leaves to translation:

- $\text{H}_2^+ + e$  and  $\text{HeH}^+ + e$  leave one H atom in  $n = 2$  at the collision energies of this layer (Takagi, 2002; Giusti-Suzor et al., 1983; Guberman, 1994), so 10.199 eV goes to the  $\text{H}(n = 2)$  population (or radiates, when that population is not tracked) and the gas receives 0.749 and 1.554 eV instead of 10.948 and 11.753.
- $\text{H}_3^+ + e \rightarrow \text{H}_2 + \text{H}$  leaves the  $\text{H}_2$  fragment vibrationally excited around  $v = 5-6$  (Kokoouline et al., 2001; Strasser et al., 2001), so 2.4795 eV of the 9.250 is internal energy and branches through the vibrational quench fraction rather than being deposited promptly.
- The three-body association  $\text{H} + \text{H} + M$  releases 4.478 eV into a nascent vibrational distribution, and the same quench fraction decides how much of it is collisionally thermalized before it radiates.

The quench fraction itself (`h2_vibrational_relaxation.f90`) is the collisional rate against the radiative one. The radiative side is the *all-level* maximum spontaneous rate over the code's own 302-level  $\text{H}_2$  ladder,  $5.5939 \times 10^{-6} \text{ s}^{-1}$ , which is 6.56 times the  $v = 1$  total decay rate the earlier form used and agrees with the independent Wolniewicz et al. (1998) set to  $1.1 \times 10^{-4}$ ; the collisional side is one published quantum calculation per collider, Lique (2015) for H, Jóźwiak et al. (2024) for He and Le Bourlot et al. (1999) for  $\text{H}_2$ , in place of two 1979 fits with helium absent. In the LHS 1140 b layer the fraction is one to a part in  $10^6-10^7$ , so the layer is in the thermalized limit and these corrections act mainly through the recipients and the association rate; a shallower or cooler layer is not, and the form is stated with that validity where it is used. `EXHALE_REACTION_HEAT_RECIPIENTS=0` restores the prompt-deposit behavior for comparison.

## 5.2 The cooling channels

`eval_cool` (`util_ion_eq.f90`) assembles the cooling rate of every cell, the atomic, ionic and metal channels through the one routine `radiative_cooling_of_cell` (`Cool_coeff.f90`) that the post-process temperature root also calls, as

$$\Lambda = n_e (\Lambda_{\text{ff}} + \Lambda_{\text{cx}} + \Lambda_{\text{rec}} + \Lambda_{\text{ci}}) + \Lambda_{\text{metal}} + \Lambda_{\text{H}_3^+} + \Lambda_{\text{H}_2} + \Lambda_{\text{H}_2\text{O}} + \Lambda_{\text{CO}}, \quad (71)$$

in  $\text{erg cm}^{-3} \text{ s}^{-1}$ , with  $n_e$  the same charge sum the ionization balance uses (`calc_ne`), molecular donors included, since inside a deep molecular base  $\text{H}_3^+$  is the dominant ion. The breakdown written to `output/Cooling_breakdown.txt` is read straight off the arrays (71) is built from, so the columns sum to  $\Lambda$  exactly.

**Recombination.**  $\Lambda_{\text{rec}} = \lambda_{\text{HIII}} n_{\text{HIII}} + \lambda_{\text{HeII}} n_{\text{HeII}} + \lambda_{\text{HeIII}} n_{\text{HeIII}} + \sum_i \lambda_i n_{m,i}$ , each the kinetic energy the captured electrons carry off, the last sum over the recombining metal ions. For captures of rate coefficient  $\alpha(T) = \langle \sigma v \rangle$  in a Maxwellian gas it is exactly

$$\lambda(T) = kT \alpha(T) \left( \frac{3}{2} + \frac{d \ln \alpha}{d \ln T} \right), \quad (72)$$

which Table 1 of Hummer (1994) obeys to 0.1 per cent, and it is applied to the *same* coefficient the ionization balance removes the ions with: the net coefficients (58) of H II, He II and He III (case B plus the escaping ground captures, whichever `Atomic rate set` gives  $\alpha_B$ ), and for each metal ion the Badnell radiative plus dielectronic (or

Huang et al. 2023 iron) coefficient of its balance; the relation holds for dielectronic capture as well, where it charges the resonance energy. The escape weights depend on the densities and the cell width only, so they are frozen while  $T$  varies. The logarithmic slope is a centered difference of half step  $10^{-3}$  in  $\ln T$  for H II, He II and He III, and is written analytically from the published forms for the metal ions (metal\_recombination\_energy\_rate). The kinetic energy of a ground capture whose photon is absorbed by another species is returned with the photoelectron (recombination\_radiation\_absorbed).

**Collisional ionization.**  $\Lambda_{ci}$  charges each ionization its measured potential in erg,

$$\Lambda_{ci} = E_{th}^{HI} \beta_{HI} n_{HI} + E_{th}^{HeI} \beta_{HeI} n_{1^1S} + E_{th}^{HeII} \beta_{HeII} n_{HeII} + E_{th}^{2^3S} \beta_{2^3S} n_{2^3S} + \sum_i E_{th}^i \beta_i n_{m,i}, \quad (73)$$

the He I term acting on the ground *singlet*  $n(1^1S) = n(\text{He I}) - n(2^3S)$ , because the metastable sits 19.8 eV up and carries its own 4.8 eV threshold ( $\beta_{2^3S}$  from Black 1981, eq. 12). The metal term charges each collisional ionization the measured potential of the ion; with it and the metal recombination term above, the electron gas, the ionization energy and the recombination radiation close for the metals as for H and He. The same subtraction is made in the photoheating and in the collisional excitation below, so the metastable is never charged the ground-state coefficients as well.

**Collisional excitation.**  $\Lambda_{cx}$  is the coronal (every excitation radiated) sum over the upper levels of the CHIANTI v11.0.2 models of H I and He II and over the 28 upper terms  $n \leq 5$  of Bray et al. (2000, Table 2) for He I (the strengths the metastable network uses, Sec. 4.2), with observed level energies, contracted with  $n_{HI}$ ,  $n(1^1S)$  and  $n_{HeII}$ . The CHIANTI strengths are descaled with the natural cubic spline CHIANTI's own `descale_scups` prescribes. For H I the  $n = 2$  part is 10.2 eV ( $q_{1s2s} + q_{1s2p}$ ) with the same rates the H( $n=2$ ) model uses (Sec. 4.6); the  $n \geq 3$  part and the He sums are fits  $CT^{-1/2} [a + \ln(1 + T/T_0)] e^{-T_{ex}/T}$  printed by `cooling_data/fit_hydrogen_helium_excitation.py`, accurate to 1–2 per cent. The H I term is the Ly $\alpha$ -dominated hydrogen line cooling, and it is the optically thin coefficient as written: *no escape probability multiplies the Ly $\alpha$  cooling channel*.  $\lambda_{HI}$  is `lambda_coex_HI` of `Cool_coeff.f90` and `eval_cool` contracts it with  $n_{HI}$  and  $n_e$  with no such factor anywhere. The two escape probabilities the code does compute enter elsewhere:  $\beta$  of `lya_rt.f90` (69) enters the H( $n=2$ ) populations and the pumping field, and  $\beta_{ul}$  of `fine_structure_line_transfer` enters the metal ground-term statistical equilibrium (74); neither is read by any cooling coefficient. The asymmetry is deliberate: multiplying  $\lambda_{HI}$  by an escape probability would leave the trapped fraction with no owner while keeping the de-excitation heating channel 8 as an independent source, that is it would double count. A ledger carrying both is the stated target and is not built.

With the metastable tracked, the He I sum leaves out the  $1^1S \rightarrow 2^3S$  transition, which is the network's own rate  $q_{13}$ , and the exchange between the two levels is charged with that rate and its detailed-balance reverse  $q_{31g}$ ,  $E(2^3S) [q_{13}n(1^1S) - q_{31g}n(2^3S)]$ , negative (a heating) wherever the metastable is overpopulated against its Boltzmann ratio to the ground. The excitations  $1^1S \rightarrow n^3L$  into the higher triplets, which feed  $2^3S$  through their cascade (in  $q_{13}$  of the triplet row), are charged their full energy by the He I sum, of which  $E(n^3L) - E(2^3S)$  is radiated in the cascade and  $E(2^3S)$  is carried by the metastable they make. Further terms scale with  $n(2^3S)$ : the excitation of  $2^3S$  within the triplets, radiated back to  $2^3S$  (10830 Å and the higher triplet lines; a CHIANTI v11 coronal sum, of which the Black (1981) coefficient it replaces is 0.81 at  $10^4$  K and 0.79 at 5000 K), and the  $2^3S \rightarrow 2^1S$  (0.796 eV),  $2^3S \rightarrow 2^1P$  (1.398 eV) and  $2^3S \rightarrow n^1L$  ( $n \geq 3$ ) collisional conversions, each removing its excitation energy from the electron gas.

**Free-free.**  $\Lambda_{\text{ff}} = 1.426 \times 10^{-27} \sqrt{T} \sum_{\text{ions}} Z_{\text{ion}}^2 \bar{g}(T, Z_{\text{ion}}) n_{\text{ion}}$ , with  $\bar{g}$  the thermally averaged Gaunt factor of van Hoof et al. (2014) as a function of  $\gamma^2 = Z^2 \text{Ry}/kT$  (their eq. 21), interpolated linearly in  $x = \log_{10} \gamma^2$  and clamped to the tabulated range. Ry is the infinite-mass Rydberg unit of energy, as their section 2 defines it,  $hcR_{\infty}$  with the CODATA 2018  $R_{\infty}$  and the exact SI  $h$ ,  $c$ ,  $k$  the code carries, so  $\text{Ry}/k = 157887.5 \text{ K}$  (`Cool_coeff.f90`, `Ry_over_kB`). Until 2026-09-26 the code used 157807 K, the constant of the Hui & Gnedin (1997) recombination fits, which moved the regression outputs by at most  $1.4 \times 10^{-4}$  relative. Only  $Z_{\text{ion}} = 1$  and 2 occur. The molecular ions  $\text{H}_2^+$ ,  $\text{H}_3^+$  and  $\text{HeH}^+$  do donate to  $n_e$  but are deliberately left out of this charge sum: they exist only in the  $T \sim 10^3 \text{ K}$  molecular base, where this hot-plasma expression is an extrapolation whose emission comes out at frequencies the atmosphere is not thin to, and where  $\text{H}_3^+$  and metal lines dominate by far.

**Metal lines.**  $\Lambda_{\text{metal}} = n_e \sum_i n_{\text{m},i} c_i(T, n_e)$ , every  $c_i$  built from CHIANTI v11.0.2 (collision strengths descaled with the natural cubic spline) where the ion is in it. The C/N/O coefficients are the default (`cno_chianti = .true.;` `cno_cool 0` in `metals.inp` restores the legacy set ported from AIOLOS). Mg II is a closed-form fit to the coronal sum of every excitation out of  $3s^2\text{S}$  with the observed level energies. Ca II and Mg I are  $(T, n_e)$  statistical-equilibrium tables with every channel (the Ca II  $3d^2\text{D}$  metastables and the Mg I  $3s3p^3\text{P}$  term besides the resonance lines; at  $n_e = 10^9 \text{ cm}^{-3}$  they are 1.7–4 and 5.8–480 times the resonance line alone over  $10^4$ –3000 K). Two entries are not in CHIANTI v11 and are built otherwise: the Na I D doublet with a  $\bar{g}$  factor 0.2 calibrated against the CHIANTI Mg II and Ca II values, and the Fe I permitted multiplets from NIST oscillator strengths with the lower levels in Boltzmann equilibrium over the even-parity metastable manifold, the high-density limit, which is an upper limit at most 2.0 times the low-density value over 2000– $3 \times 10^4 \text{ K}$  (the data for a density-resolved Fe I solve are not available). Above the top density of a  $(T, n_e)$  table ( $10^{14} \text{ cm}^{-3}$ ) the coefficient continues with the slope of the last interval clipped to  $[-1, 0]$ , the bounds of a thermalized and a coronal channel.

Two further density-dependent replacements are made on top of coronal curves, because a coronal fit assumes  $\Lambda \propto n_e$  and no level saturation while the base of an irradiated atmosphere runs at  $n_e \sim 10^9$ ,  $n_{\text{HI}} \sim 10^{14} \text{ cm}^{-3}$ , decades above the critical densities of the lines involved.

- Fe II takes a two-dimensional statistical-equilibrium table  $\Lambda_{\text{eff}}(T, n_e) = (\sum_u n_u A_{ul} \Delta E_{ul}) / n_e$ , whose  $1/n_e$  cancels the  $n_e$  of the assembly and so leaves the correct saturated emission; at low  $n_e$  it reduces to the coronal rate. Without it the forbidden  $a^6\text{D}$  lines overestimate the base cooling by about four decades.
- *Ground-term fine structure* of C I ([C I] 609.1, 370.4  $\mu\text{m}$ ), C II (157.7  $\mu\text{m}$ ), N II (205.3, 121.8  $\mu\text{m}$ ) and O I (63.2, 145.5, 44.1  $\mu\text{m}$ ) is replaced by the exact statistical equilibrium of the ground term,

$$\sum_j f_j R_{ji} = f_i \sum_j R_{ij}, \quad \sum_i f_i = 1, \quad R_{ul} = C_{ul} + \beta_{ul} A_{ul}, \quad R_{lu} = C_{ul} \frac{g_u}{g_l} e^{-E_{ul}/kT}, \quad (74)$$

with  $C_{ul} = n_e k_{e,ul}(T) + n_{\text{HI}} k_{\text{H},ul}(T)$  and  $k_{e,ul} = C \Upsilon / (g_u \sqrt{T})$  ( $C = 8.629132 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1} \text{ K}^{1/2}$ , the Maxwellian constant of the triplet row (49)), the effective collision strengths  $\Upsilon$  being quartic fits in  $x = \log_{10}(T/10^4)$  to the CHIANTI .scups values over  $10^3$ – $10^5 \text{ K}$  (largest error 8.2 percent, at the  $10^3 \text{ K}$  end on C I) and the H-atom rates taken from Yan & Babb (2023) for C I and N II, Abrahamsson et al. (2007) for O I and Barin et al. (2005) for C II. The emitted power per ion is  $W_{\text{FS}} = \sum_{u>l} f_u \beta_{ul} A_{ul} k_B E_{ul}$  and the coefficient returned is  $\Lambda_{\text{eff}} = W_{\text{FS}}/n_e + \Lambda_{\text{rem}}$ .

$\Lambda_{\text{rem}}(T, n_e)$  being the statistical-equilibrium table of the channels that leave the ground term (`cool_logLrem_*`; the metastable upper levels of those channels have critical densities of  $10^4$ – $10^9 \text{ cm}^{-3}$ ).

The line trapping  $\beta_{ul}$  of these fine-structure lines is the line-center escape probability formed from the columns above and below the cell (`fine_structure_line_transfer`), and it enters *inside* the statistical equilibrium as  $A_{ul} \rightarrow \beta_{ul} A_{ul}$  rather than as a factor on the result: in the subcritical limit the cooling is set by the collisional excitation rate and is independent of  $\beta$ , and in the saturated limit the escaping flux is proportional to it.  $\beta = 1$  reproduces the optically thin result exactly, and that is the value every other metal ion keeps. With `Base IR field: True` (default off) the same lines also *absorb* the thermal infrared of the lower atmosphere, a blackbody at  $T_0$  covering the sky fraction the base subtends, through the photon occupation number  $\bar{n}_{ul}$ ; this gives the layer a radiative equilibrium floor instead of letting it cool into vacuum.

**Molecular infrared.**  $\Lambda_{\text{H}_3^+}$  is the Miller et al. (2013) emission of one  $\text{H}_3^+$  molecule in LTE,

$$\Lambda_{\text{H}_3^+} = 4\pi n_{\text{H}_3^+} E(T) s(T, n_{\text{H}_2}), \quad \ln E(T) = \sum_n C_n T^n \text{ [W molecule}^{-1} \text{ sr}^{-1}\text{]}, \quad (75)$$

their Table 5 (the “using  $z(t)$ ” column) with the non-LTE departure factor  $s$  of their Table 6, whose only collider is  $\text{H}_2$  through the proton-hopping reaction with the Oka & Epp (2004) coefficient  $2 \times 10^{-15} \text{ m}^3 \text{ s}^{-1}$ . Table 6 covers  $n(\text{H}_2) = 10^6$ – $10^{14} \text{ cm}^{-3}$  in steps of two decades. Between its columns  $s$  is interpolated as a power law in  $n(\text{H}_2)$ : the emission per  $\text{H}_3^+$  ion can grow at most as fast as the collision rate that populates the emitting levels, so the local index  $d \ln s / d \ln n$  lies between 0 and 1, as the node-to-node indices of the table do (0.00–0.89), whereas an interpolant linear in  $s$  against  $\log n$  reached 12.6 just above the  $10^8 \text{ cm}^{-3}$  column at 2000 K. Between the temperature rows the interpolation stays linear, a form the paper does not constrain. Below the table the code adopts the linear low-collider limit  $s(T, n) = s(T, 10^6) n / 10^6$  and records the cell as evaluated outside the tabulated range. This is the dominant coolant of a molecular base: the atomic channels above are all exponentially suppressed at  $T \sim 10^3 \text{ K}$ , so leaving it out leaves that gas with no radiative loss at all.

With `Molecular IR bands: True` (default off) three further net exchanges are added (`molecular_infrared_cooling.f90`): the  $\text{H}_2$  quadrupole and magnetic-dipole line spectrum, and the  $\text{H}_2\text{O}$  and  $\text{CO}$  vibration-rotation bands. Each is a *net* rate,

$$\Lambda_X^{\text{net}} = n_X [E_X(T) - W_{\text{dil}} E_X^{\text{abs}}(T; T_{\text{rad}})], \quad E_X(T) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T) d\nu, \quad (76)$$

so each vanishes at its own radiative equilibrium temperature instead of running the layer down to nothing; a column is negative wherever that channel heats. The dilution  $W_{\text{dil}}$  is the same  $0.5 f_{\text{sky}}$  the  $\text{H}_3^+$  closure uses and is zero when `Base IR field` is off, which reduces these channels to pure emitters. Without them a converged molecular layer radiates itself down to 190–400 K against an equilibrium temperature of 1100–1400 K.

### 5.3 Advancing the temperature

**In the marching loop.** The temperature of a cell is advanced by `solve_energy_semi_implicit` (`energy_semi_implicit.f90`), the inner step of the coupled temperature-composition source step: the marching loop alternates it with the composition sweep until the pair stops moving, so the composition is fixed for one pass

and not for the whole step. With the composition and the photoheating frozen over the step,

$$\rho e(T) - \rho e_{\text{old}} = \Delta t [\mathcal{H} - \Lambda(T)], \quad \text{i.e.} \quad R(T) = \varepsilon(T) - \varepsilon_{\text{old}} - a[\mathcal{H} - \Lambda(T)] = 0, \quad a = \frac{\Delta t}{n_{\text{tot}} + n_e}, \quad (77)$$

with  $\varepsilon = \rho e / (n_{\text{tot}} + n_e)$  the internal energy per particle, which is what `internal_energy_per_particle` returns. The caloric internal energy  $\rho e$  is that of (14). The root is found by a safeguarded Newton-bisection hybrid on a physical bracket whose ends are the floor of the equation-of-state validity and a ceiling above the heating-only update: the search starts from the explicit-Euler point, brackets the root as it goes, and once both ends are known takes a Newton step whenever that step stays strictly inside the bracket and the bracket keeps halving, bisecting otherwise. The Newton derivative is exact in the equation-of-state term (`heat_capacity_per_particle`) and a secant estimate in  $d\Lambda/dT$ , which affects only the path, the safeguard owning the convergence.

What is tested before the update is returned is the residual at the *returned* temperature against  $|R|/s$  with  $s = |\rho e_{\text{old}}| + \Delta t(|\mathcal{H}| + |\Lambda|)$ , and admissibility; the cooling written back is  $\Lambda(T_{\text{returned}})$ , evaluated there and not at a previous iterate. There is no clamp: a cell that does not meet the tolerance, that has no bracket, or that reaches the equation-of-state floor is a named failure, nothing is assembled from it, and the step controller retakes the step at half  $\Delta t$  as in Sec. 7.5. The floor is not a reservoir: reaching it means the balance asks for a temperature below the range in which the equation of state and the rate fits are defined, and a floor representing real external heating would have to be a specified reservoir with its own budget. `use_semi_implicit_energy` is on by default.

**In the advection post-process.** `T_equation.f90` is *not* the marching loop's temperature solve. Its module `equation_T` is used by `post_process_adv.f90` (Sec. 10.1) and by test drivers, and by nothing else; what it solves is the *steady* internal-energy balance of the profile, differenced by the same variable-step BDF2 step as the ionization systems (Sec. 4.5): every  $r$ -derivative at  $r_j$  is  $(f_j - a_1 f_{j-1} + a_2 f_{j-2}) / (gh_j)$  and every other factor is the value at  $r_j$ ,

$$\nabla \cdot (u\mathbf{v}) + p \nabla \cdot \mathbf{v} = \rho v \frac{de}{dr} - p v \frac{d \ln \rho}{dr} + h \nabla \cdot (\rho \mathbf{v}) = \mathcal{H} - \Lambda, \quad (78)$$

with  $e$  the internal energy per unit mass and  $h = e + p/\rho$  the enthalpy per unit mass. The history of the step carries each upstream cell's specific internal energy formed at that cell's composition and temperature, not this cell's caloric state evaluated at  $T_{\text{up}}$ : the rovibrational heat capacity belongs to the gas that holds the molecules, so across a dissociation front two cells at the same temperature store different energy. With the metals on, the root is bracketed rather than guessed (`solve_T_brent`, on by default through `use_brent_tsolve`): 80 logarithmically spaced probes from  $\max(0.05T_{\text{guess}}, 1 \text{ K})$  to  $4T_{\text{guess}}$  locate the *first* sign change, i.e. the lowest root, and Brent's method closes the bracket to  $2\epsilon|b| + \text{tol}/2$  with  $\text{tol} = 10^{-10}$  in  $T/T_0$ , which is what keeps the solve off the hot spurious root; the cooling it balances is `radiative_cooling_of_cell`, the assembly `eval_cool` reports; a cell with no bracket keeps its converged equilibrium temperature and is counted as rejected. Without the metals, or with `Brent solver: False`, the root is MINPACK `hybrd1` on the same residual (`post_process_adv.f90`). The metal densities and the fine-structure escape probabilities are held fixed while the root finder varies  $T$ , the latter because the optical depth is a column over the whole atmosphere and not a function of this cell's trial temperature.

**The heating efficiency.** The diagnostic  $\eta_{\text{heat}}$  of (38), written as column `eta` by `write_output.f90`, is a property of the composition the sweep returned and is not a free parameter anywhere in the code; the closed-form parameterizations

of the same quantity that the literature uses are reproduced offline by `eta_approx.py` for comparison only.

## 6. The molecular layer, the trace metals and element transport

Everything in this section is *off* by default. A run acquires the molecular layer by asking for it (`Molecular chemistry: True`, which also switches `Molecular base` on so that the base particle count and the chemistry describe one gas), the metals by the presence of `metals.inp` or of elemental abundances in a handoff file, and the element transport by `He_diffusion`. An atomic, metal-free run never enters any of the code described here.

### 6.1 The H<sub>2</sub>/He reaction network

`mol_rates.f90` carries the reaction rate coefficients of Koskinen et al. (2022) Table 1 as published, in cgs ( $\text{cm}^3 \text{s}^{-1}$ , and  $\text{cm}^6 \text{s}^{-1}$  for the two three-body entries R13 and R15; R13 is multiplied by the total heavy-particle density and R15 by a collider-resolved one, below). EXHALE carries one temperature, so the heavy-particle and the electron temperature of that table are the same variable. The photorates P1–P5 of the paper are marked “SC” there, computed from cross sections, stellar flux and columns; in EXHALE they are the integrals of Sec. 3.4 and are not part of this module.

One entry is deliberately *not* transcribed. R12, the thermal dissociation of H<sub>2</sub>, is built by detailed balance from the R15 recombination,

$$k_{12}(T) = \frac{k_{3b}(T)}{K_{\text{eq}}(T)}, \quad (79)$$

with  $K_{\text{eq}}$  the  $\text{H} + \text{H} \rightleftharpoons \text{H}_2$  equilibrium constant built from the same internal partition function  $Q_{\text{H}_2}(T)$  the caloric equation of state uses (Sec. 6.4), because the two Table-1 entries are the same channel in opposite directions and their source measurements do not overlap in temperature. The third body of R13 is the total gas-particle density excluding electrons, its source (Miller et al. 1968) being a single coefficient with no collider dependence measured. R12 and R15 are collider-resolved (corrected 2026-09-17, item L7g): the association rate is the sum  $\sum_M k_1(M) n_M$  over  $M = \text{H}_2, \text{H}$  and He with the published coefficients of Cohen & Westberg (1983), carried in the code as an H<sub>2</sub>-equivalent third-body density so that R12 keeps following R15 by (79). Applying the  $M = \text{H}_2$  coefficient to the total density, which is what the code did before, over-counts the association: at the base of the LHS 1140 b hot-Uranus column it made the R15 rate 22 per cent too large and the molecular chemical heat 21.6 per cent too large. No measurement of He as a third body for this reaction exists, so He is given argon’s coefficient, and that entry remains an upper bound: a monatomic third body has no internal modes to take up the released energy. Two further limits in a helium-dominated gas, the R16, R18 and R19 differences of order unity to ten with García Muñoz (2025), are recorded at the code site rather than patched.

**Two entries of Table 1 are not transcribed as printed.** The first is R17,  $\text{He}^+ + \text{H}_2 \rightarrow \text{H}^+ + \text{H} + \text{He}$ , the dissociative charge transfer. Table 1 gives it as  $10^{-9} \exp(-5700/T)$  (Moses & Bass, 2000), an Arrhenius form with a 0.49 eV barrier, and that form alone is  $2 \times 10^4$  below the directly measured rate at 300 K and 200 below it at 400 K. The reaction has two mechanisms: at thermal energy it proceeds by tunnelling out of a long-lived  $\text{He}^+ - \text{H}_2$  complex, because reactants and products do not correlate adiabatically, and only above  $\sim 400$  K does an over-barrier branch take over (Böhringer & Arnold, 1986). The drift-tube measurements give the two-body total as  $k_2 = 1.1 \times 10^{-13} (300/T)^{0.24} \text{ cm}^3 \text{s}^{-1}$  over 18–408 K (Böhringer & Arnold, 1986), consistent with  $(1.1 \pm 0.1) \times 10^{-13}$  at 330 K and  $(1.5 \pm 0.15) \times 10^{-13}$  at 78 K

(Johnsen et al., 1980). The code therefore carries the sum of the two mechanisms, with the tunnelling branch taken as the measured total less the radiative branch the network carries separately as R23, so that the coded channels sum to the measured total:

$$k_{R17}(T) = \left[ 1.1 \times 10^{-13} (300/T)^{0.24} - k_{R23} \right] + 10^{-9} \exp(-5700/T).$$

Against the only measurement of the rising branch (Johnsen et al., 1980, their Fig. 3, carried to an effective temperature of 700 K) the sum is inside a factor two from 300 to 700 K. The three-body channel both papers measure,  $k_3 = 1.6 \times 10^{-30} (100/T)^{1.27} \text{ cm}^6 \text{ s}^{-1}$ , is omitted because  $k_3 n = 6 \times 10^{-19} \text{ cm}^3 \text{ s}^{-1}$  at 1000 K and  $n = 7 \times 10^{12} \text{ cm}^{-3}$ , five decades below the two-body rate.

**The second is R20.** Table 1 gives  $\text{He}^+ + \text{H}_2 \rightarrow \text{HeH}^+ + \text{H}$  as  $4.2 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ , citing Schauer et al. (1989). That paper measures the radiative ( $\text{He}^+ + \text{H}_2 \rightarrow \text{He} + \text{H}_2^+ + h\nu$ , Table 1's R23) and the dissociative ( $\text{He}^+ + \text{H}_2 \rightarrow \text{He} + \text{H} + \text{H}^+$ , R17) charge transfer over  $15 \leq T \leq 40 \text{ K}$  and no  $\text{HeH}^+$  channel; it states that the  $\text{HeH}^+$  channel “apparently does not become allowed until the collision energy approaches 9 eV”, and its  $\text{H}_3^+$  signal is the *sum* of the radiative and the  $\text{HeH}^+$  channels, which bounds the latter at  $\leq 1.0 \times 10^{-14}$  – 42 times below the value Table 1 assigns it. R20 is therefore not carried. The  $\text{HeH}^+$  source the network uses instead is the one Table 1 omits,  $\text{H}_2^+ + \text{He} \rightarrow \text{HeH}^+ + \text{H}$  at  $3.0 \times 10^{-10} \exp(-6717/T)$  (Black, 1978), whose coefficient is a Maxwellian average over a thermal population of  $\text{H}_2^+$  vibrational levels.

Measured on LHS 1140 b at  $\text{He}/\text{H} = 2.13$  with the  $\text{H}_2$  row written out reaction by reaction, the repair rewires the base layer's  $\text{HeH}^+$  and  $\text{H}^+$  budgets – at the first cell  $n(\text{HeH}^+)$  falls by a factor 487 at 1023 K and  $8 \times 10^4$  at 629 K, and  $n(\text{He}^+)$  rises by 1.11 and 4.4 – and leaves the net  $\text{H}_2$  destruction there within a per cent, because the helium ions of that layer are sink-limited. The same measurement shows what sets the  $\text{H}_2$  of a helium-rich base: the helium-ion and proton channels, with the thermal dissociation R12 nine decades below the first cell's total  $\text{H}_2$  loss at 1023 K and twenty-two decades below it at 629 K, and nowhere in that layer reaching 0.4 per cent of the loss. The helium-ion channels, not the temperature and not the photons, set the  $\text{H}_2$  content of a helium-rich base in this network.

The eight balance rows the network makes are (50)–(53) together with the  $\text{H}^+$ ,  $\text{He}^+$ ,  $\text{He}^{2+}$  and  $\text{He } 2^3\text{S}$  rows of Sec. 4.2.

## 6.2 $\text{H}_2$ photoabsorption resolved into channels

`h2_photo_channels.f90` splits the one measured cross section  $\sigma_{\text{H}_2}(E)$  into four mutually exclusive final states whose sum is identically  $\sigma_{\text{H}_2}$ , so no consumer of the opacity changes and the total  $\text{H}_2$  destruction rate is what it was; what changes is which fragments the source terms are told to make.

$$\sigma_N = f_n(E) \sigma_{\text{H}_2}(E) \quad \text{H}_2 + h\nu \rightarrow \text{H} + \text{H} \text{ (33–41 eV window),} \quad (80)$$

$$\sigma_M + \sigma_S + \sigma_D = [1 - f_n(E)] \sigma_{\text{H}_2}(E) \quad \text{the three ionizing channels,} \quad (81)$$

with  $f_n$  the neutral fraction of Chung et al. (1993) Table 1, carried as a *fraction* so that it rides on this code's  $\sigma_{\text{H}_2}$  normalization rather than importing theirs. Inside the ionizing part the split follows the Chung et al. Table II ratio column  $r(E) = \sigma(\text{H}^+)/\sigma(\text{H}_2^+)$ , a ratio of time-of-flight detector signals and therefore a ratio of *events*, one count per event and not one per proton; `frac_H2_dissociative_ionization` is that column converted to  $\sigma(\text{H}^+)/[\sigma(\text{H}^+) + \sigma(\text{H}_2^+)]$  and tabulated at 69 energies from 18.076 to 124 eV, linearly interpolated, zero at and below threshold and held flat above 124 eV. The thresholds are 15.4 eV (M, leaves  $\text{H}_2^+$ ), 18.076 eV (S, leaves  $\text{H} + \text{H}^+$ ), 51.4 eV (D, two protons), and none

of its own for N. The D channel is resolved by default (H2 double ionization: chung80, the 20 per cent of the  $H^+$  events at and above 80 eV that Chung et al. (1993) state, ramped linearly from the 51.4 eV threshold, the ramp not being in the source); yan\_rho builds it from the double-to-single ratio of Yan et al. (1998) instead (at 110 eV chung80 gives 15 per cent more double ionization than yan\_rho), and off folds it back into S, which reproduces the code before the channel existed. The N window is resolved by default as well (H2 neutral dissociation: True).

The energy of one absorbed photon is accounted channel by channel by h2\_channel\_energy\_recipients, the single statement of what each channel charges, and the recipients of each sum to  $h\nu$  once. Two entries are fixed with no free parameter: the double channel's vertical threshold 51.4 eV (Yan et al., 1998) exceeds the asymptotic product energy  $D_0 + 2I(H) = 31.675$  eV by exactly the kinetic energy release of the two receding protons, 19.725 eV; and the neutral window leaves both fragments in  $n = 2$  (the  $Q2 \ ^1\Pi_u$  state whose fluorescence curve overlaps this cross section), so  $2E(n=1 \rightarrow 2)$  departs as prompt  $Ly\alpha$  and two-photon continuum and is not heat.

### 6.3 Lyman-Werner photodissociation

lyman\_werner.f90 adds the Solomon process,  $H_2(X) + h\nu(11.2\text{--}13.6\text{ eV}) \rightarrow H_2(B, C) \rightarrow H + H$ , which the Koskinen et al. Table-1 network leaves out (their photorates start at the 15.4 eV edge). The band is 912–1201 Å: 912 Å is the H Lyman edge, and 1201 Å is both the blue edge of band B2 and the red end of the Lyman and Werner line list the shielding table is built from. Draine & Bertoldi's own 912–1110 Å is not used, because a planetary base runs at 700–3200 K where the vibrationally excited levels pump in lines longward of 1110 Å that carry about a third of the dissociations at saturation. One band, one line list, one normalization.

The optically thin rate is formed from the band-integrated *energy* flux at the planet and the mean photon energy of a flat  $F_\lambda$  band,  $\langle h\nu \rangle = 2hc/(912 + 1201 \text{ Å}) = 11.7354$  eV:

$$k_{LW}^{\text{thin}} = \sigma_{LW} \frac{\xi_{\text{day}} F_{LW}}{\langle h\nu \rangle}. \quad (82)$$

The band-mean  $\sigma_{LW}$  is not the effective cross section  $2.263 \times 10^{-18} \text{ cm}^2$  that Draine & Bertoldi (1996) Table 1 and 2 imply for a flat- $F_\lambda$  field restated per 912–1201 Å photon: the rate is the level-resolved table of h2\_self\_shielding\_table.f90, normalized per photon of the same band, and that number is only the limit it approaches in optically thin interstellar conditions.

That table is a line-by-line calculation with *overlapping* lines, and it stores three different counts because three are needed:  $\sigma_{\text{diss}}$ , dissociations per unit incident band photon fluence, which carries the self-shielding, the line overlap and the trapping of the fluorescent photons and sets the rate and the fragment heating;  $p_{\text{single}} = D/(D + \sum A_{ul})$ , dissociations per pump with every fluorescent photon free to escape, which sets the vibrational heat return because a trapped decay plus its re-absorption moves one molecule down and another up and deposits nothing; and  $p_{\text{eff}} = D/(D + \sum A_{ul} P_{\text{loss}})$ , dissociations per photon removed from the beam, which is the band photon ledger. They are related by  $\sigma_{\text{diss}} = p_{\text{eff}} \sigma_{\text{pump}}$ , and the beam's own loss is served from the same table, so the photons the beam loses and the dissociations the rate spends are one absorption with one normalization. Line overlap is not a correction at these columns: at  $n_H = 10^{13} \text{ cm}^{-3}$  the surviving pumping relative to a one-line-at-a-time calculation is 0.45 at  $N_{H_2} = 10^{20} \text{ cm}^{-2}$  and 900 K, and  $4.8 \times 10^{-5}$  at  $4.4 \times 10^{21} \text{ cm}^{-2}$  and 2700 K, which is where the base of a molecular layer sits. The rate enters (50) as  $g_{LW}$  and is zero when no band flux is supplied.

## 6.4 The caloric equation of state

The caloric equation of state that gives the bound  $H_2$  its rovibrational heat capacity is (14) of Sec. 2.5, where it is stated with its validity and its Caloric EOS: `monatomic switch. caloric_eos.f90` supplies the four maps the hydrodynamics needs,  $(\rho, \rho e) \rightarrow T$ ,  $(\rho, \rho e) \rightarrow p$ ,  $(\rho, p) \rightarrow \rho e$  and  $(\rho, p) \rightarrow \gamma_{\text{eff}}$ , and  $\gamma_{\text{eff}}$  itself for the wave speeds.

## 6.5 Carrier transport

Everywhere else in the code the composition of a cell is the root of a local algebraic system. A *carrier* is a species the code does not fix that way: one whose chemical time can exceed the flow time, so that its density in a cell is what the flow brought there and not what the cell's own equilibrium would give. `diffusive_photochemistry.f90` solves, for the carriers, the continuity equation of the neutral carriers  $H_2$ ,  $OH$ ,  $H_2O$  and  $CO$  (and, with `Ionization transport`, the proton),

$$\frac{\partial n_i}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} [r^2 (n_i v + J_i)] = P_i - L_i, \quad J_i = -n_{\text{tot}} (D_i + K_{zz}) \frac{\partial f_i}{\partial r} - n_i D_i \left( \frac{1}{H_i} - \frac{1}{H_{\text{atm}}} \right), \quad (83)$$

with  $f_i = n_i/n_{\text{tot}}$ . Between them these four carry every H nucleus that is not atomic or in a molecular ion and every O and C nucleus that is not a free atom.  $H_2^+$ ,  $H_3^+$ ,  $HeH^+$  and  $O(^1D)$  are left local; on the HD 209458 b molecular example the three molecular ions together carry at most  $1.2 \times 10^{-9}$  of the H nuclei, and the H-nucleus budget is closed against the element total rather than against a sum of transported species, so whatever they hold is simply not available to the carriers. All four transported species are neutral, so the ambipolar field term of the general drift-diffusion equation is absent rather than neglected.

$D_i$  is Blanc's law over the background carriers (H I, H II,  $H_2$ ,  $H_2^+$ ,  $H_3^+$ , He I, He II, He III,  $HeH^+$ ), each pair from the rigid-sphere coefficient of Banks & Kockarts (1973). The ion-neutral polarization enhancement is not added: the transported species only exist where the gas is neutral (electron fraction per H nucleus at most  $10^{-7}$  over the cells that hold  $H_2$ ), so the ion pairs carry that share of the Blanc friction sum, and for  $H_2O$  the induced-dipole formula would be the wrong one anyway, a permanent dipole of 1.85 D dominating the capture rate.  $K_{zz}$  is read from `kzz_cell` and nowhere else; that array is filled once, from the scalar `He_Kzz` or from the lower-atmosphere profile's own  $K_{zz}$  column. Settling is  $G_i = (m_i - \bar{m})g/(kT)$  with  $\bar{m} = \rho/n_{\text{tot}}$  and  $g$  from the same potential the hydrodynamics uses; thermal diffusion is not carried ( $\alpha_T = 0$ ), there being no evaluated  $\alpha_T$  for  $H_2O$ ,  $OH$  or  $CO$  in an  $H_2/He$  bath.

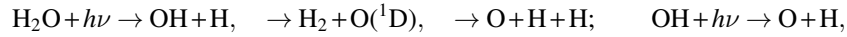
$P_i - L_i$  is not written in this module: it is obtained by calling the very rows the local equilibrium solve uses, `mol_heh_rows` row 4 for  $H_2$  and `oxygen_carrier_rows` for  $OH$  and  $H_2O$ , at the trial densities with the background frozen, and the Jacobian is a forward difference of those same calls, so a rate that changes there changes here.  $CO$  has no chemical source and only its transport moves it, with a ceiling from `limit_to_element_budget` that keeps it from carrying the oxygen and the carbon out of the wind, where the molecule cannot survive and where the O I and C II lines are the coolant.

The discretization is cell-centered  $f_i$ , face-centered fluxes on `r_edg`, one backward-Euler step per call, Newton with a backtracking line search, block tridiagonal with  $4 \times 4$  blocks. Faces 0 and  $N$  carry zero diffusive flux; the drift term switches from central to donor cell on a Péclet test. The element reservoirs at the base are Dirichlet, imposed through `melem_ab`, the base density and the equation of state, but the *partition* of each element among its carriers is not imposed anywhere, which is what makes the comparison against a photochemical handoff a test rather than a tautology.

The material advection of a carrier is *not* in these rows. It is the divergence of the hydrodynamic face mass flux,  $F_p(j)Y_c(j)/m_c$ , taken with the mass row itself inside the Runge-Kutta stages (107) on the same faces, areas, volumes and time step (`species_face_flux.f90`), so a cell can only lose the carrier the mass row says it loses and a uniform partition is preserved to round-off. That this half must ride on the *face* mass flux is a numerical statement about the front: a first-order donor-cell difference on the cell velocity carries a numerical diffusivity  $|v|\Delta r/2$ , which on the 500-cell hot Uranus grid is 8.5, 20 and 22 times the molecular  $D$  of  $H_2$  at 1.15, 1.30 and 1.50  $R_p$ . The key is Molecular carrier transport, default off.

## 6.6 The oxygen option

With `thereis_oxychem` (default off) three neutral carriers OH,  $H_2O$  and CO join the network. `oxygen_rates.f90` carries the rate coefficients, each transcribed from a named publication and each checked against two independent ports of the same literature that are on this machine (the Photochem `zahnle_earth.yaml` network and the VULCAN `NCHO_photo_network.txt`), every disagreement being recorded at the coefficient. `water_photolysis.f90` supplies the photolysis,



in the four far-ultraviolet bands of Sec. 3.1 (the balance rows these rates enter are (54) and (55)), whose edges are fixed by the branching ratios rather than chosen: the three-body branch opens only in the interval containing  $Ly\alpha$  and is 12 percent there, so  $Ly\alpha$  needs its own band on branching grounds alone. The first branch alone carries 54.5 percent of the  $H_2O$  loss at the HD 189733 b base, which makes the OH/ $H_2O$  cycle catalytic rather than a one-way sink. `co_photodissociation.f90` destroys CO on the same Lyman-Werner beam, using a band-mean dissociation cross section times the band photon flux and the Visser et al. (2009) shielding function  $\Theta(N_{CO}, N_{H_2})$  rather than their  $k_0\Theta$  form, because EXHALE has a band flux and not an interstellar field and  $k_0$  is a statement about a particular field. The other CO destruction channel is the  $He^+$  charge transfer  $He^+ + CO \rightarrow C^+ + O + He$  at the Langevin value  $1.6 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ , which appears in the  $He^+$  row of (51)'s system and whose +2.2117 eV appears in heating channel 18, one event counted once.

## 6.7 The lower-atmosphere handoff

The base of the escape model sits at a stated pressure level, and three routes supply the state there.

**The analytic column.** `lower_column.f90` integrates the hypsometric relation from the 1-bar level up to the base pressure (default 1  $\mu\text{bar}$ ), isothermal at  $T_{eq}$  with spherical gravity, using the mean molecular weight of the chemical-equilibrium  $H_2/H/He$  partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022) eqs. (11)–(13),

$$q_{H_2} = \frac{1.9845 + 10^u - \sqrt{10^u(3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p [\text{bar}] + 6.2645, \quad (84)$$

with  $q_{He}$  from the elemental He/H and  $q_H = 1 - q_{H_2} - q_{He}$ . It returns the radius of the base level and the base mixing ratios, and also a fully atomic bracket, because in chemical equilibrium  $q_{H_2}(1 \mu\text{bar})$  stays large up to  $T \sim 2000 \text{ K}$  while photochemistry dissociates  $H_2$  there: a large  $q_{H_2}$  on a hot planet is the signal that a photochemical handoff is needed.

The validation case is the Koskinen et al. (2022) Model A Uranus-mass planet at 0.05 au, for which the column gives  $r_0(1 \mu\text{bar}) = 1.34 R_{\text{bar}}$  with  $q_{\text{H}_2} \simeq 0.84$ .

**base.inp, the scalar handoff.** An optional file of “key value” lines, written by `src/utls/run_lower.py` or `src/utls/vulcan_to_base.py`. Every key belongs to one of five categories and the category states what the value may do to the wind: *provenance* (comments only); *EOS boundary* ( $T_{\text{base}} \rightarrow T_0$ ,  $r_{\text{base}} \rightarrow R_0$ ,  $p_{\text{base}}$ , and  $q_{\text{H}_2\text{base}}$ , the  $\text{H}_2$  volume mixing ratio of the base gas); *elemental reservoir* ( $\text{HeH}_{\text{base}}$  and  $\langle \text{El} \rangle_{\text{H}_{\text{base}}}$  for the ten elements, overriding `metals.inp` for that element); *initial guess* (no key today); and *boundary constraint* ( $K_{\text{zz}_{\text{base}}} \rightarrow \text{he}_{\text{kzz}}$ ).  $q_{\text{H}_2\text{base}}$  is the  $q_{\text{H}_2}$  of (15) and is imposed on the  $\text{H}_2$  partition of the inflowing ghost species as well; it is refused above  $0.5/(0.5 + \text{He}/\text{H})$ . Species mixing ratios other than  $q_{\text{H}_2\text{base}}$  stay comments: no part of the code consumes them, so presenting them as physics would be false. A `base.inp` that states  $p_{\text{base}}$  fixes the base level and  $n_0$  is derived from it as in Sec. 1.2, instead of being read from `input.inp`.

**Lower atmosphere profile, the tabulated handoff.** `lower_atmosphere_profile.f90` reads the lower model’s solution over an *interval* of pressure that reaches above the level where EXHALE places its base. It supplies the state at the matching level  $p_{\text{match}}$  ( $T$ ,  $r$ ,  $q_{\text{H}_2}$ , and the elemental ratios  $\text{He}/\text{H}$  and  $\text{El}/\text{H}$ ), which is what the base equation of state and the elemental reservoirs are built from, and  $K_{\text{zz}}$  as a *profile*, because the homopause, the only consequential thing  $K_{\text{zz}}$  does, is where  $K_{\text{zz}} = D_{12}$ , a property of two profiles, and the molecular coefficient rises by nearly five decades between the base and  $1.2 R_p$ . Every intensive quantity is interpolated linearly in  $\log p$ , the variable both models solve on; extrapolation is never performed. The values go in through exactly the doors the scalar keys use, so nothing in the base equation of state changes;  $n$  and  $\rho$  carried by the file are *not* imposed, the base density remaining  $n_0$ .

## 6.8 Trace metals

`metals.inp` (`metals_input_read.f90`) sets the elemental abundances  $n_X/n_{\text{H}}$  by number, one ion per line with the neutral label naming the element; a second statement of the same element stops the run, since two abundances for one element are two answers to one question. The same file also carries `cno_cool`, `eos_metals`, `cx_full` and `cx_O2p_H`. The ionization of each element is solved in the same coupled system as hydrogen and helium (`System_HeH_metals.f90` and its triplet and molecular counterparts), through the rows (48); the code carries no parallel decoupled metal solver.

`eos_metals` (default *on*) puts the metals into the bulk gas budget consistently: their mass in  $\rho$  and hence in  $\mu$ ,  $v_0$ ,  $p_0$ ; their electrons in  $n_e$ ; and their nuclei in  $n_{\text{tot}}$  and in the ghost pressure, matching the charge balance the ionization systems already use. The legacy trace approximation (metals in the ionization and cooling but not in the bulk gas) is `eos_metals 0`. A metals-off run is identical either way, every metal sum vanishing.

## 6.9 Binary element transport

`binary_element_diffusion.f90` treats the gas as two components moving through each other with the single mass-averaged velocity of the hydrodynamics: component 1 is the hydrogen carriers together with the trace metals and the heavy nuclei bound in the molecular carriers, and component 2 is helium in all stages. Since  $\rho_1 + \rho_{\text{He}} = \rho$  exactly, the

two diffusive mass fluxes close,  $J_1 = -J_{\text{He}}$ , and the composition update creates no mass. The transported variable is the helium *mass* fraction  $X = \rho_{\text{He}}/\rho \in [0, 1]$ , which is bounded at both ends unlike the ratio  $n_{\text{He}}/n_{\text{H}}$  that the earlier trace kernel carried and that diverges exactly in the helium-rich limit this operator exists for. The equation is

$$\frac{\partial(\rho X)}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} [r^2 (\rho X v + J)] = 0, \quad J = -\rho(D_{12} + K_{zz}) \frac{\partial X}{\partial r} - \rho D_{12} G X(1-X), \quad (85)$$

$$G = \frac{(m_{\text{He}} - m_{c1})m_{\text{HG}} - (\bar{Z}_{\text{He}} - \bar{Z}_{c1}) eE}{kT} + \alpha_T \frac{d \ln T}{dr} - \frac{d \ln \psi}{dr} \quad [\text{cm}^{-1}]. \quad (86)$$

The masses and charges in  $G$  are *carrier* quantities:  $\psi = n_{1c}/n_{\text{H}}$  is the number of collision partners each hydrogen nucleus is spread over (1 atomic, 1/2 fully molecular) and  $m_{c1} = m_1/\psi$ . The gradient coefficient collapses to  $\rho D_{12}$  for any  $\psi$ , so the Fickian term is the same in the molecular and the atomic region; the piece of the mole-fraction gradient the chemistry carries does not cancel and rides in  $G$  as  $-d \ln \psi / dr$ , which vanishes identically wherever  $\psi$  is constant. The ambipolar field is computed and not assumed, following Koskinen et al. (2013) section 2.1,

$$eE = -\frac{1}{n_e} \frac{d(n_e kT)}{dr} = -kT \frac{d \ln(n_e T)}{dr}, \quad (87)$$

in the logarithmic form, which is exact for the barometric stratification of the base and the same second-order central difference elsewhere. It is gated by `he_ambipolar` (default on);  $\alpha_T$  is 0 by default (`He_alphaT`).

The equation is split. The advection of the element mass fraction is a divergence of the same face mass flux the density rides on and is carried inside the Runge-Kutta stages (107), so each element's nucleus total over the domain changes only by its boundary fluxes to round-off. What is left for the implicit step here is  $\rho(X^{\text{new}} - X^{\text{adv}})/\Delta t = -r^{-2} \partial_r(r^2 J)$ , taken at the composition the stages advected; the operator advects only when it is *given* the face mass flux, which happens in the fixed-wind relaxation that has no stages to ride on.

The discrete operator is one backward-Euler step with frozen coefficients, faces carrying  $r^2$  areas, Newton on a tridiagonal Jacobian (the step is nonlinear in  $X$  through the product  $X(1-X)$ ). The gradient term is central; the drift term keeps a Péclet hybrid, central where  $|B|\Delta r \leq 2(A+E)$  and donor cell otherwise. The two factors of  $X(1-X)$  come from *opposite* sides of the face, because the drift is a counter-flow and a donor-cell rule must take each element's mass fraction from the cell that element leaves:

$$J_{\text{drift}}(f) = \begin{cases} -B_f X(j+1) [1 - X(j)], & B_f \geq 0 \text{ (helium inward)}, \\ -B_f X(j) [1 - X(j+1)], & B_f < 0. \end{cases} \quad (88)$$

A donor at  $X = 0$  then sends nothing and an acceptor at  $X = 1$  receives nothing, so a cell on either end of the axis cannot be pushed past it; with the Péclet switch the same holds in the central branch, and the row argument (time term, gradient flux, upwind advection and drift all of one sign at a cell that first touches an end) gives  $0 \leq X \leq 1$  on the residual itself rather than through a comparison principle. Measured, the worst excursion outside  $[0, 1]$  before the assertion clip is  $-1.05 \times 10^{-1}$ .

`project_elements` then rebuilds the species vector on the new element totals: helium-bearing species are scaled by  $r_{\text{He}} = n_{\text{He}}^{\text{new}}/n_{\text{He}}^{\text{old}}$ , component-1 species by  $r_1$ , and a species carrying a nucleus of each ( $\text{HeH}^+$ ) by  $\min(r_1, r_{\text{He}})$ , the nuclei this undercounts for the element with the larger factor being deposited into that element's neutral ground species. Every operation is a multiplication by a nonnegative factor or an addition, so no negative intermediate can arise and both element totals are met exactly. The invariant is that the species handed back carry the mass the species handed in

carried, to round-off, which requires  $r_1$  to be the ratio of the two component-1 *masses* read from the cell. The result is the initial guess for `ioniz_eq`, which owns the split *within* an element while this step owns the element totals.

**The base of the operator, and what crosses it.** The gradient and drift coefficients are built on the faces 1 to  $N-1$ , and faces 0 and  $N$  are left at zero, so **the diffusive flux at the base face is zero by construction and not by the state**: an element crosses either end of the column with the gas alone. The flux the Dirichlet reservoir composition drives sits at the first solved face, which is face 1 while the operator solves rows 2 to  $N$  against the reservoir, and face 0 where a closed base makes cell 1 an unknown. The record `element_base_flux` carries both faces rather than leaving either to be assumed: the advective and the diffusive half at the base face and at the first solved face, in  $\text{gcm}^{-2}\text{s}^{-1}$  positive outward (toward increasing  $r$ ), together with the index of the first solved face and a count of the operator evaluations that produced a composition. A report reads the evaluation the numbers belong to, and a count of zero means no operator has run, which is not a flux of zero. The evaluation matters because the base entry is not steady while the base is: MEASURED on the `lower_profile` configuration, the advective half at the base face reads  $+4.47 \times 10^{-8}$  at evaluation 1079,  $-3.35 \times 10^{-10}$  at 11220 and  $+6.29 \times 10^{-10}$  at 12000, as the base sound wave turns the face flux over in sign (item P6e).

The whole operator is gated by `He_diffusion`, default off. With `he_metal_diffusion` (default off) each trace element additionally carries its own mass fraction and its own binary coefficient; otherwise the metals are frozen to hydrogen, i.e. they ride in component 1 at the metal/H ratio the cell holds, and what `project_elements` preserves for them is that ratio and not the density.

The Dirichlet composition sits at the *center* of cell 1, half a cell above the base level, which since 2026-09-27 is the lower face of the grid (Sec. 8.1). That offset is a first-order term in the cell width which is recorded and not changed; the He/H change near the base is  $1.5 \times 10^{-4}$  (md/Update\_EXHALE\_stage3.md section 29.11).

## 6.10 The interdiffusion enthalpy flux

Particles that move relative to the mass-averaged velocity carry their enthalpy with them. In a mixture whose species move with velocities  $v + w_s$ , the energy equation is

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot [(\mathcal{E} + p)\mathbf{v}] = \dots - \nabla \cdot q_d, \quad q_d = \sum_s h_s J_s, \quad J_s = \rho_s w_s, \quad \sum_s J_s = 0, \quad (89)$$

(Cook, 2009, eqs. 11–13, the Dufour flux and the kinetic energy of the relative motion left out there and here). Because  $J_s$  is a flux relative to the *mass*-weighted velocity,  $h_s$  is the specific enthalpy and not the internal energy. With the binary element model of Sec. 6.9 every species moves with its element and the electrons with their ions, so  $J_1 = -J_{\text{He}}$  and

$$q_d = J_{\text{He}} (h_{\text{He}} - h_1) + \sum_X J_X (h_X - h_{\text{H}}), \quad (90)$$

the metal sum present only where `He_metal_diffusion` moves each metal against hydrogen (taken against  $h_{\text{H}}$ , component 1 without its metals, the group that recoils) and the metals carry mass in the mixture. The  $h_c$  are *sensible* specific enthalpies built from the caloric equation of state (`component_specific_enthalpies` in `binary_element_diffusion.f90`): per particle  $h = e + kT = kT \gamma / (\gamma - 1)$  for every atom, ion and molecular ion, each ion counted with the  $Z$  electrons the ambipolar field makes follow it, and  $\text{H}_2$  adding its rovibrational energy of Sec. 6.4;

$\text{HeH}^+$  is shared between the two components in the proportion of its nuclei. Formation energy is not in them: the energy update books chemical energy through the formation-energy density, whose reference in the source step is the composition the element transport has just written, so the formation energy of the moved nuclei rides with them already, and the stationary energy row holds chemical energy only as the reaction heating and cooling inside  $\mathcal{H} - \Lambda$ .  $J$  is the operator's whole diffusive flux, gradient, eddy and settling drift together, since eddy mixing carries enthalpy as it carries the elements. Face values of  $h_c$  are the arithmetic means of the two cells, the face rule of every coefficient of the operator, and the flux is zero wherever  $J$  is, in particular at faces 0 and  $N$ .

The divergence is taken on the face areas and volumes of the element rows,  $[A_+ q_d(j) - A_- q_d(j-1)]/V_j$  for  $j = 2 \dots N$ , so an interior face cancels between its two cells and the energy the term moves telescopes over the column; cell 1, the Dirichlet reservoir of the element operator, carries no element equation and no interdiffusion energy. Both routes carry it: the marching step subtracts  $\Delta t \nabla \cdot q_d$ , formed from the element fluxes the accepted element step itself carried, from the thermal energy before the coupled source step starts (EXHALE\_main.f90); the stationary energy row adds  $\nabla \cdot q_d$  formed from the element fluxes of the state itself (123).

**The transported carriers.** A molecular carrier moved by the carrier operator of Sec. 6.5 ( $\text{H}_2$ , and OH,  $\text{H}_2\text{O}$ , CO with the oxygen chemistry) moves relative to the rest of its own elements as well: its diffusive, eddy and settling number flux  $\Phi_c$  moves nuclei that the element fluxes do not, and the nuclei of the same elements held by the species that are not carriers move the other way, nucleus for nucleus, so the element fluxes are left as they are. The same identity then gives

$$q_{\text{car}} = \sum_c \Phi_c \left[ h_c - \sum_{\text{el}} \nu_{c,\text{el}} \bar{h}_{\text{el}} \right], \quad (91)$$

with  $h_c$  the enthalpy of one carrier particle and  $\bar{h}_{\text{el}}$  the enthalpy of the non-carrier species of element el per nucleus (for hydrogen HI,  $\text{H}^+$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$  and the hydrogen share of  $\text{HeH}^+$ ; for O and C their atomic stages where the metals are in the mixture), each particle with the electrons its charge gave up; an element with no non-carrier species in a cell recoils with its neutral atom. An  $\text{H}_2$  molecule rising through atomic hydrogen at zero hydrogen flux thus carries  $h_{\text{H}_2} - 2h_{\text{H}} = k(u_{\text{rv}} - 5T/2)$ , about  $-1.5kT$  at 1000 K.  $\Phi_c$  is the carrier operator's own face flux for the state in hand (carrier\_enthalpy\_face\_flux in diffusive\_photochemistry.f90), with the temperature and particle count of that state; both ends of the carrier column are closed, so  $q_{\text{car}}(0) = q_{\text{car}}(N) = 0$  and the divergence, taken over cells  $1 \dots N$  on the carrier rows' geometry, sums to zero over the column. The stationary energy row adds  $\nabla \cdot q_{\text{car}}$  beside  $\nabla \cdot q_d$ ; the marching step subtracts  $\Delta t \nabla \cdot q_{\text{car}}$  after an accepted carrier step, from the fluxes of the composition that step returned (its own fluxes to first order in  $\Delta t$ , exactly at a fixed point). The ionization stages carried by Ionization transport are not covered: their face flux contains the element's advection, and the relative part is not formed. MEASURED on the certified well-mixed He/H = 0.19 state of LHS 1140 b before the term existed:  $|\nabla \cdot q_{\text{car}}|$  is a median 1.8 per cent of the heating over  $1.001 < r/R_p < 1.03$ , up to 10 per cent, comparable to the net heating where heating and cooling cancel, and that state is no longer stationary with the term (the carrier row stands at  $7 \times 10^{-3}$  on the first pass). The key Interdiffusion enthalpy flux: is **on by default** whenever He\_diffusion moves helium or a molecular carrier is transported, and controls both terms, because the term belongs to the energy equation of any mixture whose components move relative to one another; False exists to reproduce published escape models that omit it (Koskinen et al., 2013, eq. 3), (Koskinen et al., 2022, eq. B3). With He\_diffusion off the element term is identically zero with

no arithmetic, and so is the carrier term without transported carriers. MEASURED on the LHS 1140 b He\_diffusion states: the divergence reaches 13 to 26 per cent of the local energy budget at  $1.1\text{--}5 R_p$ , and on the molecular He/H = 1.8 state (24 outer passes with the term, not yet certified) it lowers  $T$  by a median 166 K over  $1.3\text{--}2 R_p$  and He/H by 3.1 per cent, and moves  $\log_{10} \dot{M}$  by +0.002.

## 7. Numerical scheme

### 7.1 The radial grid

The domain above the base level  $r = 1$  is discretized into  $N$  finite-volume cells (Grid cells:, default 500) with  $N_g = 2$  ghost cells at each end; all grid arrays run over  $1 - N_g \leq j \leq N + N_g$ . `define_grid.f90` builds the cell centers  $r_j$ , the interfaces  $r_{j+1/2}$  as their midpoints, and the widths  $\Delta r_j = r_{j+1/2} - r_{j-1/2}$ . The last is an exact identity of the discretization and not an approximation: it is the same face pair that forms  $\Delta V_j$  in (5), the same pair the gravitational source differences, and the same width the column integrals  $n_j \Delta r_j$  and the cell optical depths use.

Three grid types are available, and since 2026-09-27 every one of them places the first face exactly on the base level,  $r_{\text{edg}}(0) = 1$ , so that the reservoir stated there is the state of that face (Sec. 8.1); the two lower ghost cells lie below the level. Uniform takes  $\Delta r = (r_{\text{max}} - 1)/N$  and  $r_j = 1 + (j - \frac{1}{2})\Delta r$ , so its faces are  $1 + j\Delta r$  and the outer face of cell  $N$  is  $r_{\text{max}}$ . Stretched places the centers in geometric progression,  $r_j = 2x^j/(1+x)$ , so that the face between  $r_0$  and  $r_1$  is the level, with  $x$  the root of  $(N + N_g)\ln x - \ln[(1+x)/2] - \ln r_{\text{max}} = 0$  (the center of the outermost ghost at  $r_{\text{max}}$ ), found by Newton's method from  $x = r_{\text{max}}^{1/(N+N_g)}$ , which climbs to it monotonically because the function is increasing and concave. Mixed, the type every shipped configuration uses, lays  $N_{\text{low}}$  uniform cells of width  $\Delta r_{\text{base}}$  on the base and then stretches the remaining  $N_{\text{up}} = N - N_{\text{low}}$  cells by a factor  $k$  that solves

$$\frac{1 - k^{N_{\text{up}}}}{1 - k} = \frac{r_{\text{max}} - r_{N_{\text{low}}}}{\Delta r_{\text{base}}} \quad (92)$$

by Newton-Raphson to  $10^{-8}$ , which is Eq. (10) of Caldiroli et al. (2021). The two numbers are the key Base grid [dr,cells]: and default to  $\Delta r_{\text{base}} = 2 \times 10^{-4} R_p$  and  $N_{\text{low}} = 50$ , the values ATES hard-codes. The widths are then smoothed once by a 1-2-1 pass running downward from the junction to the base, the centers are rebuilt from the smoothed widths, and the grid is mapped by the affine map that sends its first face to 1 and the center of its outermost ghost to  $r_{\text{max}}$ , which multiplies every width by one factor and so keeps the smoothed shape. For all three types the first ghost center is then restated as the mirror of the first cell center,  $r_0 = 2 - r_1$ , which moves it by rounding only and makes the first face exactly 1.0; the run stops if that face is not 1 or if a ghost cell reaches  $r \leq 0$ . Until 2026-09-27 the level was a ghost center ( $r_0 = 1$  on the Mixed grid,  $r_{1-N_g} = 1$  on the other two), so every cell center has since moved (by half a cell on the Mixed grid, by one and a half on the other two) and a state written before then does not load on the present grid (Sec. 8.4). The requirement behind  $\Delta r_{\text{base}}$  is that the pressure scale height  $H = k_B T / (\bar{m}g)$ , with  $\bar{m} = \rho / n_{\text{tot}}$  the mean particle mass, be resolved by several cells at the base; a configuration that violates it is the one that produces a base-layer momentum imbalance no solver removes.

Two windows are located once, at grid construction:  $j_{\text{min}}$ , the first cell with  $r \geq r_{\text{esc}}$  (Escape radius [R\_p]:), which the marching convergence measure of Sec. 7.7 is taken over, and  $j_{\text{flux}}$ , the first cell with  $r \geq r_{\text{flux}}$  (default  $1.2 R_p$ ), which the stationary flux gate uses. Both are clamped to mid-domain, with a message, where the requested radius lies outside

the domain: a deep Roche-lobe-filling case can have  $r_{\max}$  smaller than the default escape radius, and an empty window would make the measure  $-\infty$  and the state converged at step zero.

## 7.2 Reconstruction

Both schemes reconstruct the *primitive* variables  $W = (\rho, v, p)$  to the two sides of every face, and both write the left state of face  $j$  from cell  $j$  and the right state of face  $j$  from cell  $j+1$ .

**PLM.** `PLM_rec.f90` builds a limited slope from the three cell averages  $j-1, j, j+1$  and evaluates it at the two faces,

$$W_{L,j+1/2} = W_j + \frac{1}{2}\sigma(r_{j+1} - r_j), \quad W_{R,j-1/2} = W_j - \frac{1}{2}\sigma(r_j - r_{j-1}), \quad (93)$$

with  $\sigma$  the generalized minmod (monotonized central) slope of the right, left and central difference quotients at  $\theta = 2$ ,

$$\sigma = \text{minmod}\left(\theta \frac{\Delta W_{j+1/2}}{r_{j+1} - r_j}, \theta \frac{\Delta W_{j-1/2}}{r_j - r_{j-1}}, \frac{W_{j+1} - W_{j-1}}{r_{j+1} - r_{j-1}}\right), \quad (94)$$

the slope being zero whenever the three arguments do not share a sign. This is the generalized minmod limiter as Käppeli & Mishra (2016) state it, at the least diffusive value for which the scheme stays total-variation diminishing; ATES uses the same limiter and attributes it to Kurganov & Tadmor (2000). The difference from ATES is that the quotients here are formed on the actual center spacing of a stretched grid. The same scalar limiter is exported as `PLM_rec_scalar`, so that a transported mass fraction rides on the same discretization as the density whose face flux carries it.

**WENO3.** `Reconstruction.f90` uses the third-order WENO reconstruction with the nonlinear weights of Yamaleev & Carpenter (2009). With  $\Delta W_{j\pm 1/2}$  the two interface jumps bounding cell  $j$ ,

$$\tau = \Delta W_{j+1/2} - \Delta W_{j-1/2}, \quad S_0 = 1 + \frac{\tau^2}{\Delta W_{j+1/2}^2 + \Delta r_j^2}, \quad S_1 = 1 + \frac{\tau^2}{\Delta W_{j-1/2}^2 + \Delta r_j^2}, \quad (95)$$

which is their  $1 + \tau/(\epsilon + \beta_r)$  with their already-squared  $\tau$  (their Eq. 22) and the floor  $\epsilon = \Delta r_j^2$  folded in. The face states are

$$W_{L,j+1/2} = W_j + \frac{S_0 C_2(j) \Delta W_{j+1/2} + D_1(j) S_1 C_1(j-1) \Delta W_{j-1/2}}{S_0 + D_1(j) S_1}, \quad (96)$$

$$W_{R,j-1/2} = W_j - \frac{D_2(j) S_0 C_2(j) \Delta W_{j+1/2} + S_1 C_1(j-1) \Delta W_{j-1/2}}{D_2(j) S_0 + S_1},$$

with the purely geometric coefficients  $C_1(j) = \Delta V_{j+1}/(\Delta V_j + \Delta V_{j+1})$ ,  $C_2 = 1 - C_1$ ,  $D_1(j) = \Delta V_{j+1}/(\Delta V_j + \Delta V_{j-1})$  and  $D_2(j) = \Delta V_{j-1}/(\Delta V_j + \Delta V_{j+1})$ . Each candidate is the linear interpolant of two cell averages *in the volume coordinate*, so the jump to a neighbor is weighted by the cell's own volume share; these shares, and not the nonlinear weights, are what set the order on a stretched grid. The ideal weights  $d_0 = 2/3$ ,  $d_1 = 1/3$  of the uniform-grid scheme enter only through their ratio, which is what  $D_1$  and  $D_2$  carry: exactly  $1/2$  on a grid uniform in the volume coordinate and  $1/2 + O(h)$  otherwise.

Two limits of the scheme are stated at the routine. What runs is a third-order WENO reconstruction *with* those weights and not the energy-stable ESWENO scheme of the same paper, whose extra dissipation flux (their Eq. 48) is not computed here. And the floor  $\epsilon = \Delta r_j^2$  has the required  $h^2$  scaling but drops the solution scale of the published prescription, so it is not invariant under a rescaling of the solution; what that costs is the stencil biasing in the tenuous outer domain and not the order, which is flat over four decades of the coefficient.

**Positivity of the face states.** Neither scheme has a positivity property: a face value of  $\rho$  or  $p$  can cross zero while every cell average on the stencil is positive, and at Mach 60 the thermal pressure is  $2 \times 10^{-4}$  of the total energy density, so a reconstruction across a jump tips it negative and the sound speed takes the square root of it. `positivity_limited_faces` therefore scales each offending face state *continuously* toward its own cell average,

$$W_{\text{face}} \leftarrow W_{\text{avg}} + \theta(W_{\text{rec}} - W_{\text{avg}}), \quad \theta = \min_{q \in \{\rho, p\}} \frac{q_{\text{avg}} - \varepsilon}{q_{\text{avg}} - q_{\text{rec}}}, \quad \varepsilon = \epsilon_{\text{mach}} q_{\text{avg}}, \quad (97)$$

which is the linear scaling idea of Zhang & Shu (2010a) applied to the primitive vector, and not their limiter: they scale the conserved vector in two steps with  $\theta$  the root of a quadratic, one  $\theta$  per cell against one per face state here, and their floor is absolute against one unit in the last place of the cell average here. What carries over is their Lemma 2.5, that a convex combination of an admissible average with the reconstruction is admissible; their Theorem 2.1 does not, its hypotheses being a Gauss-Lobatto quadrature set and a CFL condition nothing here imposes. Continuity is the point: a hard switch at the crossing is a step discontinuity of the residual and no stationary solve can descend across one. A face whose reconstruction is admissible has  $\theta = 1$  and is not rewritten.

**The PLM to WENO3 continuation.** The two-stage recipe `Reconstruction` scheme: `PLM+WENO3` marches with PLM until the convergence measure falls below the first `du_th` value (or its plateau is detected) and then continues with WENO3. Replacing one discretization by another between two time steps is a jump of the right-hand side, so the key `Reconstruction` continuation: `interpolates` instead. `reconstruction_continuation_rhs` (`steady_residual.f90`) evaluates the homotopy

$$R_{\lambda}(u) = (1 - \lambda)R_{\text{PLM}}(u) + \lambda R_{\text{WENO3}}(u), \quad (98)$$

returning its flux-difference and source parts separately, and blends the stored interface fluxes, the face pressures and the three momentum terms on the same  $\lambda$ . Everything the two schemes disagree about is inside the two evaluations: the face reconstruction, the extrapolated reconstructed-face boundary states, and the pressure split of Sec. 2.3. At  $\lambda \leq 0$  and  $\lambda \geq 1$  a single evaluation is taken and the two pure operators are reproduced exactly; only the open interval costs two right-hand sides. Because the admissible set  $\rho > 0$ ,  $p > 0$  is convex and each stage is then a convex combination of two forward Euler stages, the continuation cannot manufacture a positivity failure that neither scheme has. The ramp is adaptive:  $\lambda$  advances by  $\Delta\lambda$  where the relative state change of the step stays below a tolerance and  $\Delta\lambda$  is halved where it does not.

### 7.3 Approximate Riemann solvers

`Num_Fluxes.f90` carries three fluxes, selected by the mandatory key `Numerical_flux`. Each side's total energy and sound speed are built with its *own* heat capacity, which is what the caloric equation of state makes different across a face inside the  $\text{H}_2$  front; the physical flux is

$$F(W) = (\rho v, \rho v^2 + p, v(\mathcal{E} + p)), \quad (99)$$

the bracketed pressure being present under PLM without the well-balanced option, as Sec. 2.3 requires.

**HLLC.** The three-wave solver of Toro et al. (1994), with the Davis/Einfeldt speed bounds of Batten et al. (1997),

$$S_L = \min(0, v_L - a_L, v_R - a_R), \quad S_R = \max(0, v_L + a_L, v_R + a_R), \quad (100)$$

$$S_* = \frac{p_R - p_L + \varphi_L v_L - \varphi_R v_R}{\varphi_L - \varphi_R}, \quad \varphi_K = \rho_K(S_K - v_K), \quad (101)$$

and the star states

$$u_K^* = \frac{\rho_K(S_K - v_K)}{S_K - S_*} \left( 1, S_*, \frac{\mathcal{E}_K}{\rho_K} + (S_* - v_K) \left[ S_* + \frac{p_K}{\varphi_K} \right] \right)^\top, \quad (102)$$

the flux being  $F_K$ ,  $F_K + S_K(u_K^* - u_K)$  or the mirror of those according to the signs of  $S_L$ ,  $S_*$ ,  $S_R$ . Because  $S_L$  and  $S_R$  are clamped at zero, a face with every physical wave moving outward returns  $F(W_L)$  and never forms an expression in the right state. The face pressure  $p_{\text{out}}$  that the momentum assembly of (7) needs is the pressure of whichever side the branch took. HLLC is the flux every shipped configuration under examples/ and backup/regression/ states; the key is mandatory and has no default. It is not the flux of every run: the seven lowest-XUV cases of the LHS 1140 b catalog state ROE, because on a 500-cell grid whose base is subsonic to  $M \sim 10^{-7}$  the HLLC flux does not reach the stationary root there. That is a resolution statement about the flux and not about the wind: the same unmodified HLLC flux certifies the same case on 1000 cells, and the state it reaches is the 500-cell Roe state to 0.1–3 per cent above  $1.005 R_p$  (measured 2026-09-17).

**Roe.** The linearized solver of Roe (1981), with the averages

$$\tilde{\rho} = \sqrt{\rho_L \rho_R}, \quad \tilde{v} = s v_L + (1-s) v_R, \quad \tilde{H} = s H_L + (1-s) H_R, \quad s = \frac{\sqrt{\rho_L}}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \quad (103)$$

$\tilde{a} = \sqrt{(\tilde{\gamma} - 1)(\tilde{H} - \tilde{v}^2/2)}$ , and the flux

$$F_{j+1/2} = \frac{1}{2} (F_L + F_R) - \frac{1}{2} \sum_{i=1}^3 \tilde{\alpha}_i |\tilde{\lambda}_i| \tilde{K}^{(i)}, \quad (104)$$

with the wave strengths  $\tilde{\alpha}_{1,3} = (\Delta p \mp \tilde{\rho} \tilde{a} \Delta v) / 2 \tilde{a}^2$  and  $\tilde{\alpha}_2 = \Delta \rho - \Delta p / \tilde{a}^2$ . The acoustic strengths  $\tilde{\alpha}_{1,3}$  may be formed with the velocity jump scaled by  $\min(\text{Ma}, 1)$  after Rieper (2011), which is the published low-Mach cure for a flux whose dissipation does not vanish with the Mach number; the option is Low Mach velocity jump:, it is off by default, and it is not recommended on a near-hydrostatic column. The factor is exactly zero where two neighboring velocities cancel in the Roe average, so it removes the damping of the odd-even ( $2\Delta r$ ) velocity mode at precisely the faces that mode lives on; measured on the LHS 1140 b low-XUV base it turned a solve unmodified Roe completes into a nearly pure odd-even mode. Rieper's premise, that the local and the global Mach number are comparable, does not hold on a column running from  $M \sim 10^{-8}$  at the base to  $10^{-2}$  at the top (item L25, 2026-09-17). The single index  $\tilde{\gamma}$  is the arithmetic mean of the two face gammas, exact wherever the two sides agree; a run whose caloric equation of state makes the index vary across the front, that is molecular chemistry with the ladder heat capacity, is refused this flux at startup rather than being given a wave speed with no bound property. A Harten-Hyman entropy correction is applied to the acoustic eigenvalues from the star state that speed\_estimate\_ROE.f90 builds (the primitive-variable, two-rarefaction and two-shock estimates of Toro (2009) chapter 9, selected by the pressure ratio). Where that estimate has no admissible star state, two rarefactions separating fast enough leave vacuum, or no branch returns a positive star

pressure and positive star densities, the face is given the HLLC flux instead,

$$F_{j+1/2} = \frac{S_R F_L - S_L F_R + S_L S_R (u_R - u_L)}{S_R - S_L}, \quad \begin{aligned} S_L &= \min(v_L - a_L, \tilde{v} - \tilde{a}), \\ S_R &= \max(v_R + a_R, \tilde{v} + \tilde{a}), \end{aligned} \quad (105)$$

which is positively conservative under those bounds (Einfeldt et al., 1991), where the Roe flux is not. The number of such faces is counted and reported.

#### Local Lax-Friedrichs.

$$F_{j+1/2} = \frac{1}{2} [F_L + F_R - \alpha (u_R - u_L)], \quad \alpha = \max(|v_L| + a_L, |v_R| + a_R), \quad (106)$$

the coefficient being the spectral radius of the flux Jacobian (Rusanov, 1961). It is the most diffusive of the three and resolves no contact wave; it is also the flux the positivity repair below substitutes at a single face, because with cell-average input states it maps positive density and positive internal energy to positive density and positive internal energy under  $\Delta t(|v| + c)/\Delta r \leq 1$  (Perthame & Shu, 1996; Zhang & Shu, 2010b).

### 7.4 Positivity repair of a stage

A Runge-Kutta stage can leave the admissible set  $\rho > 0$ ,  $\rho e > 0$  even with admissible face states, because the source terms are bounded by no flux choice. `positivity_limited_fluxes` (RK\_rhs.f90) repairs such a stage by replacing, at the offending cells only, the high-order interface fluxes with the first-order Lax-Friedrichs flux of the neighboring cell averages and rebuilding those cells' update from the mixture; the replacement is swept until no cell is inadmissible or until both faces of a cell are already first order, in which case no flux choice repairs it and the caller halves the time step. Replacing one face rather than all of them is the flux correction of Hu et al. (2013). The stored face fluxes are written back after the repair, because the transported species rows ride on  $F^\rho$  and a species flux built on a discarded high-order flux would not sum to the mass update the density took. Every count is reported at the end of a run, separately for repaired faces belonging to accepted steps, so that a run in which nothing fired is identifiable as the run an unguarded build gives.

### 7.5 Time integration and the time step

The marching loop of EXHALE\_main.f90 advances the hydrodynamics with the three-stage strong-stability-preserving Runge-Kutta method (Gottlieb & Shu, 1998),

$$\begin{aligned} u^{(1)} &= u^n - \Delta t (dF - S)|_{u^n}, \\ u^{(2)} &= \frac{1}{4} \left[ 3u^n + u^{(1)} - \Delta t (dF - S)|_{u^{(1)}} \right], \\ u^{n+1} &= \frac{1}{3} \left[ u^n + 2u^{(2)} - 2\Delta t (dF - S)|_{u^{(2)}} \right], \end{aligned} \quad (107)$$

each stage followed by the boundary condition, by the admissibility test of Sec. 7.4, and by the species advection sub-step that rides on the face mass fluxes that stage built. A stage that cannot be repaired causes the whole step to be discarded and retaken from  $u^n$  at half  $\Delta t$ , up to a bounded number of bisections.

The time step is the explicit-stable interval of the scheme actually in use, derived at `eval_dt.f90` for the spherical finite-volume divergence the right-hand side assembles,

$$dF_j = \frac{A_{j+1/2}F_{j+1/2} - A_{j-1/2}F_{j-1/2}}{V_j}, \quad A = r^2, \quad V_j = \frac{r_{j+1/2}^3 - r_{j-1/2}^3}{3}. \quad (108)$$

Every stage of (107) is a convex combination of forward-Euler steps of the same length (the SSP coefficient is 1), so the restriction is the forward-Euler one of (108). Each face flux is a Riemann flux of the two states bounding that face, and the fastest signal of that face's Riemann problem is bounded by  $S_{j+1/2} = \max(|v| + c_s)$  over the two cell averages bounding it. Godunov's condition is that the fan opened at a face has not left the cell it enters when the step ends; written with the geometry (108) uses, the face carries signal into cell  $j$  at the rate  $A_f S_f$  per unit volume  $V_j$ , so

$$\Delta t = C_{\text{CFL}} \min_{1 \leq j \leq N} \frac{V_j}{\max(A_{j-1/2} S_{j-1/2}, A_{j+1/2} S_{j+1/2})}, \quad C_{\text{CFL}} = 0.6 \text{ by default}, \quad (109)$$

with  $c_s = \sqrt{\gamma_{\text{eff}} p / \rho}$  from the cell's own composition. In the Cartesian uniform limit  $A_{j-1/2} = A_{j+1/2}$  and  $V_j = A \Delta r$ , and (109) returns  $C_{\text{CFL}} \Delta r / (|v| + c_s)$ , so the number the CFL: key carries keeps its usual meaning and nothing is recalibrated. The stricter condition that the two fans of one cell may not meet inside it replaces the maximum by the sum of the two face terms, which is the same restriction at  $C_{\text{CFL}}/2$ .

Two properties of (109) are worth stating, because the cell-centered form  $\Delta r_j / (|v_j| + c_{s,j})$  has neither. In spherical geometry  $V_j / A_{j+1/2} < \Delta r_j < V_j / A_{j-1/2}$ , so the cell-centered width is not an upper bound of the interval on either face; on the production base grid the two differ by  $\Delta r / r = 1.96 \times 10^{-4}$  (MEASURED, item P6b). And the minimum is taken over the evolved cells  $1 \leq j \leq N$  alone. A ghost row is overwritten by the boundary condition after every stage, so its own width is not a stability requirement, and the lowest ghost does not even carry a width of its own; the ghosts nevertheless bound the two boundary faces  $r_{\text{edg}}(0)$  and  $r_{\text{edg}}(N)$ , which are faces of cells 1 and  $N$ , so the base Riemann problem restricts cell 1 with the reservoir's own signal speed although no ghost cell is evolved. Letting the ghosts carry widths of their own is not conservative in the other direction either: an outer ghost given a sound speed a thousand times the column's shortened the step of a 500-cell domain by 89.7 per cent (MEASURED, item P6b).

$S_f$  is formed from the cell averages bounding the face and not from the reconstructed face states, which inside a steep cell can carry a larger  $|v| + c_s$ ; that margin is the one a Godunov number below unity leaves. With Time stepping: Local each cell instead takes its own  $\Delta t_j$ , the same expression without the minimum; since the fixed point  $dF = S$ ,  $\mathcal{H} = \Lambda$  is independent of the step, this is a legitimate acceleration toward a steady state, and it makes the trajectory meaningless as a time history. In the global mode the local array is uniformly the global step, which keeps the update arithmetic identical to the scalar form. A ghost row, which is not evolved, is given the interval of the evolved cell it borders so that a stage formed over the padded range is arithmetic on a physical interval.

(109) is not only a marching quantity: it supplies the stationary route's initial pseudo-time and the floor of the JFNK pseudo-transient continuation (Sec. 9.5), so its value moves solver behavior as well as a time step. The no-step evaluation route does not call it.

## 7.6 Operator splitting of one step

One accepted marching step applies the following operators, in this order, with the state after each one carried into the next (EXHALE\_main.f90):

1. the three Runge-Kutta stages (107), each with its reconstruction, flux assembly, boundary fill, positivity test and species advection sub-step;
2. the conversion to primitive variables, the composition-dependent particle and electron counts, and the temperature (13);
3. the advective half of the element transport written into the species vector, followed by the implicit diffusive half (binary\_element\_diffusion.f90, Sec. 6.9), active only with He\_diffusion;
4. the vertical transport of the molecular carriers with their chemistry (diffusive\_photochemistry.f90, Sec. 6.5), active only with the carrier transport option;
5. the refresh of the lagged  $H(n=2)$  populations and their Balmer source (excited\_hydrogen.f90, Sec. 4.6);
6. the coupled source step: a bounded fixed-point iteration alternating the composition equilibrium at the current temperature (ioniz\_eq, Sec. 4.4) with the semi-implicit energy update (77) at the returned composition, exiting when the temperature and the composition both stop moving;
7. the boundary condition;
8. the Crank-Nicolson viscous and conductive stage (20), where those options are on, and the boundary condition again;
9. the periodic Shapiro filter, where it is on, and the boundary condition again.

Steps 3, 4, 8 and 9 are no-ops unless their key is present. The Shapiro filter is the 1-2-1 low-pass pass  $u_j \leftarrow u_j + \frac{1}{4}\epsilon(u_{j-1} - 2u_j + u_{j+1})$  applied every shapiro\_every steps, the way CETIMB damps the base sound wave (Koskinen et al., 2013). It is off by default and is recorded as a validity state of every step it runs on, because unlike the low-Mach term of Sec. 2.7 it alters the marched state with no source term behind it and therefore has no counterpart in the stationary residual.

## 7.7 The marching convergence measure and the stop conditions

The measure the marching stops on is the radial spread of the mass flux over the escape window,

$$du = \frac{\max_{j \geq j_{\min}} F_j - \min_{j \geq j_{\min}} F_j}{|\bar{F}|}, \quad F_j = \rho_j v_j r_j^2, \quad (110)$$

with  $\bar{F}$  the mean over the same window. A steady spherical wind carries one mass flux at every radius, which is the stop ATES uses ( $\Delta \dot{M} / \dot{M} < 10^{-3}$ ) and the constancy of  $F_c = \rho v r^2$  CETIMB asks for. The sign is kept and the normalization is the mean: taking  $|F|$  first would read a window whose flux reverses, an inflow at one radius and an outflow at another, as flat. It is evaluated once per step and read by every consumer, so the stop, the trigger arming, the staged secondary-ionization activation of Sec. 3.5 and the handoff to the steady solver all speak about one number. This is emphatically *not* a velocity change, and a single-stage PLM run stalls at a few percent spread.

A second measure, the largest relative change of the conserved state over a step,

$$dtu = \max_{k \in \{1,2,3\}} \max_{j_{\min} \leq j \leq N} \left| 1 - \frac{u_{k,j}^{n+1}}{u_{k,j}^n} \right|, \quad (111)$$

says that the state has stopped changing; a row that was exactly zero at the start of the step has no relative change defined and the measure is taken as unbounded rather than dropped.

A run stops when one of the following holds, and reports which:  $du < du\_th$  (default  $10^{-3}$ ) with the trigger armed and the flux level stable;  $dtu < dtu\_th = 10^{-8}$ ;  $du$  settled on a plateau over  $N\_stall$  steps (default 2000) within  $stall\_tol$  (default  $10^{-6}$ ); the steady gates of Sec. 9.2 where `Resid tol:` was stated; the Newton finish returning a state; or the hard cap `count_max` ( $10^6$ ). The triggers are *armed* the first time the measure is seen at or above its threshold, so only a descending crossing can fire them and a run whose initial condition already carries a flat mass flux, which a transonic one does, cannot stop at step zero. With `Solver: Newton` the marching is a warm-up only: the  $du$  and  $dtu$  stops are disabled and the loop hands over once  $du$  falls below `newton_du_switch` (default  $10^{-2}$ ) or plateaus near it. Both measures can be small at a premature balance of the operator split while the energy residual is of order 30, so for a quantitative mass-loss rate the stationary finish is not optional.

## 8. Boundary and initial conditions

### 8.1 The lower boundary

The lower boundary is a characteristic condition imposed *at the base face*  $r_{-1/2}$ , with the number of conditions set by the sign and magnitude of the Mach number there (`base_boundary.f90`). At that face the eigenvalues are  $v - c$ ,  $v$  and  $v + c$ , and a wave enters the domain when its eigenvalue is positive, the radial coordinate increasing outward:

|              |                    |                                     |       |
|--------------|--------------------|-------------------------------------|-------|
| $0 < v < c$  | subsonic inflow    | 2 reservoir conditions, 1 interior  |       |
| $-c < v < 0$ | flow reversal      | 1 reservoir condition, 2 interior   |       |
| $v > c$      | supersonic inflow  | 3 reservoir conditions, 0 interior  |       |
| $v < -c$     | supersonic outflow | 0 reservoir conditions, 3 interior. | (112) |

The interior relation is the linearized  $v - c$  compatibility condition in its LODI form (Thompson, 1987; Poinot & Lele, 1992),

$$p_b - \rho_i c_i v_b = p_i - \rho_i c_i v_i, \quad (113)$$

with the subscript  $i$  denoting the interior state continued to the face. The closed-form Riemann invariant  $v - 2c/(\gamma - 1)$  is deliberately not used: it assumes a single constant  $\gamma$ , and the caloric equation of state gives every cell its own  $\gamma_{\text{eff}}(x_{\text{H}_2}, T)$ , so that invariant is wrong wherever the  $\text{H}_2$  fraction or the temperature varies across the boundary stencil, which is the whole molecular base. The sound speed  $c_i$  is the interior one, because the outgoing wave is an interior wave.

**The reservoir is  $(p, s)$ .** The two reservoir conditions are the pressure and the specific entropy of the lower atmosphere at the base *level*, not the pressure and the density. The distinction matters as soon as the base composition changes: with  $(p, s)$  the temperature follows the composition, with  $(p, T)$  the entropy does. The entropy is carried implicitly, as the isentrope through  $(p_{\text{base}}, T_{\text{base}})$  at the base composition, which is what `base.inp` and the lower-atmosphere profile reader state.

**The face is the level.** The level is  $r = 1$ , and every grid type places its first face exactly there,  $r_{-1/2} = r_{\text{edg}}(0) = 1$  (Sec. 7.1; `base_boundary_states` stops the run if it is not). The reservoir's  $(p, s)$  is therefore the state of the face side it owns, with no transport along any isentrope, and the contact between the reservoir and the domain neither moves with the cell width nor encloses a slab of reservoir gas; the two lower ghost cells lie below the level. Until 2026-09-27 the level was the center of a ghost cell, the face lay half a cell above it on the Mixed grid, and the reservoir was carried up to the face along its own hydrostatic isentrope. That put a slab of reservoir gas of thickness  $\Delta r/2$  under the domain and made the base pressure of the domain  $p_0[1 - \Delta r/(2H_{\text{res}})]$ , a first-order error in  $\Delta r$ : MEASURED, the molecular lower layer of the LHS 1140 b He/H = 1.8 state sat 2.2 and 1.15 per cent low in density on 500 and 1000 cells, and the pressure gap between the interior at the level and the reservoir, 2.93/1.44/0.70 per cent on 500/1000/2000 cells, is 0.114/0.067/0.038 per cent with the face at the level (boundary model v4, `face_at_level_v4`). The wind quantities stay first order in  $\Delta r$  with errors 1.6–1.9 times smaller; the remaining first-order error is likely the thermal layer at the contact, which the grid does not resolve (cell-1 temperature 800/695/604 K against the 226 K reservoir), and two first-order terms were found and left: the element-diffusion Dirichlet He/H at the cell-1 center (Sec. 6.9) and the conduction bath at the ghost center  $r(0)$  (`viscous_conduction.f90`).

**The interior state at the face.** The interior state  $(\rho_i, v_i, p_i)$  that (113) reads is built from the cell average of cell 1 alone (`interior_state_at_level_face`), never from the reconstruction stencil, because that stencil reaches into the ghost and a boundary condition that reads its own output is circular. How cell 1 is carried to the face depends on the reconstruction the boundary has to agree with.

With the well-balanced reconstruction (Well balanced: True, Sec. 2.6; boundary model `face_at_level_discrete_equilibrium_v5`) it is cell 1's own constant-density equilibrium at the face,

$$\rho_i = \rho_1, \quad p_i = P_{\text{dn}}(1) = p_1 + \rho_1 [\Phi_c(1) - \Phi_i(0)], \quad (114)$$

the pressure the reconstruction itself extrapolates cell 1 to at the same face, formed by the same expression from the same cell averages and potential arrays, with  $T_i = p_i/(\hat{n}_1 \rho_1)$  at cell 1's particle count. The boundary and the reconstruction then measure the pressure of cell 1 at the face from one discrete equilibrium. That equilibrium is mechanical, the discrete form of  $dp/dr = -\rho d\Phi/dr$ , and says nothing about the temperature or the entropy (Käppeli & Mishra, 2016, Sect. 2.1); it holds for a column of any entropy stratification, and the reservoir is not changed by it: it still states the pressure and the entropy of the gas it supplies. The isentropic continuation below, which this replaces under the well-balanced option, differs from  $P_{\text{dn}}(1)$  by about  $(1/2\gamma)(\Delta r/2H)^2$  of  $p$ , a truncation difference between two second-order constructions with no physics in it. MEASURED (`src/tests/grid_and_gates`, `base_reservoir_equilibrium` and `low_mach_face_flux`): on a static column built in the reconstruction's discrete equilibrium and meeting the reservoir's pressure at the level, that difference left a face-0 flux of  $1.46 \times 10^{-4}$  of  $(\rho c)_{\text{res}}$ ,  $v_b = -2.6 \times 10^{-4} c_{\text{res}}$  and a cell-1 momentum row of  $3.1 \times 10^{-3}$  of the weight,  $1.47 \times 10^4$  times the wind flux against a manufactured wind at Mach  $10^{-8}$ ; with (114) and the ghosts below, all three are at the rounding level (below  $10^{-14}$ ) on the isothermal column, on a column warming upward and on a column with a pressure-balanced contact at the level, and the face flux of the manufactured wind is within  $2.1 \times 10^{-3}$  (isothermal) and  $2.5 \times 10^{-3}$  (warming) of the wind at every Mach number from  $10^{-3}$  to  $10^{-9}$ , falling at first order with the base cell width.

Without it the reconstruction carries no discrete equilibrium to share, and cell 1 is continued along its own hydrostatic isentrope by `continue_hydrostatic_isentrope` (boundary model `face_at_level_v4`), which uses the fact that at rest and at constant entropy the Bernoulli function is uniform,

$$u(T) + T - \frac{\Phi(r)}{\hat{n}} = \text{const} \implies u(T_b) + T_b = u(T_a) + T_a - \frac{\Phi(r_b) - \Phi(r_a)}{\hat{n}}, \quad (115)$$

with  $u$  the internal energy per particle of the caloric equation of state and  $\hat{n}$  the particles per unit mass, solved by Newton with  $dh/dT = c_v + 1$ ; the density then follows from  $d \ln \rho = c_v d \ln T$  as  $\rho_b = \rho_a (T_b/T_a)^{c_v(T_{\text{mid}})}$ . The midpoint heat capacity makes this exact to third order in  $\ln(T_b/T_a)$ , and the routine is only ever used over half a cell and over the two ghost cells, where that logarithm is below 0.1 and the density error is below  $10^{-7}$  relative. It is not a general isentrope integrator over a scale height and is not used as one. The same routine gives, under both models, the temperature of the reservoir continued below the level, which the ghost cells carry and the conduction operator reads as the bath's temperature. The isentropic continuation balances an atmosphere that is isentropic between the level and the center of cell 1 and no other rest state exactly: hydrostatic balance does not imply constant entropy, and the continuation starts from a point state while cell 1 holds an average.

In both models the interior velocity is carried to the face at constant  $\rho v r^2$ ,  $v_i = \rho_1 v_1 r_1^2 / (\rho_i r_b^2)$ , rather than copied.

**What is held, and the order of the three steps.** The acoustic matching comes first: the pressure is a reservoir condition in every subsonic branch,  $p_b = p_{\text{res}}$ , and the face velocity then follows from (113) as  $v_b = v_i + (p_b - p_i) / (\rho_i c_i)$ , so  $v_b$  is a *result* of the matching and does not depend on  $\rho_b$ . The direction of the contact is read off the matched state next, and only then is the contact upwinded: the entropy and composition of the face are the reservoir's where gas enters the domain and the interior's where it leaves,

$$\rho_b = (1 - w_{\text{rev}}) \rho_{\text{res}} + w_{\text{rev}} \rho(s_i, p_b), \quad w_{\text{rev}} \in \{0, 1\}. \quad (116)$$

**The upwind choice is discrete and has no width.** A contact carrying different entropies on its two sides has direction-dependent traces; a face density between them is an interpolation and not a law of thermal contact, and the average of the two isentropes that a symmetric handover returns at zero has no physics behind it. This is what distinguishes boundary model v2 from v1, whose entropy source was handed over by a cubic smoothstep in the face Mach number.

**The interior's mass flux is carried at the face density (well-balanced model).** Through a stationary face the mass flux is continuous: a contact cannot stand at a face gas flows through, so in a steady state the gas just inside the level has the density  $\rho_b$  of the face state the contact selects. Under the well-balanced model  $\rho_i = \rho_1$  is the density of a mechanical model of cell 1 and not of the gas at the face, so once  $\rho_b$  is chosen the advective part of the face velocity is the interior's mass flux at that density,

$$v_b = \frac{\rho_1 v_1 r_1^2}{\rho_b r_b^2} + \frac{p_b - p_i}{\rho_i c_i}, \quad (117)$$

and the face state carries the interior's own mass flux wherever the pressures balance. MEASURED on the manufactured winds above, dividing by  $\rho_1$  instead made the base face carry 3.9 (isothermal) and 7.8 (warming) per cent more than the wind at every Mach number, and the ghosts, which continue  $\rho_b v_b r^2$ , the same excess. The direction of the contact is read from the velocity at  $\rho_i$ ;  $\rho_1 v_1$  has the same sign at any density, so the two differ only where the two terms of  $v_b$  cancel to  $O(\Delta r/H)$ , a face that carries no resolved flow either way.

**At rest and at inflow the reservoir owns the level.** At a pressure-balanced contact at rest the two sides may carry two entropies and characteristic theory alone does not choose between them; this model chooses the reservoir, and the physical assumption is stated rather than carried implicitly: the lower atmosphere below the base level is dense and radiatively controlled, so on the time scales of a stationary solution it is a heat bath and a column standing at rest above it takes the level's own temperature. Its validity is a subsonic base at a level lying inside the radiatively controlled lower atmosphere, and it says nothing about a supersonic base or about a level above that region. Only a *reverse* flow hands the trace to the interior, and then the reservoir keeps exactly the one condition its single entering characteristic allows, the pressure.

**Which quantity the direction of the contact is read from.** The contact moves with the gas, so the side that owns the face is the side the level's mass flux comes from. The first interior cell's own product  $\rho_1 v_1 r_1^2$  is not that flux at a base carrying a collocated odd-even velocity mode: MEASURED on the certified LHS 1140 b atomic state, the Riemann face flux is  $+1.00000 F_{\text{wind}}$  at every face of the grid while the cell-centered product reads  $-2.00 F_{\text{wind}}$  at cell 1, and the hot-Uranus base reads  $-196 F_{\text{wind}}$  against a face flux of  $+0.98$ . The matched face velocity  $v_b$  is the face's own velocity and is what characteristic theory names, but it carries  $(p_{\text{res}} - p_i)/(\rho_i c_i)$ , and a converged state leaves that pressure difference at about  $10^{-4}$  of  $p$ , which at a base Mach number of  $3 \times 10^{-7}$  is a velocity two orders above the flow's own. The direction is therefore taken from the mass flux the state carries, read where the cell-centered product *is* a flux: the wind window  $r \geq r_{\text{flux}}$  the convergence gate is already defined on. The face flux itself is not available here, being produced by the Riemann solve this boundary feeds.

The window has standing only when it carries a flux at all and carries *one* flux, both properties of its own profile:

$$M_{\text{wind}} = \frac{F_{\text{wind}}}{\rho_i r_b^2 c_i}, \quad d_{\text{window}} = \frac{\sqrt{\langle F^2 \rangle - \langle F \rangle^2}}{|\langle F \rangle|}, \quad F = \rho v r^2, \quad (118)$$

the moments taken over the physical cells of the window. Where  $|M_{\text{wind}}| \geq 10^{-12}$  and  $d_{\text{window}} \leq 2 \times 10^{-3}$  the window states the direction; otherwise the matched face velocity does, and then a velocity below the rounding of the cancellation it is formed from,  $4\epsilon[(|p_i| + |p_b|)/(\rho_i c_i) + |v_i|]$ , counts as zero and the reservoir owns the level. A relative standard deviation is used rather than a range because the range is not a differentiable function of the state: its extrema are carried by particular cells, and MEASURED with one window cell swept through the argmax the two one-sided derivatives of the face density are  $-242.2$  and  $+0.0128$ , a slope jump of  $1.9 \times 10^4$  at a point where the value itself is continuous. The spread threshold is the geometric middle of the measured gap between the two populations, to one digit: 302 certified stationary states run from  $4.06 \times 10^{-5}$  to  $1.82 \times 10^{-4}$  and a stalled transient reads  $1.90 \times 10^{-4}$ , while the tightest state that is not a wind reads  $1.39 \times 10^{-2}$ . The amplitude floor acts on no state the code meets: the smallest window Mach number measured anywhere is  $3.7 \times 10^{-9}$ , at a cold start.

**The direction is held through each steady solve.** The separation above is one of *written* states. The iterates of a steady solve are not written states, and they do cross the threshold. MEASURED on three stalled LHS 1140 b molecular continuations (the well-mixed He/H = 0.3169 and 0.3347 rungs and the  $K_{\text{zz}} = 10^9$  He/H = 1.09 rung, stage exports of every pass): every solve that converged returned  $d_{\text{window}} = 8.4 \times 10^{-5}$  to  $1.4 \times 10^{-4}$  with the reservoir owning the level, and every solve that failed returned  $d_{\text{window}} = 2.000 \times 10^{-3}$  to  $4.1 \times 10^{-3}$  with the interior owning it, several at

$2.000 \times 10^{-3}$  to four digits. A solve entered at a written state is pushed across the threshold by its first steps, and to reach the root it has to cross it again through a residual that steps there; the outer passes then alternated between a converged and a failed solve for 60 to 80 passes. The upwind side of the contact is a property of a solution, so while a steady solve runs the side is the one the rule selects at the solve's first evaluation, its entry state, and it does not change inside the solve. The rule is applied again to the state the solve returns; where a converged solve returns a state whose rule selects the other side, one more solve from that state holds that side, so a root is accepted only on the side its own rule selects, the condition every written state was judged by before. Only the path to the root changes. The marching route and every evaluation outside a steady solve apply the rule at every evaluation, and EXHALE\_BASE\_CONTACT\_PER\_EVALUATION=1 does so inside the solves as well, for a control experiment. On the well-mixed He/H = 0.3169 rung, which had alternated for 80 passes, holding the side certified the state in six passes.

Holding the incoming acoustic invariant  $p_b + \rho c v_b = p_{\text{res}}$  instead of the static pressure is present in the source as a weight that is zero. Both forms are exactly well balanced at rest; on a moving solution the invariant makes the static face pressure fall below the reservoir's by a fraction  $M$  of itself where a real reservoir's Bernoulli drop is a fraction  $M^2$ , which is why the weight is zero.

**The supersonic branches.** A supersonic *outflow* takes every characteristic out of the domain, so nothing may be prescribed and the whole face state passes to the interior. That transition is a change in the number of outgoing characteristics and its width is a declared numerical regularization, a  $C^1$  cubic smoothstep in  $M_i$  of half width  $\text{base\_face\_mach\_blend} = 10^{-8}$  about  $M_i = -1$ ; it is the module's one designed-in width and it does not touch the entropy source. A supersonic *inflow* would need a third reservoir condition which a  $(p, s)$  reservoir does not have, so the state is not a solution of a well-posed problem there; the run reports the crossing and does not invent a velocity. A compatibility relation is linear and far from a solution can ask for any velocity, so the face state is capped at Mach 5 and the events are counted.

**The ghost cells.** `base_ghost_averages` fills the two lower ghosts from the face state downward, with the velocity carrying a constant  $\rho v r^2$  so the reconstruction sees no spurious velocity gradient. Without the well-balanced reconstruction they are the *volume averages* of the hydrostatic isentrope through the face state, continued below it, evaluated by 8-point Gauss-Legendre quadrature weighted by  $r^2$ . With it they continue the reconstruction's discrete equilibrium below the level,

$$P_{\text{up}}(0) = p_b, \quad P_{\text{up}}(-1) = P_{\text{dn}}(0), \quad p_j = \frac{P_{\text{up}}(j)}{1 - [\Phi_i(j) - \Phi_c(j)]/\chi_j}, \quad \rho_j = \frac{p_j}{\chi_j}, \quad (119)$$

each ghost keeping the ratio  $\chi_j = p_j/\rho_j$  of the isentropic average, that is the reservoir-side temperature at the particle count the reservoir states, so that only its pressure and density move, together, by a factor  $1 + O[(\Delta r/H)^2]$ . The equilibrium mismatch of face 0 is then  $P_{\text{dn}}(1) - p_b$ , the same pressure difference (113) closes, and it vanishes with the WENO3 departures of faces 0 and 1 (whose stencils read the two ghosts) and the jump of the face below the ghosts, whose state is  $P_{\text{dn}}(0)$  on both sides, on any column at rest in discrete equilibrium with the reservoir. A ghost wider than two scale heights of the gas it holds,  $[\Phi_i(j) - \Phi_c(j)]/\chi_j \geq 1$ , has no such equilibrium and keeps the isentropic average; the events are counted. Writing the face value into a ghost cell average instead of either construction is a half-cell error: the two differ by  $\Delta r/2H$ , which is first order in  $\Delta r$  in the state and *zeroth* order in the momentum residual it produces, an

imbalance that no resolution removes. Rec\_BC then puts the face state itself into the left slot of the base face; the right slot is the interior's reconstruction of cell 1 down to that face and is deliberately left alone, the Riemann solver being what mediates the two.

A column sampled from a continuum hydrostatic profile (an analytic isothermal or isentropic column as cell averages) is not a rest state of the discrete scheme: its cell averages lie  $O[(\Delta r/H)^2]$  of  $p$  from the discrete equilibrium through the same reservoir pressure, and under the well-balanced model that distance appears at the base face as  $P_{\text{dn}}(1) - p_{\text{res}}$  (MEASURED on the analytic isothermal column, face-0 flux  $3.1 \times 10^{-4}$  of  $(\rho c)_{\text{res}}$  on the coarsest base grid, falling at order 2.0). A seed has to be projected onto the discrete equilibrium for its lower layer to be at rest.

The base Mach number is recorded every step and a first crossing into supersonic inflow is warned about, as a validity check on the boundary that is independent of the chemistry (Apply\_BC.f90).

**The ghost composition is a fixed point of the sweep that produced it (boundary model v3).** The boundary is built on the two lower ghost cells, whose composition is not prescribed but solved: the composition sweep is a map from the composition it is entered at to the composition it returns, the entry state setting the columns, the cell depths, the electron and particle counts, the secondary-ionization partition, the shielding and the ionization the base handoff's partition is stated against. A ghost that is merely one application of that map is therefore a boundary that depends on the composition the run happened to be entered at, and MEASURED it does: on one frozen physical column, reservoir, spectrum, options and executable, two entry compositions returned ghosts  $4.7 \times 10^{-3}$  to  $5.5 \times 10^{-2}$  apart in their trace ions, both accepted by every test the sweep applies at reaction residuals of about  $3 \times 10^{-26}$ , and the base continuity row of the first physical cell read  $8.335 \times 10^{-8}$  in one and  $2.923 \times 10^{-9}$  in the other, against the same tolerance. Neither state is a fixed point, and the difference does not close with the inner solve: over six decades of the cell solver's stopping accuracy the two states stay put to  $10^{-15}$  while remaining that far apart.

Two statements close it, and together they are what distinguishes boundary model v3 from v2. *The seed is part of the model:* the ghost solve starts from the composition of the gas the reservoir holds at the base level, whose partition within each element is the first physical cell's, at the first sweep of a run, and from the ghost the previous sweep returned at every sweep after it. The first is the only seed that is a function of the physical column and the declared inputs alone, which is why the loader installs it in place of the ghost rows a restart pair carries. *The returned ghost is a fixed point of the map that produced it:* `ioniz_eq` applies its own map again at the ghost it returned, restoring the cells above the ghosts to the composition they were handed at each application, until the largest absolute change of any ghost species density, in the cell's own units, is at most  $10^{-11}$  and the ghost's heavy-particle plus electron count moves by at most  $10^{-6}$  of itself. A boundary that cannot reach that in twenty applications stops the run by name, with both moves and the accuracy, exactly as the ghost's imposed  $\text{H}_2$  partition closure does. What the routine returns for the physical cells is still one application of one map, so the stationary residual is still one application of one operator.

The accuracy is set by the base continuity row, which is a near-cancellation of two face fluxes and therefore reads the ghost composition with a large gain: MEASURED, a ghost difference of  $4.7 \times 10^{-3}$  takes that row from 3.90 to 111.3 of its own rounding floors, so one floor of the row is bought by  $4.4 \times 10^{-5}$  of ghost composition and  $10^{-11}$  is a quarter of one floor. **The norm is absolute and not per species.** The sweep pins the composition absolutely, in the cell's own units, and not stage by stage, and no quantity the boundary builds reads a trace stage against its own

value: MEASURED on a molecular carrier configuration, a ghost settled into an exact two-cycle whose H I moved by  $6.5 \times 10^{-12}$  and whose  $H_3^+$  moved by  $2.2 \times 10^{-12}$  of the cell's composition, worth 0.06 of one rounding floor of the base row, while against their own values those same moves read  $4.1 \times 10^{-9}$  and  $5.1 \times 10^{-3}$ . A relative statement cannot be made for a stage sitting at  $4 \times 10^{-10}$  of the cell. Species below  $10^{-18}$  of the cell's own density are left out of the comparison. The bracket is stated rather than tuned:  $10^{-11}$  sits a factor 1.5 above the lowest plateau measured, so a configuration whose composition solve plateaus higher stops the run, and the answer there is to measure that plateau. The temperature leg closes with the composition. The ghost's heavy-particle plus electron count is what turns its temperature into the pressure base\_ghost\_averages builds and into the internal energy and sound speed the caloric map returns, so a ghost whose composition has stopped moving but whose count has not is a ghost whose rates belong to one gas and whose thermodynamic state to another; the count is measured across every application and the rates are rebuilt from the composition each application is entered at. Because the applications are *one* sweep, nothing that separates one sweep from the next may turn over between them; the frozen background's readiness is held at the value the sweep was entered with and takes the sweep's own value only when the sweep ends.

The contract covers the lower ghosts of a run whose network is molecular and carries helium, which is where the boundary's composition holds the trace ions the base row reads. A run outside it is marked reached before the first traversal and leaves after it, so not one number of such a run moves. The atomic ghost is not under the contract and its seed dependence is not measured. On the state that motivated the contract, one refused molecular generation, the evaluation now gives one answer whatever the ghost solve is started from: three seeds  $9.7 \times 10^{-1}$  apart certify with the same cell-1 row  $2.923 \times 10^{-9}$  and stand at most  $8.0 \times 10^{-11}$  apart over every column of both state files (MEASURED, item P1 step 2).

The model identity, the seed and what the fixed point reached are printed in the boundary report and the identity travels in every state file (Sec. 8.4).

## 8.2 The upper boundary

No characteristic enters the domain at an outflowing outer face, so the boundary imposes no condition and the ghost is a continuation of the interior. The rule is stated once, for every reconstruction: which characteristics cross a face is a property of the flow and not of the scheme that reads the ghost. The continuation (free\_outflow\_ghost) is the isothermal hydrostatic one,

$$\rho_g = \rho_N \exp \left[ -\frac{\Phi_c(N+k) - \Phi_c(N)}{p_N / \rho_N} \right], \quad p_g = p_N \exp[\dots], \quad v_g = v_N, \quad (120)$$

the state of rest the last physical cell carries outward under the code's own potential with the isothermal sound speed squared  $p/\rho$  held at cell  $N$ 's value. Since  $\rho$  and  $p$  are scaled by the same factor, the ghost is at the *same* temperature as cell  $N$ , so the ionization sweep, the column integral and the sound speed of the time step all see the temperature they saw before, at the density the stratification gives.

A zero-gradient copy would assert  $dp/dr = 0$  at the outer face, a hydrostatic imbalance of the order of the local weight that no resolution removes, and under PLM it is worse: the forward difference entering the limiter is then exactly zero, so the limited slope of cell  $N$  vanishes and that cell carries no pressure gradient at all. On an analytic hydrostatic column the momentum residual of cell  $N$  in units of its own weight converges at order 1.80 for (120), 1.42 for a linear

extrapolation and not at all (0.006) for the zero-gradient copy. The velocity stays a copy: the outgoing characteristics carry it out of the domain and the boundary has nothing of its own to say about it; a gradient-carrying ghost velocity either drives the run into a base-breathing limit cycle or divides by a ghost density the stratification can make arbitrarily small. A ghost that leaves the admissible set falls back on the zero-gradient copy and is counted.

For the reconstructed face states, Rec\_BC extrapolates outward with a zero-gradient copy under PLM and a linear extrapolation under WENO3.

### 8.3 Initial conditions

set\_IC.f90 builds four families, selected by IC mode: (or by the older Transonic IC and Hot Parker IC keys, which take precedence).

**Cold hydrostatic (cold, the default).** An isothermal hydrostatic atmosphere at the base temperature,

$$\rho(r) = \rho_{bc} \exp[-b_{\text{eff}}(\Phi_c(r) - \Phi_c(0))], \quad p = \frac{\hat{n}_{bc}}{\rho_{bc}} \rho, \quad T = 1, \quad (121)$$

with a small linear velocity seed  $v = \frac{1}{2}(r - r_0)$  and a nearly neutral composition. The factor  $b_{\text{eff}}$  starts at 1 and is raised in steps of 0.2 until the mid-domain density falls below  $10^{-8}$ , so that a low-gravity planet does not start with a domain at uniform density; the value used is reported. This is the ATES initial condition.

**Transonic (transonic).** The steady isothermal-wind profile obtained from the algebraic Bernoulli integral of the isothermal wind equation in the code's own potential,

$$\frac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \Phi(r) = B, \quad (122)$$

with  $B$  fixed at the sonic point, where  $v = c$  and  $d\Phi/dr = 2c^2/r$ . The subsonic root is taken inside the sonic radius and the supersonic root outside it, and the density follows from  $\rho v r^2 = \dot{M}$  with  $\dot{M}$  pinned to the base density. It exists only where the potential admits an interior sonic point; where it does not, the cold hydrostatic profile is kept. It is needed for deep Roche-lobe-filling cases whose sonic point sits near L1 and which launch no wind from a hydrostatic start.

**Hot Parker warm seed (hot\_parker).** Seeding the full isothermal Parker wind, its density as well as its velocity, destabilizes the run: the hot, low-density Parker column is globally inconsistent with the code's non-isothermal equilibrium and drains onto the base. The warm seed therefore keeps the stable cold hydrostatic *density* and overlays only a warm temperature and ionization ramp and the Parker *velocity* as a head start, both anchored to the cold static base through a cubic smoothstep window that is zero at  $r = 1$  and unity above a stated transition radius. This removes the inverted base pressure gradient that otherwise drives an inflow and a thermal runaway.

**Automatic selection (auto).** One exact probe decides between the cold and the transonic family with no tunable threshold: does the *cold* base sound speed admit an interior sonic point in the actual potential? If it does the transonic profile is taken, which catches both the deep Roche-lobe cases, where the Roche  $d\Phi/dr \rightarrow 0$  toward L1 always crosses the critical condition, and the low-gravity cases whose sonic radius falls inside the domain; if it does not, the cold hydrostatic profile is taken. The hot Parker seed is deliberately not auto-selected.

**Wind-AE (windae).** The ported Wind-AE solver in `src/modules/wind_ae/` is run in process to build a warm-start profile on the code's own grid, which is then loaded through the restart path.

Every family fills the whole species vector, and a molecular run seeds the  $H_2$  column from the same two functions the inflowing ghost uses: the handoff value where a lower atmosphere states one (Sec. 6.7), and the chemical-equilibrium mixing ratio (84) at  $(p_{\text{base}}, T_0)$  otherwise, clipped at the element-ratio ceiling  $q_{\text{max}} = 0.5/(0.5+y)$ . The reason is that the optically thick base equilibrium is strongly molecular, and a zero- $H_2$  start makes the base composition solve land on the spurious atomic root.

Two of the families set `force_start`, which suppresses the convergence test for the first iterations: a transonic profile already carries  $\rho v r^2 = \text{const}$ , so the marching measure of (110) would be near zero at step 0 and the run would exit before the cold wind had heated to its steady state.

## 8.4 Restart

`Load IC?` True restores the state from `Hydro_ioniz_IC.txt` and `Ion_species_IC.txt` (`load_IC.f90`). Files carrying the # columns schema header are read by label, so every species the file holds, metal ionization stages included, is restored; a headerless legacy file takes the original fixed seven-column read and the metals are rebuilt from the abundances, which is reported. An element the file does not carry, or carries as identically zero, is initialized from its abundance exactly as a cold start does. An element whose reservoir a lower-atmosphere handoff states has its loaded column renormalized onto that reservoir by one factor common to all its ionization stages. The file's own He/H ratio is compared against the input's, and a state written at a different element ratio is rescaled or refused rather than silently adopted, the input file being the authority on composition.

`Restart intent`: states what the loaded state is continued as, and is refused when no state is being loaded. `trajectory` continues the history, including its elapsed time; `relaxation` treats the state as a numerical iterate; `stationary` asks for a stationary solve from it, and takes a second word, `evaluate` to only measure the residual of the loaded state, or `equilibrate` to bring the composition to equilibrium before the solve. `Restart option change`: names the entries of the state file's option line that are allowed to differ from this run's, and a token that decides how many unknowns the state has is refused: naming it cannot make the file's rows this run's rows.

Every state file carries the lower boundary as well as the state: one line names the boundary model that built it and one states the prescribed reservoir with its version,  $(p, T, \hat{n}, r_{\text{level}})$  at round-trip precision. The reservoir is the only boundary input the physical column cannot reconstruct, which is why it travels; the ghost rows the file holds are the boundary's own output and are rebuilt at the read from the column and this run's reservoir rather than adopted. A state written under an older boundary model is *reported* and loaded, not refused: the loader names the model that wrote it, names the model this run solves, and says that the boundary is rebuilt from the physical column and this run's reservoir and that its ghost composition is solved to this model's own seed and fixed point, so the state's residual is not the residual it was written with. A file written before the identity was recorded says so as well. A state written before 2026-09-27 (boundary model v3 or older) is nevertheless refused, by the grid check and not by the boundary model: every cell center has moved since the face was put on the level (by half a cell on the Mixed grid), and the loader compares the file's centers with the run's to  $10^{-10}$ . Such a state enters a run only as an initialization seed mapped onto the new grid by `src/utls/map_state_to_grid.py --ic`, which writes `mode=init t_phys=0 certified=F`

cert\_reason=mapped\_seed into it, and the run has to solve it again.

A stationary run that alternates a wind solve with a composition relaxation also leaves a restartable pair of its last completed outer pass, the generation output/pass\_state/<id> that output/pass\_state/current names (Sec. 9.6); copied to the \_IC names it is loaded like any state file, and it is never a certified state.

## 9. The stationary state

Marching to a steady state is only one of the two ways EXHALE reaches one. The other is to solve for it directly, as the root of the stationary residual of the very same discrete operator. This section states that residual, the measure by which a state is judged steady, and the two solvers that drive it to zero.

### 9.1 The stationary residual

assemble\_residual (steady\_residual.f90) evaluates the right-hand side of one forward Euler step of the marching operator and subtracts the radiative source from its third row,

$$\begin{aligned} R_1 &= dF_1 - S_1, \\ R_2 &= dF_2 - S_2 - S_\mu, \\ R_3 &= dF_3 - S_3 - (\mathcal{H} - \Lambda) - S_\mathcal{E} + \nabla \cdot q_d, \end{aligned} \tag{123}$$

with  $\nabla \cdot q_d$  the divergence of the interdiffusion enthalpy flux of Sec. 6.10, formed from the element and carrier fluxes of the state itself and identically zero, with no arithmetic, unless He\_diffusion moves the elements or a molecular carrier is transported; the composition is therefore an argument of every assembly.  $dF$  and  $S$  are those of (6) to (11), evaluated through the same reconstruction\_continuation\_rhs the marching stages call, so that the fixed point of the marching and the zero of the residual are the same state by construction. The lowest ghost cell has no lower face and therefore no balance to state; its row is explicitly zero, so that no reader of  $R$  finds a heating rate standing alone in a row that has no fluxes.

A residual row is a dimensional quantity and has to be divided by something before it can be compared with a tolerance. Every row is divided by the largest term *that row itself contains* (residual\_row\_scale):

$$\begin{aligned} s_1(j) &= \frac{\max(|F_{j-1/2}|A_-, |F_{j+1/2}|A_+)}{\Delta V_j}, \quad s_2(j) = \max(|\text{ram}|, |\partial_r p|, \rho |\partial_r \Phi|, |S_\mu|), \\ s_3(j) &= \max(|dF_3|, |S_3|, \mathcal{H}, \Lambda, |S_\mathcal{E}|, |\nabla \cdot q_d|). \end{aligned} \tag{124}$$

Since  $\Delta V_j$  is exactly the volume the flux difference was divided by,  $R_1/s_1$  is the fractional change of the mass flux across the cell, and the mass row having no source, that fraction is the whole of what the row says. The momentum scale uses the three gathered physical terms of Sec. 2.3 and not  $dF_2$  and  $S_2$ , for the reason given there.

A signal-speed bound  $D_k(|v| + c_s)/\Delta r$  is not usable as the scale of these rows: nothing in any of them moves at  $c_s$ , so in a quasi-hydrostatic layer that bound stands far above the terms the row holds and  $R_1$  divided by it is the fractional flux error per cell *times* the local Mach number, which at Mach  $5 \times 10^{-5}$  is a blindness of four decades.

These are the scales of the *measure* only. The Newton system and its merit keep the state scales  $D_k$  of state\_scales\_of\_cell, so the merit and the acceptance test are deliberately not one expression apart.

## 9.2 The acceptance test

`steady_gates_met` is the only definition of an accepted steady state in the code: every route that may stop on “the equations are satisfied”, the two stationary solvers and the marching loop under `Resid_tol:`, calls it. Four gates:

1. the *residual gate*,  $\|R\| < \text{resid\_tol}$ , with  $\|R\|$  the maximum over physical cells and rows of  $|R_k(j)|/s_k(j)$ ;
2. the *flux gate*, the spread of the Riemann face mass flux  $F_{f+1/2} r_{f+1/2}^2$  over  $r \geq r_{\text{flux}}$  below `flux_spread_th` (default  $2 \times 10^{-5}$  at  $r_{\text{flux}} = 1.2 R_p$ ). A residual norm does not say this: a slowly varying flux has a small local derivative, and two states that differ by a factor 25 in the spread differ by only 1.9 to 5.7 in every residual norm tried. It is taken on the *face* flux and not on the cell-centered  $\rho v r^2$ , because the finite-volume update moves the face flux and the centered product is a reconstruction of the state;
3. the *chemistry gate*, no cell carrying a composition that is not a root of the chemical network, that is no cell in acceptance class 4 or 6 of Sec. 4.4. Heat and cool in the energy row are evaluated on the accepted composition, so on such a cell the row is a number computed on a state the chemistry does not describe;
4. the *carrier gate*, where  $n(\text{H}_2)$  is among the unknowns: the carrier row balanced to `carrier_resid_th` of its own largest terms.

A caller that cannot supply the third makes no statement about the chemistry and the test rests on the others; the marching stop is such a caller.

## 9.3 Certification

The four gates above judge the hydrodynamic rows and at most one species row. A configuration that also carries an elemental transport balance, a level population or a set of eliminated species would reach its answer with nothing said about those equations. `certification.f90` closes that: it lists every equation the configuration makes independent and returns, for each, one of three statuses, *not applicable*, *evaluated* with its residual measured on *this* state, or *unavailable*. The third is the point of the construction: a routine that returns zeros when its frozen background is not ready is indistinguishable at the call site from a balanced equation, and an active equation with no evaluator returns nothing at all. Both now say so, and a state carrying either is written and not certified.

The inventory covers the three hydrodynamic rows, one balance for each transported carrier, the ionization stage nucleus sums of H and He, the He/H partition and the trace elemental transport balances, the  $\text{He} 2^3\text{S}$  and  $\text{H}(n=2)$  level balances, and the eliminated-species closure of the coupled cell system. The stage sums are the identity  $\sum_k F_k(f) + F_{\text{close}}(f) = N_{\text{el}}(f)$  at every face, summed over all stages of one element, read from the stationary evaluation that formed the stage fluxes rather than re-formed, so the number the report carries is the one the rows were built with; one entry per element, because the identity is a statement about one element’s own nucleus flux. It gates, at the floating-point bound of the identity: the identity is algebraic and has no truncation error, so the whole of what a tolerance on it can clear is the rounding of the sums,  $2(5n_k + 2)\epsilon g$  with  $g$  the ratio of magnitude sums measured at the face the measure was taken at, and at its algebraic ceiling ( $3/2$  for one carried stage, 2 for two) where the operator reports no  $g$ . Constructions that break the identity on purpose stand ten decades above that bound and the states this

code produces at 0.07 to 0.08 of it. Every row measure is the *maximum* over physical cells of  $|\text{res}|/\max(\text{scale}, \text{floor})$ , never a volume average: a front or a thin layer must not be averaged away by the wind that carries most of the volume. The volume average is reported beside it and is never decisive. The measurement has no side effects; every module array the carrier evaluator writes is held aside around the call, allocation status included, because a certification is a measurement of a state and not an operation on it.

**The capacity of a report is the size of the inventory.** A report holds `cert_max_entries` entries, and that number is not a round constant: it is formed from the counts that set it, three hydrodynamic rows plus one entry for each transported carrier, two stage nucleus sums, the He/H partition, one for each trace element, two level balances and one for each coupled cell system, which is 32 today. It follows the inventory when a carrier, a trace element or a cell system is added. The capacity is exactly consumed: MEASURED over the regression logs of the binary of record, a configuration with the ionization stages transported and metals on prints active equations 11 of 32 in the inventory, so no free slot is left. The entry writers therefore address only the entry their own call created, and an entry that does not fit stops the run before any field is written, naming the equation that did not fit, the capacity and the entry that stands last and unchanged. Both halves matter. What an overfilled report costs is that a new equation's measure, tolerance and verdict are attached to the name of the equation before it, and a certification then states a verdict about a set of equations it does not name; raising the constant would leave that available one equation later.

The tolerances are one per kind of row, so that the report says which number refused a state:  $3 \times 10^{-12}$  for mass,  $10^{-8}$  for momentum,  $10^{-6}$  for energy,  $10^{-5}$  for an elemental or carrier balance in the wind, and the composition solve's own root tolerance for the closure and the level balances. One number for the three hydrodynamic rows is not possible: on one converged state the three reach values seven decades apart and their one-ulp arithmetic floors two decades apart.

Two of these are functions and not constants. A species row is gated only in the wind,  $r \geq 1.20 R_p$ , and reported without gating below it: the balance is a cancellation between terms whose ratio is set by the flow, the advective divergence carrying the row in the wind and the eddy and settling terms carrying it below the homopause, each of the latter orders of magnitude larger than their difference, so the precision the same equation can be resolved to is not one number. And the mass row's tolerance is a function of the cell,

$$\text{tol}_1(j) = \max \left[ 3 \times 10^{-12}, \min(1, c_{\text{round}} f(j)) \right], \quad c_{\text{round}} = 10, \quad (125)$$

anchored on the rounding floor of that cell's own continuity row,

$$f(j) = \epsilon_{\text{mach}} \frac{\max_{\text{faces}} [\rho(|v| + c_s)A]}{\max_{\text{faces}} |AF|}. \quad (126)$$

The reasoning is that each interface flux is assembled from two reconstructed states and from their jumps, and where the column is nearly hydrostatic those jumps stand many decades below the states, so a correctly rounded interface flux carries the last bit of the  $O(\rho c_s)$  quantities the Riemann solve is built from and not the last bit of its own value  $\rho v$ . The smallest flux difference the arithmetic can resolve is then  $\epsilon \rho(|v| + c_s)A/\Delta V$ , which is one ulp in the wind and about  $\epsilon/\text{Mach}$  in a layer whose flux is a tiny residue of the momentum. Measured at the base of three reference configurations that floor spans four decades,  $5.7 \times 10^{-14}$  to  $3.2 \times 10^{-11}$ , so a single fixed number would hold one state to eight ulps of its own base layer and another to a tenth of one ulp. The margin  $c_{\text{round}} = 10$  stands a factor 3.7 above the largest

measured ratio between the step the row takes when one ulp is added to every density and the estimate (126). The ceiling of unity is not an operative branch on any state measured: the row measure is the fractional change of the mass flux across a cell, so a state whose estimated floor reaches 1 is one whose mass row cannot be judged at all.

## 9.4 The Jacobian-free Newton-Krylov solve

The unknowns of the direct solve are the conserved state on the physical cells,

$$Y = (\rho_j, (\rho v)_j, \mathcal{E}_j, \{y_{s,j}\})_{j=1}^N, \quad n_{\text{var}} = 3 + n_{\text{spec}}, \quad (127)$$

where the optional species rows  $y_{s,j}$  are the elemental mass fractions of the diffusing elements and the transported molecular carriers, registered by `set_transported_species_rows` from the configuration. The ghost cells are not unknowns: the boundary condition fills them from the interior at every residual evaluation. The ionization state and the temperature are eliminated *locally* inside the residual, that is, given  $Y$  the equilibrium sweep of Sec. 4.4 re-solves the composition and returns the consistent heating and cooling of Sec. 5., so that  $F(Y)$  is a residual of size  $n_{\text{var}}N$  and not of the full chemical system.

The solve is a Jacobian-free Newton-Krylov iteration in the sense of Knoll & Keyes (2004): the Krylov method needs only the action of the Jacobian on a vector, which is taken as a forward difference of the residual, so the nonlocal radiative coupling never has to be assembled.

The outer iteration of `solve_steady_jfnk` is, per step: evaluate the full residual  $F$  at the iterate; build the banded finite-difference Jacobian of the *frozen-radiation* residual as a preconditioner; solve  $(I/\Delta\tau + J)\delta Y = -F$  by right-preconditioned GMRES with the matrix-vector product taken matrix free, so that the nonlocal radiative coupling is captured; take a step along a controlled direction; and ramp  $\Delta\tau$ .

**The banded preconditioner.** WENO3 reaches two cells on either side and within a cell every variable couples, so in the flat ordering  $Y(n_{\text{var}}(j-1)+k)$  a column couples to rows within  $|\text{row} - \text{col}| \leq 3n_{\text{var}} - 1$ ; the half-bandwidths are that number and a graph coloring with stride  $n_{\text{color}} = k_l + k_u + 1 = 6n_{\text{var}} - 1$  makes the row supports of same-color columns disjoint, so that the frozen-radiation residual, which is strictly local, is captured exactly by that many finite-difference probes. For  $n_{\text{var}} = 3$  this is 8 and 17. The band is factored by LAPACK `dgbrtf` and applied as the right preconditioner of the Krylov cycle. The finite-difference and Krylov probe steps are taken in the diagonally scaled variables of `cell_state_scales`, the characteristic magnitude of each unknown in its own cell; a probe point that leaves the set of states the code can describe, a chemical balance row going to NaN or overflowing, is answered by halving the step and sampling closer to the iterate, at most six times, and never by accepting a non-finite residual or inventing a derivative.

**The step control.** Where the system carries no species row the step is the Krylov direction with a backtracking line search on the merit  $\|F/D\|_2$ , with a Grippo non-monotone reference, the worst true merit of the last five iterates, so that sideways steps are admissible and the solve can leave a bad neighborhood. Where the system carries a species row the step control is a scaled trust region with a Powell dogleg: the path runs from the origin to the Cauchy point  $s_U$ , the minimizer of the linear model along the steepest descent direction, and from there to the Krylov point  $s_N$ , and the step

is where that path first meets the ball of radius  $\Delta$ ,

$$s = a_U s_U + a_N s_N, \quad \begin{cases} a_U = 0, a_N = 1 & \|s_N\| \leq \Delta, \\ a_U = \Delta / \|s_U\|, a_N = 0 & \|s_U\| \geq \Delta, \\ a_U = 1 - \tau, a_N = \tau & \text{otherwise,} \end{cases} \quad (128)$$

with  $\tau$  the root of  $\|s_U + \tau(s_N - s_U)\| = \Delta$  in  $[0, 1]$ . Returning the two coefficients rather than the vector is deliberate: the image  $As$  follows from  $As_U$  and  $As_N$  by linearity and costs no further matrix-free product. The Krylov leg is solved without the pseudo-transient shift,  $As_N = -r_0$  rather than  $(I/\Delta\tau + A)s_N = -r_0$ , because the dogleg falls monotonically along its path only when that leg is the model's own minimizer. A species unknown also carries a box, the fraction-to-the-boundary rule holding the step inside the physical range of a mass fraction, and an element with no reservoir is excluded from the registry altogether: its unknown would sit exactly on its own lower bound with no room below, and the boundary rule then returns zero for any direction with a component pointing there, so the whole Krylov direction would be refused.

**The ledger and the stop.** The best iterate reached so far is kept, with its composition, its flux spread and whether it met the gates. The non-monotone search accepts sideways steps, which is what lets the solve leave a bad neighborhood and also what lets it circle, so when the best iterate has not improved for 20 outer iterations the solve returns to it and switches to a monotone Armijo acceptance, and if that buys nothing in another 20 it stops and reports that it stagnated, a different statement from “no descent direction exists”. The loop-top stop requires the certified rows of Sec. 9.3 and not only the merit: the two are different functionals, and a window on the merit alone bounds nothing about the quantity the state is judged by. The return flag is 0 when the returned state met the acceptance gate and every judged row, 1 at the iteration limit, 2 when the solve ended without a state meeting the gate, and 3 at an iterate where no feasible descent direction exists. A flag of 2 is not a solver failure: the solve converged and the certification refused the state it hands back, naming the row, and that state is written with the refusal recorded rather than marched on.

## 9.5 Pseudo-transient continuation

`solve_steady_ptc` is the same residual solved with the banded Jacobian used as the *operator* and not only as a preconditioner: per iteration  $(I/\Delta\tau + J)\delta Y = -F$  is solved by LAPACK banded LU, the step is taken with a backtracking line search and a positivity test, and  $\Delta\tau$  follows the switched-evolution-relaxation ramp  $\Delta\tau \leftarrow \Delta\tau \|R\|_{\text{old}}/\|R\|_{\text{new}}$ , cut on a failed step. As  $\Delta\tau \rightarrow \infty$  this is Newton's method; at small  $\Delta\tau$  it behaves like explicit relaxation, which is the robust startup the continuation exists for. The route is not available with a carrier unknown: it builds its Jacobian from the frozen residual, which by construction performs no equilibrium sweep and therefore has no carrier row, and giving it one would mean assembling that row from a background nothing in the routine refreshes. The caller is told so rather than being handed a silently three-row answer.

The starting  $\Delta\tau$  matters: the marching handoff starts the continuation from  $\Delta\tau_0 = 1$  and the two direct routes from the CFL step.

## 9.6 The partitioned outer iteration

A species the run moves by an operator-split transport step instead of by a Newton row, which is the diffusing elements always and the molecular carriers whenever Coupled carrier solve is off, has a stationary balance no hydrodynamic solve touches; a single steady solve would freeze it at whatever state it was handed. `steady_wind_with_element_diffusion` in `EXHALE_main.f90` alternates the two: solve the wind at fixed composition, relax the composition to its own steady state in that wind, solve again. “Fixed composition” means the *transported* part of it: the element fractions the diffusion operator moves and the carrier densities. The local part, the ionization state of every cell and the molecular reaction balance at the carriers’ densities, is re-solved inside every evaluation of the hydrodynamic residual (Sec. 9.1), so the wind solve carries its own chemistry and only the transport of composition is alternated. A run with neither element diffusion nor carrier transport therefore has one outer pass: the wind solve alone. The composition relaxations are the fixed-wind element relaxation of `binary_element_diffusion.f90` (Sec. 6.9), under-relaxed with a factor  $\omega$  starting at 0.5, and the fixed-wind carrier relaxation of `diffusive_photochemistry.f90` (Sec. 6.5), bounded by a movement trust rather than damped. The advecting flux handed to the element operator is the face mass flux of the state just solved, read from the row terms that state assembled, so the operator is given its wind and does not reach into the residual for one.

What is accepted is only a state on which the certification of the *full* set of active equations passes, evaluated on the refreshed state the routine would hand back. A hydrodynamic solve returning 0 while an alternated species row refuses is not an accepted state: that flag is a statement about the rows in the Newton registry, and the alternated rows stand outside it by construction. Conversely a nonzero hydrodynamic flag does not end the loop by itself, because the composition update that follows it still makes progress.

**The progress control.** A pass makes progress when the joint distance of the state to the certification falls, or, where that distance is carried by a row that is not hydrodynamic, when the composition’s own residual falls; three consecutive passes without progress end the loop rather than spend the budget (20 passes, `EXHALE_OUTER_PASSES`) on a fixed point the alternation is not approaching. The endings are named: the joint certification passes; a row of the state is not finite; the element composition update is refused; the joint measure stops falling; the pass budget ends uncertified.

The two step lengths of the composition half, the element under-relaxation  $\omega$  (start 0.5, `EXHALE_DIFF_OMEGA`; floor 0.125) and the carrier movement bound, the largest relative change of one cell’s particle count  $n_{\text{tot}} + n_e$  one pass may make (start  $10^{-2}$ , `EXHALE_CARRIER_TRUST`; floor  $10^{-3}$ ; ceiling 0.5), answer to the hydrodynamic solve that consumes the update (revised 2026-09-27). *A pass whose carrier update ended on its movement bound is not a pass without progress* as long as the solve that consumed it converged and the joint distance rose by no more than a factor 1.1: the state moved by the whole bound, and the relative measures cannot show it because residual and scale fall together. MEASURED on the LHS 1140 b He/H = 1.8 state with transported ionization stages, every pass ended on the bound while the gated measure stood at  $2.17$  to  $2.20 \times 10^{-2}$ ; the earlier control read that as stagnation, halved the bound to its floor and ended the loop, while the same state with the bound held at 0.5 certified in 14 passes. Such a pass is neither counted toward the ending nor allowed to shorten a step, and the bound doubles, up to 0.5. The bound halves when a bound-limited update raised the joint distance of a solved wind by more than 1.1, when it preceded the first failed solve after a converged one, and when a pass without progress moved the carriers short of the bound on a solved wind;

a relaxation that kept no step leaves it where it is.  $\omega$  halves on a pass without progress whose wind was solved and on the first failed solve after a converged one, and doubles back toward its starting value after a converged solve on a pass with progress. A failed solve that follows another failed solve did not start from a solved wind, so the update is not what failed it and neither step is shortened. EXHALE\_CARRIER\_TRUST\_HOLD freezes the bound for every pass. Recorded as open: the starting bound for transported stages (the state above certifies from 0.5 and not from  $10^{-2}$ ), and the molecular  $\text{He/H} = 0.083$  continuation, where a growing bound makes the  $\text{H}_2$  row of the outermost cells overshoot and neither control certifies.

**The state of every pass, kept.** Every pass after which the iteration goes on publishes the state the next pass starts from, rebuilt from  $(u, f_{\text{sp}})$  in the order of the checkpoint restoration with every side effect on the caloric mixture undone, as one generation output/pass\_state/<id>/ (Hydro\_ioniz.txt, Ion\_species.txt, manifest.txt) written by the writers of the final state, with <id> = r<run identity>\_p<n>, so a run stopped from outside keeps its last completed pass. The pair carries certified=F cert\_reason=pass\_snapshot\_p<n> in both halves, because the certification of a pass is taken before its composition update and the snapshot is the state after it, and a # pass\_snapshot line with the movement bound and  $\omega$  of the next pass, which a restart does not inherit. The generation is written in a temporary directory, renamed onto its name, and published by renaming a one-line file onto output/pass\_state/current; an interruption at any point therefore leaves current naming a complete generation. Both halves state state\_id=<id> on their coupling line, and the loader refuses a pair whose halves state two identities or only one. Nothing the run reads afterwards changes: the final state files of a run are bitwise those of the binary before the snapshot existed.

Two rules keep the state self-consistent. The progress control stands *ahead* of the composition update, so the pass that ends the iteration takes no update and what is handed back is the very state the certification of that pass was evaluated on; and the last pass of the budget takes no update either, an update existing only to be consumed by the next hydrodynamic solve. Taking it would replace a stationary wind by a state whose hydrodynamic rows the update itself spoiled, by eleven decades in the mass row on a molecular configuration.

With Coupled carrier solve: True the alternation does not arise for the carriers at all:  $n(\text{H}_2)$  becomes a fourth Newton unknown per cell, one solve returns a wind and a carrier partition that are steady states of each other, and the outer loop runs once. A third value, On\_stall (2026-09-17), runs the alternation and hands the state to that block at the pass where the alternation gives up: the joint distance has not fallen in three consecutive passes AND the carrier relaxation of this pass and of the two before it ended on the movement bound. What is known about the two routes where they have been compared: on a certified molecular state the alternation certifies at outer pass 12 while the block moves its residual by 0.6 per cent in the same wall clock, every one of its linear solves ending with the Krylov subspace exhausted, so the block as it stands is limited by its preconditioner and not by the species box.

**What the alternation cannot do, measured.** The movement bound on the carrier relaxation is not what holds the refusing molecular states: relocating the cells that control it only moves the control to the edge of whatever region is exempted, and the refusing row is worse afterwards. What the measurements point at is the splitting itself. The bound exists because the wind is held while the carriers relax, and a composition good enough for the carrier rows is already outside the range in which holding the wind is admissible: the one relaxation that reached the carriers' own fixed point

moved the base particle count almost seven times the bound, and the hydrodynamic solve that had to follow it did not converge, its mass rows outside tolerance in every cell over a hundred iterations. With no alternated species at all, which is what an atomic run and a fully coupled run both give, the body is one steady solve followed by the state refresh and the flag is the solve's own.

## 10. Output and post-processing

### 10.1 The advection-corrected profiles

For the reason given in Sec. 4.5, `post_process_adv.f90` re-solves the profile once, at the end of the run, with the ionization advected: the systems (62)–(64) are solved cell by cell from the base outward, each cell taking its history  $x^{\text{up}}$  from the solved fractions of the two cells below it, and the temperature is then re-solved on the steady internal-energy equation (78) with the Brent bracket of Sec. 5.3. In a run that solved heat conduction or the enthalpy flux of the element and carrier fluxes, the heating of that equation holds them too, evaluated with the corrected temperature: conduction (21) couples each cell to both neighbors, so after the marching sweep the corrected cells are solved together by a Newton iteration on the column (banded Jacobian, conductivity and enthalpy flux divergence re-evaluated at each iterate); cells that keep the run's temperature are fixed values of that solve. That iteration stops when no temperature moves by more than  $10^{-10}$  of itself (or after 100 iterations), and the column is then judged on its residual: the energy row of every unknown, formed again at the final profile with the transport terms of that profile, must be within  $10\epsilon \sum |\text{terms}| + 10^{-6}S_E$ ,  $S_E$  the largest of its physically grouped terms and  $10^{-6}$  the energy tolerance of the stationary certification. It is rejected when the banded solve fails, a residual, step or term is not finite, a temperature is not positive, or the residual test fails; a rejected column keeps the marching profile and every one of its cells is written failed, and the outcome, the reason, the triggering cell and the worst residual are recorded in the `# adv_derived_state:` line of the `_adv` headers. The ten outer passes are not tested for convergence, so an accepted column is recorded as `converged_on_residual` and the outer iteration as `unverified`. Without these terms the corrected temperature of a conduction-dominated layer answers another equation (7900 K against the run's 5500 K at  $1.2 R_p$  on the fiducial LHS 1140 b atomic case).

**The step.** Both recursions are one-step integrations of an ODE in  $r$ ,  $dy/dr = F(y, r)$ , with the variable-step second-order backward differentiation formula (BDF2),

$$y_j - a_1 y_{j-1} + a_2 y_{j-2} = g h_j F(y_j, r_j), \quad a_1 = \frac{(1+w)^2}{1+2w}, \quad a_2 = \frac{w^2}{1+2w}, \quad g = \frac{1+w}{1+2w}, \quad (129)$$

$h_j = r_j - r_{j-1}$ ,  $w = h_j/h_{j-1}$ : the left side over  $g h_j$  is  $dP/dr$  at  $r_j$  of the quadratic  $P$  through the three points, so the step is exact for quadratics and the recursion second order in the cell width (the formula of Li & Liao 2022, eq. (1.4) of the preprint). Every  $r$ -derivative of a step, the known density profile of the energy equation included, is taken at  $r_j$  by this rule, and every other factor (rates, heating and cooling, the velocity of the residence time) is the value at  $r_j$ . The recursion starts, and restarts after every cell it does not integrate, with backward Euler ( $a_1 = g = 1$ ,  $a_2 = 0$ ); a cell whose step ratio exceeds the zero-stability bound  $w \leq 1 + \sqrt{2}$  is integrated by backward Euler, and a BDF2 step whose populations come out negative or whose temperature has no root is retaken by backward Euler, which keeps the

populations of a linear rate system nonnegative; the run log counts all three. BDF2 is L-stable, like backward Euler, so in the stiff limit a population that relaxes within the cell returns its local equilibrium; the trapezoid rule, A- but not L-stable, would alternate about it from cell to cell. Until 2026-09-27 every step was backward Euler with the rates at  $r_j$  and the residence time at  $r_{j-1}$ , a first-order recursion that was the main source of the first-order grid convergence of the He 10830 equivalent width. MEASURED on nine certified LHS 1140 b states (He/H 1.6, 1.8 and 2.13 on 500, 1000 and 2000 cells) with the state held fixed, the post-process share of the equivalent-width error falls from +3.0/+1.0 per cent to below  $3 \times 10^{-4}$ , the  $\_adv$   $2^3S$  column converges at order 1.98–2.14 (1.00–1.01 before), and a closed-form column test gives 1.99; the equivalent width itself moves by –3.9 per cent (500 cells) to –0.97 per cent (2000 cells) at He/H = 1.6 and still converges at first order, what remains being the solved state’s own base closure and the transit integration.

Three properties of that pass are worth stating.

- The third term of (78), the enthalpy flux carried by the divergence of the mass flux,  $hr^{-2}d(\rho vr^2)/dr$ , vanishes for a stationary mass flux and for nothing else, so a correction built without it would describe a converged wind and no other state. The divergence is the mass row of the state itself, the face mass fluxes of its Riemann solve differenced over the cell,  $(A_{j+1/2}F_{j+1/2} - A_{j-1/2}F_{j-1/2})/dV_j$ , so the term is an exact zero wherever the two faces carry the same flux. (Until 2026-09-27 it was the same operator applied to the cell-centered  $\rho vr^2$  of two neighboring cells, which at the base of a quasi-hydrostatic layer carries the odd-even pattern of the cell-centered velocity; its difference set the temperature of the lower layer.)
- A cell is corrected only where the correction means something. Above the Damköhler number  $10^2$ , formed with the *slowest* relaxing species of the solved system, the gas reaches local equilibrium many times over while it crosses the cell and the equilibrium solution already solves the advection equation, so the cell is pinned to it. Below an equilibrium ion fraction of  $10^{-6}$  the residual, which carries the neutral fraction, cannot represent the ion density to better than about a percent. Where the mass row of the supplied state exceeds `adv_conditional_tol` as a fraction of its own largest term, the cell keeps the run’s own value; each cell’s verdict travels in the output file as `adv_T_status`, `adv_comp_status` and `adv_mass_row`.
- The reconstruction treats the gas as H/He plus trace metals and *excludes* the molecular species:  $H_2$ ,  $H_2^+$ ,  $H_3^+$  and  $HeH^+$  are not passed in and are not re-solved, so inside a molecular layer the  $\_adv$  particle count omits the neutral  $H_2$  and the molecular-ion electrons. The output file says so in its own header rather than only in the source. The same holds for the energy balance: the post-process lacks the molecular chemistry heat and the  $H_3^+$  and molecular band cooling, so wherever the heating and cooling it omits, measured on the state handed in as the difference between the run’s own channels and its own at the run’s temperature and composition, exceed the ones it carries, the row keeps the run’s temperature with `adv_T_status` = 3 (unsupported); MEASURED on the LHS 1140 b molecular states, 196/389/776 rows on 500/1000/2000 cells, up to 1.134–1.137  $R_p$ , with an equivalent-width change of at most  $1.2 \times 10^{-4}$  of itself.

The first of those gates has a fourth companion: wherever  $v \leq 0$  in the cell or the one below it, the cell keeps the converged equilibrium ionization and temperature, because the step takes its history from the cells below, which are

not upstream when the gas moves inward, and the residence time  $h/v$  is then negative as well. The relaxation rate entering the Damköhler number is the total rate at which each solved population is destroyed and re-formed, and the *minimum* over the species the system solves is taken, because He( $2^3\text{S}$ ) relaxes orders of magnitude more slowly than hydrogen ( $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$  sets the floor) and the vector is only as local as its slowest member.

## 10.2 The mass-loss rate

The quantity the run reports as `Log10 of steady-state Mdot` is measured at one radius, in `EXHALE_main.f90`:

$$\dot{M} = 4\pi \rho(j) v(j) r(j)^2 n_0 v_0 m_{\text{H}} R_0^2 \quad \text{at} \quad j = \max(N-20, 1), \quad (130)$$

the offset of 20 cells keeping the evaluation far enough from the outer boundary, and the floor at  $j = 1$  keeping it out of the base ghosts where the boundary closure and not the wind sets  $\rho$  and  $v$ . A logarithm is taken only where the flow there is directed outward:  $\rho v \leq 0$  is an inflow, which is not a mass-loss rate, and the run then reports the signed flux instead of a logarithm. The dayside convention of Table 1 is applied here and only here:  $\log_{10} \dot{M} \rightarrow \log_{10} \dot{M} - \log_{10} 2$  for `Rate/2 + Mdot/2`, and  $-\log_{10} 4$  for `Mdot/4`.

A *quantitative*  $\dot{M}$  must come from a state the steady solver has finished. A run stopped on the `du` threshold (110) alone carries a path-dependent spread of several percent in (130); the two-stage marching of Sec. 7.2 followed by the stationary solve of Sec. 9.4 is the working recipe, and it is what the regression matrix uses.

## 10.3 Output files

**Which state the products are built from (2026-09-17).** A solved state's advection-corrected products are made by `Restart intent: stationary evaluate`, which loads the state, derives its pressure and temperature from the loaded conserved state, runs the ionization sweep, evaluates the certification and writes every product from that work state without entering the time integration. The route this replaced took one time step at `CFL: 1.0e-12` with `Do only PP: True`, which suppressed the movement but did not remove it, reconstructed with whatever `Reconstruction scheme: stated` rather than with the scheme the solution was reached under, and wrote the stepped state back over `Hydro_ioniz.txt`. The difference is largest where the advection operator is weakest: the operator's own weighting measure `adv_mass_row` falls from 1.57 to  $1.7 \times 10^{-9}$  at the first cell and from  $3.1 \times 10^{-5}$  to  $4.0 \times 10^{-14}$  at the last, while the He I 10830 equivalent width moves by  $5 \times 10^{-6}$  of itself.

Every file written to output/ carries a schema header, so a reader adapts instead of counting columns. `Hydro_ioniz.txt` opens with `# EXHALE schema 2` and then

```
# columns r[Rp] rho[mH/cm3] v[cm/s] p[cgs] T[K] heat[erg/cm3/s] cool[erg/cm3/s]
```

(with three further status columns on the `_adv` file). Column 2 is  $\rho n_0$ , the *mass* density in units of  $m_{\text{H}}$  per  $\text{cm}^3$  with the metals included under the `eos_metals` policy, not a number density. The first and last `Ng` rows of every file are ghost cells filled by the boundary conditions, and their  $\rho v r^2$  is a boundary extrapolation. `Ion_species.txt` carries the species fractions in the `species_table` order of Sec. 4.1; `Heating_breakdown.txt` and `Cooling_breakdown.txt` carry the channel decompositions of Sec. 5., each built from the array the energy equation itself used, so the channel sum and the heat or cool column of `Hydro_ioniz.txt` are the same number to round-off; `Excited_H.txt`,

Lyman\_Werner.txt, FUV\_bands.txt, Oxygen\_chemistry.txt and OI\_levels.txt carry the optional physics; the generation pass\_state/<id>/ named by pass\_state/current is the state of the last completed outer pass of an alternating stationary run, never certified (Sec. 9.6). The two \_adv files carry a # derived\_from: line rather than the # coupling: header of a state file (item L9, 2026-09-17): they describe an advection-corrected composition, which is not the composition any certification was made on, so the input state's certification pair is carried there as provenance and not as a claim about the numbers in the file. The Python loader examples/exhale\_io.py reads all of them through those headers (exhale\_io.Run('output/') gives .r, .T, .Mdot and the rest), and it should be used in place of any new column parser.

## 10.4 The transmission spectrum

EXHALE\_transit.py, with the library exhale\_transit\_lib.py, builds transit transmission spectra from either of the two states a run can leave, selected by EXHALE\_TRANSIT\_STATE: the \_adv profiles by default, or the solved state pair Hydro\_ioniz.txt and Ion\_species.txt with EXHALE\_TRANSIT\_STATE=solution, which is how a stationary run is read. The lines are the same either way: the He I 10830 Å triplet, Ly $\alpha$ , H $\alpha$  and H $\beta$ , and the metal doublets Mg II h&k, Ca II H&K and Na I D, the metal lines being skipped automatically for a metals-off run. It integrates the Voigt line opacity over impact parameter, averages over the stellar disk, and convolves with the instrument profile and the planet's rotation; the Balmer branch builds the non-LTE  $2s/2p$  populations with the level balance of Christie et al. (2013) and the rate coefficients of the Fortran model of Sec. 4.6 (transcribed in exhale\_transit\_lib.py, n2\_populations, and equal to them to  $3 \times 10^{-13}$  relative), but with a field of its own, because neither the solver's Ly $\alpha$  field nor its Balmer rate is in the profiles the tool reads:  $J_{\text{Ly}\alpha}$  from a file or the estimate of Huang et al. (2017), and a diluted stellar blackbody for the Balmer continuum. An optional triaxial Roche geometry comes from roche\_recon.py. Overrides are environment variables (EXHALE\_TRANSIT\_\*: resolving power, He window, rotation period, azimuthal samples) and the output keeps the historical tpm\_\*.txt names, each with its own # transit\_schema 1 metadata block. The physics of that tool is documented separately in docs/transmission\_spectrum.pdf and is not repeated here.

**Which state the spectrum is built from.** The two selections are two answers for one column and not two formats of one answer. The \_adv pair is a post-process of a marching state: a steady advective correction cell by cell, second order in the cell width (variable-step BDF2 along the flow at the bulk velocity), with no eddy term and no element drift, refused in a cell where an assumption of the steady equations fails. It is an independent discretization of the same column and no equation of the run was solved for it. The solved pair is the state the solver produced and the certification judged; where the ionization stages are transported its composition is the partition the stage transport itself produced, so an advective correction on top of it would be a second and different transport approximation of the same stages. A stationary route therefore writes no \_adv products, none are manufactured for one, and the default selection refuses by name and names the other selection rather than reading it. The two files of a selection are read together, the O I ground-term populations included: the temperature of one state with the composition of the other solves neither set of equations. Every saved curve carries the selection, the path, md5 and size of each file read, and the state's own provenance, coupling, boundary model and reservoir lines.

What the choice costs was MEASURED on wasp\_full, which carries both states, as the peak transit depth relative to

itself (before the atomic-data corrections of 2026-09-24/25, which moved both states, and before the post-processor took the solver's  $n = 2$  rates; not re-measured since): He I 10830  $-0.64$  per cent,  $H\alpha -1.28$ ,  $H\beta -3.48$ ,  $Ly\alpha$  unchanged at its saturated ceiling, and the optically thin metal lines  $+0.1$  to  $+0.8$ . The metastable helium column is 12 per cent *larger* in the solved state and its line is nonetheless shallower, because the advection correction moves that column outward (column-weighted mean radius 1.2941 against  $1.3112R_p$ ) and a saturated line's disk-averaged depth follows how far the absorbing shell reaches;  $H\beta$  moves most because it is the line still on the linear part of the curve of growth (item P6f).

## 11. The input keys that select physics

Runtime configuration is by the presence of a key, not by a compile flag (Sec. 1.1). Three companion files are read at startup and their mere presence is itself a switch: `metals.inp` turns the trace metals on, `opacity.inp` the tabulated continuum opacities, and `base.inp` the lower-atmosphere handoff.

Table 2 lists the keys that select physics or the numerical scheme, with the default the code takes when the key is absent. The defaults were read from `parameters.f90` and the parsing from `input_read.f90`; keys that only name a file, a path or an output option are omitted. Four lines of the core block are mandatory and have no default: they are marked as such.

Table 2: Physics and algorithm keys of `input.inp`, with the value the code takes when the key is absent.

| Key                                                         | Default                 | What it selects, and where                                                                                                                                                                                              |
|-------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Geometry, grid and the one-dimensional approximation</i> |                         |                                                                                                                                                                                                                         |
| 2D approximate method:                                      | mandatory               | $\dot{M}_{\text{dot}}/4$ , $\text{Rate}/2$ , $\text{Rate}/4$ or $\alpha$ : the dayside dilution of every stellar beam and the factor on $\dot{M}$ (Table 1; <code>parameters.f90</code> , <code>utilities.f90</code> ). |
| Grid type:                                                  | mandatory               | Uniform, Mixed or Stretched; every type puts the first face on the base level, $r_{\text{edg}}(0) = 1$ ( <code>define_grid.f90</code> ).                                                                                |
| Grid cells:                                                 | 500                     | Number of physical cells; ghosts are added on top ( <code>define_grid.f90</code> ).                                                                                                                                     |
| Base grid [dr,cells]:                                       | $2 \times 10^{-4}$ , 50 | Width and count of the uniform base region of the Mixed grid ( <code>define_grid.f90</code> ).                                                                                                                          |
| Domain mode:                                                | Roche                   | Spherical drops the tidal and centrifugal terms of (12) ( <code>grav_field.f90</code> ).                                                                                                                                |
| Outer radius [R_p]:                                         | L1 radius               | Outer boundary; required in spherical mode ( <code>input_read.f90</code> ).                                                                                                                                             |
| Outer shells [r_face,cells]:                                | none                    | Cells appended beyond the constructed grid out to the face $r_{\text{face}}$ , the cells below unchanged bit for bit ( <code>define_grid.f90</code> ).                                                                  |

Table 2 continued

| Key                                           | Default                  | What it selects, and where                                                                                               |
|-----------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <i>Discretization</i>                         |                          |                                                                                                                          |
| Numerical flux:                               | mandatory                | HLLC, Roe or LLF (Num_Fluxes.f90); Roe is refused with the molecular ladder equation of state.                           |
| Reconstruction scheme:                        | mandatory                | PLM, WENO3 or the two-stage PLM+WENO3 (Reconstruction.f90).                                                              |
| Reconstruction continuation:                  | off                      | $\Delta\lambda$ of the homotopy (98) between the two operators (steady_residual.f90).                                    |
| CFL:                                          | 0.6                      | CFL number of (109) (eval_dt.f90).                                                                                       |
| Time stepping:                                | global                   | Local gives every cell its own pseudo-time step (eval_dt.f90).                                                           |
| Well balanced:                                | off                      | Carries the departure from the cell's own hydrostatic equilibrium instead of the state (Reconstruction.f90, RK_rhs.f90). |
| Low-Mach damping:                             | off, $M_{th} = 10^{-3}$  | Gated fourth-difference dissipation (19) of the contact mode (low_mach_dissipation.f90).                                 |
| Shapiro filter:                               | off, every 4 steps       | 1-2-1 low-pass filter on the marched state (Apply_BC.f90).                                                               |
| Caloric EOS:                                  | ladder                   | monatomic forces $\gamma = 5/3$ everywhere (caloric_eos.f90).                                                            |
| Energy solver:                                | semi-implicit            | Explicit reverts the source update to forward Euler (energy_semi_implicit.f90).                                          |
| <i>Boundary and initial conditions</i>        |                          |                                                                                                                          |
| Base BC:                                      | density                  | pressure <microbar> sets the base level from a pressure instead of a density (input_read.f90, base_boundary.f90).        |
| IC mode:                                      | cold                     | cold, transonic, hot_parker, auto or windae (set_IC.f90).                                                                |
| Transonic IC:                                 | off                      | Older synonym of IC mode: transonic.                                                                                     |
| Hot Parker IC:                                | off, $T_{wind} = 10^4$ K | Older synonym of IC mode: hot_parker.                                                                                    |
| Restart intent:                               | relaxation               | What a loaded state is continued as: trajectory, relaxation or stationary (load_IC.f90).                                 |
| <i>Convergence and the stationary solvers</i> |                          |                                                                                                                          |
| du_th [PLM,WENO3]:                            | $10^{-3}$                | Threshold on the mass-flux spread (110); a second value makes the run two-stage (EXHALE_main.f90).                       |
| Escape radius [R_p]:                          | mandatory                | Inner edge of the window (110) is taken over (define_grid.f90).                                                          |

Table 2 continued

| Key                               | Default                                      | What it selects, and where                                                                                                                    |
|-----------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Resid tol:                        | off                                          | Residual convergence instead of the flux spread; the solvers use $10^{-5}$ when it is absent (steady_residual.f90).                           |
| Flux spread tol:                  | $2 \times 10^{-5}$ , $r_{\text{flux}} = 1.2$ | The flux gate of the acceptance test (steady_residual.f90).                                                                                   |
| Solver:                           | marching only                                | Newton hands over to the stationary solve once the spread falls below $10^{-2}$ (steady_newton.f90).                                          |
| Stall:                            | $10^{-6}$ , 2000 steps                       | Plateau detector of the marching stop (EXHALE_main.f90).                                                                                      |
| Level tol:                        | off                                          | Mass-flux level stability required beside every marching stop.                                                                                |
| Max steps:                        | $10^6$                                       | Hard cap on marching iterations.                                                                                                              |
| Newton solver:                    | on                                           | False reverts the composition solve to MINPACK hybrd1 (newton_solver.f90).                                                                    |
| Brent solver:                     | on                                           | False reverts the post-process temperature solve (T_equation.f90).                                                                            |
| <i>Atomic hydrogen and helium</i> |                                              |                                                                                                                                               |
| Include He23S?                    | on                                           | The He $2^3S$ metastable as a solved level (System_HeH_TR*.f90).                                                                              |
| ATES_photoionization_rate         | off                                          | Reverts He I photoionization to the legacy ATES fit (cross_sec.f90).                                                                          |
| Legacy_HHe_rates                  | off                                          | Reverts H/He recombination and collisional ionization to the ATES fits (Cool_coeff.f90).                                                      |
| Atomic rate set:                  | default                                      | Koskinen2022 swaps four atomic H/He rate coefficients (Cool_coeff.f90, mol_rates.f90).                                                        |
| Secondary_ionization              | on, staged                                   | Secondary ionization by fast photoelectrons; Immediate applies it from step 0 (electron_energy_degradation.f90).                              |
| He_rec_coupling                   | on                                           | He II and He III recombination radiation absorbed on the spot; off is case B for both (util_ion_eq.f90).                                      |
| H_rec_escape                      | on                                           | H II ground captures whose photons are not re-absorbed by H I in the cell are net recombinations; off is case B everywhere (util_ion_eq.f90). |
| He_H_charge_exchange              | on                                           | The He/H charge-exchange pair (charge_exchange.f90).                                                                                          |
| Photoelectron heating:            | excess                                       | Whether a photoionization deposits $h\nu - E_{\text{th}}$ or a stated fraction of $h\nu$ (util_ion_eq.f90).                                   |
| Deexc heat                        | on                                           | Collisional de-excitation heating of H( $n=2$ ) (excited_hydrogen.f90).                                                                       |

Table 2 continued

| Key                                               | Default                  | What it selects, and where                                                                                                                                      |
|---------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Radiation field</i>                            |                          |                                                                                                                                                                 |
| Spectrum type:                                    | mandatory                | Load, Power-law, Planck or Monochromatic: the spectrum of the whole photon grid ( <code>sed_read.f90</code> , <code>J_inc.f90</code> ).                         |
| Stellar Teff: + Stellar radius:                   | 0 (off)                  | Enable the $H(n=2)$ model and its Balmer-continuum coupling, whose band is read from the run's own spectrum type ( <code>excited_hydrogen.f90</code> ).         |
| Stellar Ly $\alpha$ flux:                         | 0                        | Stellar Ly- $\alpha$ at the orbit; also band B2 of the oxygen photolysis ( <code>lya_rt.f90</code> ).                                                           |
| Jlya escape-prob                                  | parameterized            | The Neufeld escape-probability Ly- $\alpha$ transfer ( <code>lya_rt.f90</code> ).                                                                               |
| Jlya RT file:                                     | none                     | Import a $J_{Ly\alpha}(r)$ profile computed elsewhere.                                                                                                          |
| Ly $\alpha$ stellar halfwidth:                    | 70 km/s                  | Half-width of the stellar Ly- $\alpha$ line.                                                                                                                    |
| Ly $\alpha$ stellar boost:                        | 5.0                      | Bounded buildup of the trapped stellar beam.                                                                                                                    |
| Ly $\alpha$ absorbing bottom                      | reflecting               | Closes the Ly- $\alpha$ domain from below with an absorber.                                                                                                     |
| Stellar LW flux:                                  | 0                        | Lyman-Werner band flux driving $H_2$ photodissociation ( <code>lyman_werner.f90</code> ).                                                                       |
| Stellar FUV B3/B4 flux:                           | 0                        | The two longer FUV bands of the oxygen photolysis ( <code>water_photolysis.f90</code> ).                                                                        |
| <i>Molecules, metals and the lower atmosphere</i> |                          |                                                                                                                                                                 |
| metals.inp present                                | metals off               | Trace metals solved in the same coupled system as H and He ( <code>System_HeH_metals.f90</code> ).                                                              |
| eos_metals                                        | 1                        | Metals in the bulk mass, electron and particle budget ( <code>composition.f90</code> ).                                                                         |
| cno_cool                                          | 1 (CHIANTI)              | 0 reverts C/N/O line cooling to the legacy analytic fits ( <code>Cool_coeff.f90</code> ).                                                                       |
| Molecular chemistry:                              | off                      | The $H_2/H_2^+/H_3^+/HeH^+$ network ( <code>System_HeH_mol.f90</code> , <code>mol_rates.f90</code> ).                                                           |
| Oxygen chemistry:                                 | off                      | OH, $H_2O$ and CO with their photolysis ( <code>oxygen_rates.f90</code> , <code>water_photolysis.f90</code> ).                                                  |
| Molecular carrier transport:                      | follows oxygen chemistry | Vertical transport of the molecular carriers ( <code>diffusive_photochemistry.f90</code> ).                                                                     |
| Composition update holds:                         | energy                   | What the composition update of the stationary alternation holds beside $\rho$ and $v$ : the conserved energy, or the pressure ( <code>EXHALE_main.f90</code> ). |
| Coupled carrier solve:                            | off                      | $n(H_2)$ as a fourth Newton unknown per cell instead of an alternation ( <code>steady_newton.f90</code> ).                                                      |

Table 2 continued

| Key                                      | Default                                                   | What it selects, and where                                                                                                                                                                                                |
|------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ionization transport:                    | off                                                       | $H^+$ carried with the flow as a fifth carrier.                                                                                                                                                                           |
| Molecular base:                          | off                                                       | Removes the H nuclei bound into $H_2$ from the base particle count (15) ( <code>composition.f90</code> ).                                                                                                                 |
| $H_2$ double ionization:                 | chung80                                                   | $H_2 + h\nu \rightarrow 2H^+ + 2e$ resolved above 51.4 eV; <code>yan_rho</code> the other model, off folds it into the single dissociative channel ( <code>h2_photo_channels.f90</code> ).                                |
| $H_2$ neutral dissociation:              | on                                                        | $H_2 + h\nu \rightarrow H + H$ over 33–41 eV ( <code>h2_photo_channels.f90</code> ).                                                                                                                                      |
| Molecular reaction heat:                 | on                                                        | Deposits the energy the collisional molecular reactions release ( <code>molecular_reaction_heat.f90</code> ).                                                                                                             |
| Molecular IR bands:                      | off                                                       | $H_2$ , $H_2O$ and CO infrared line cooling ( <code>molecular_infrared_cooling.f90</code> ).                                                                                                                              |
| Base IR field:                           | off                                                       | The infrared coolants see the thermal radiation of the atmosphere below the base instead of vacuum.                                                                                                                       |
| Lower atmosphere:                        | none                                                      | analytic or vulcan lower-atmosphere pre-step ( <code>lower_column.f90</code> ).                                                                                                                                           |
| Lower atmosphere profile:                | none                                                      | The lower atmosphere handed over as a table over pressure ( <code>lower_atmosphere_profile.f90</code> ).                                                                                                                  |
| Lower column:                            | off                                                       | 1-bar radius of the analytic column.                                                                                                                                                                                      |
| <i>Diffusion and molecular transport</i> |                                                           |                                                                                                                                                                                                                           |
| He_diffusion                             | off                                                       | He/H diffusive separation as a transported element ( <code>binary_element_diffusion.f90</code> ).                                                                                                                         |
| He_metal_diffusion                       | off                                                       | Extends the same operator to the trace metals.                                                                                                                                                                            |
| He_Kzz                                   | 0                                                         | Constant eddy diffusion coefficient where no profile states one.                                                                                                                                                          |
| He_alphaT                                | 0                                                         | Thermal-diffusion factor.                                                                                                                                                                                                 |
| He_ambipolar                             | on                                                        | Ambipolar term of the element diffusion.                                                                                                                                                                                  |
| Interdiffusion enthalpy flux:            | on (with He_diffusion or a transported molecular carrier) | The enthalpy flux $\sum_s h_s J_s$ of the diffusing elements and of the transported carriers in the energy equation, Sec. 6.10 ( <code>binary_element_diffusion.f90</code> , <code>diffusive_photochemistry.f90</code> ). |
| Viscosity:                               | off                                                       | Radial viscous force and its dissipation (20) ( <code>viscous_conduction.f90</code> ).                                                                                                                                    |
| Conduction:                              | off                                                       | Heat conduction with the mixture $\kappa$ of (21).                                                                                                                                                                        |

## References

- Abrahamsson, E., Krems, R. V., & Dalgarno, A. 2007, *ApJ*, 654, 1171
- Allgower, E. L., & Georg, K. 2003, *Introduction to Numerical Continuation Methods*, SIAM Classics in Applied Mathematics 45 (Philadelphia: SIAM)
- Backx, C., Wight, G. R., & Van der Wiel, M. J. 1976, *J. Phys. B*, 9, 315
- Badnell, N. R. 2006, *ApJS*, 167, 334
- Banks, P. M., & Kockarts, G. 1973, *Aeronomy* (New York: Academic Press)
- Barinovs, Ģ., van Hemert, M. C., Krems, R., & Dalgarno, A. 2005, *ApJ*, 620, 537
- Batten, P., Clarke, N., Lambert, C., & Causon, D. M. 1997, *SIAM J. Sci. Comput.*, 18, 1553
- Baulch, D. L., Bowman, C. T., Cobos, C. J., et al. 2005, *J. Phys. Chem. Ref. Data*, 34, 757
- Böhringer, H., & Arnold, F. 1986, *J. Chem. Phys.*, 84, 1459
- Black, J. H. 1978, *ApJ*, 222, 125
- Biassoni, F., Caldiroli, A., Gallo, E., et al. 2024, *A&A*, 682, A115
- Black, J. H. 1981, *MNRAS*, 197, 553
- Bray, I., Burgess, A., Fursa, D. V., & Tully, J. A. 2000, *A&AS*, 146, 481
- Caldirolì, A., Haardt, F., Gallo, E., et al. 2021, *A&A*, 655, A30
- Cecchi-Pestellini, C., Ciaravella, A., Micela, G., & Penz, T. 2009, *A&A*, 496, 863
- Chase, M. W., Jr. 1998, *NIST-JANAF Thermochemical Tables*, 4th ed., *J. Phys. Chem. Ref. Data Monograph* 9
- Christie, D., Arras, P., & Li, Z.-Y. 2013, *ApJ*, 772, 144
- Chung, Y. M., Lee, E.-M., Masuoka, T., & Samson, J. A. R. 1993, *J. Chem. Phys.*, 99, 885
- Cohen, J. S., & Lane, N. F. 1971, *Chem. Phys. Lett.*, 10, 623
- Cohen, J. S., & Lane, N. F. 1977, *J. Chem. Phys.*, 66, 586
- Cohen, N., & Westberg, K. R. 1983, *J. Phys. Chem. Ref. Data*, 12, 531
- Giusti-Suzor, A., Bardsley, J. N., & Derkits, C. 1983, *Phys. Rev. A*, 28, 682
- Guberman, S. L. 1994, *Phys. Rev. A*, 49, R4277
- Jóźwiak, H., Thibault, F., Viel, A., Wcisło, P., & Lique, F. 2024, *A&A*, 685, A113
- Kokoouline, V., Greene, C. H., & Esry, B. D. 2001, *Nature*, 412, 891

- Le Bourlot, J., Pineau des Forêts, G., & Flower, D. R. 1999, MNRAS, 305, 802
- Lique, F. 2015, MNRAS, 453, 810
- Rieper, F. 2011, J. Comput. Phys., 230, 5263
- Strasser, D., Lammich, L., Kreckel, H., et al. 2001, Phys. Rev. Lett., 86, 779
- Takagi, H. 2002, Phys. Scr., T96, 52
- Wolniewicz, L., Simbotin, I., & Dalgarno, A. 1998, ApJS, 115, 293
- Cook, A. W. 2009, Phys. Fluids, 21, 055109
- Dalgarno, A., Yan, M., & Liu, W. 1999, ApJS, 125, 237
- Draine, B. T., & Bertoldi, F. 1996, ApJ, 468, 269
- Drake, G. W. F. 1986, Phys. Rev. A, 34, 2871
- Drake, G. W., Victor, G. A., & Dalgarno, A. 1969, Phys. Rev., 180, 25
- Einfeldt, B., Munz, C.-D., Roe, P. L., & Sjögren, B. 1991, J. Comput. Phys., 92, 273
- García Muñoz, A. 2025, A&A, 698, A199
- Gordon, S., & McBride, B. J. 1994, Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications, NASA RP-1311
- Gottlieb, S., & Shu, C.-W. 1998, Math. Comput., 67, 73
- Harrington, J. P. 1973, MNRAS, 162, 43
- Hu, X. Y., Adams, N. A., & Shu, C.-W. 2013, J. Comput. Phys., 242, 169
- Huang, C., Koskinen, T., Lavvas, P., & Fossati, L. 2023, ApJ, 951, 123
- Hui, L., & Gnedin, N. Y. 1997, MNRAS, 292, 27
- Hummer, D. G. 1994, MNRAS, 268, 109
- Hummer, D. G., & Storey, P. J. 1998, MNRAS, 297, 1073
- Jameson, A., Schmidt, W., & Turkel, E. 1981, AIAA Paper 81-1259
- Käppeli, R., & Mishra, S. 2014, J. Comput. Phys., 259, 199
- Käppeli, R., & Mishra, S. 2016, A&A, 587, A94
- Knoll, D. A., & Keyes, D. E. 2004, J. Comput. Phys., 193, 357
- Koskinen, T. T., Harris, M. J., Yelle, R. V., & Lavvas, P. 2013, Icarus, 226, 1678

- Koskinen, T. T., Lavvas, P., Huang, C., et al. 2022, *ApJ*, 929, 52
- Krause, M. O. 1979, *J. Phys. Chem. Ref. Data*, 8, 307
- Kurganov, A., & Tadmor, E. 2000, *J. Comput. Phys.*, 160, 241
- Li, Z., & Liao, H.-L. 2022, *SIAM J. Numer. Anal.*, 60, 2253 (read as arXiv:2201.00527)
- Locci, D., Cecchi-Pestellini, C., & Micela, G. 2022, *PSJ*, 3, 1
- Loreau, J., Ryabchenko, S., & Vaeck, N. 2014, *J. Phys. B*, 47, 135204
- Millar, T. J., Walsh, C., Van de Sande, M., & Markwick, A. J. 2024, *A&A*, 682, A109
- Miller, S., Stallard, T., Tennyson, J., & Melin, H. 2013, *J. Phys. Chem. A*, 117, 9770
- Morgner, H., & Niehaus, A. 1979, *J. Phys. B*, 12, 1805
- Neufeld, D. A. 1990, *ApJ*, 350, 216
- Norcross, D. W. 1971, *J. Phys. B*, 4, 652
- Nussbaumer, H., & Schmutz, W. 1984, *A&A*, 138, 495
- Odert, P., Erkaev, N. V., Lammer, H., et al. 2020, *A&A*, 638, A49
- Oka, T., & Epp, E. 2004, *ApJ*, 613, 349
- Oklopčić, A., & Hirata, C. M. 2018, *ApJ*, 855, L11
- Pengelly, R. M. 1964, *MNRAS*, 127, 145
- Pengelly, R. M., & Seaton, M. J. 1964, *MNRAS*, 127, 165
- Perthame, B., & Shu, C.-W. 1996, *Numer. Math.*, 73, 119
- Poinsot, T. J., & Lele, S. K. 1992, *J. Comput. Phys.*, 101, 104
- Robinson, T. D., & Catling, D. C. 2012, *ApJ*, 757, 104
- Roe, P. L. 1981, *J. Comput. Phys.*, 43, 357
- Roueff, E., Abgrall, H., Czachorowski, P., et al. 2019, *A&A*, 630, A58
- Rusanov, V. V. 1961, *J. Comput. Math. Phys. USSR*, 1, 267
- Incropera, F. P., DeWitt, D. P., Bergman, T. L., & Lavine, A. S. 2007, *Fundamentals of Heat and Mass Transfer*, 6th ed. (Hoboken: Wiley)
- Salz, M., Banerjee, R., Mignone, A., Schneider, P. C., Czesla, S., & Schmitt, J. H. M. M. 2015, *A&A*, 576, A21
- Sutton, E. K., Thayer, J. P., Wang, W., Solomon, S. C., Liu, X., & Foster, B. T. 2015, *J. Geophys. Res. Space Physics*, 120, 6884

- Salz, M., Czesla, S., Schneider, P. C., & Schmitt, J. H. M. M. 2016, *A&A*, 586, A75
- Moses, J. I., & Bass, S. F. 2000, *J. Geophys. Res.*, 105, 7013
- Samson, J. A. R., & Haddad, G. N. 1994, *J. Opt. Soc. Am. B*, 11, 277
- Johnsen, R., Chen, A., & Biondi, M. A. 1980, *J. Chem. Phys.*, 72, 3085
- Schauer, M. M., Jefferts, S. R., Barlow, S. E., & Dunn, G. H. 1989, *J. Chem. Phys.*, 91, 4593
- Seaton, M. J. 1955, *Proc. Phys. Soc. A*, 68, 457
- Seaton, M. J. 1959, *MNRAS*, 119, 81
- Sekiya, M., Nakazawa, K., & Hayashi, C. 1980, *Prog. Theor. Phys.*, 64, 1968
- Shull, J. M., & van Steenberg, M. E. 1985, *ApJ*, 298, 268
- Stancil, P. C., Lepp, S., & Dalgarno, A. 1998, *ApJ*, 509, 1
- Taylor, J., Oklopčić, A., Rustamkulov, Z., et al. 2025, *ApJ*, 989, 68
- Thompson, K. W. 1987, *J. Comput. Phys.*, 68, 1
- Toro, E. F., Spruce, M., & Speares, W. 1994, *Shock Waves*, 4, 25
- Toro, E. F. 2009, *Riemann Solvers and Numerical Methods for Fluid Dynamics*, 3rd edn. (Berlin: Springer)
- van Hoof, P. A. M., Williams, R. J. R., Volk, K., et al. 2014, *MNRAS*, 444, 420
- Verner, D. A., & Yakovlev, D. G. 1995, *A&AS*, 109, 125
- Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, *ApJ*, 465, 487
- Visscher, C., Lidders, K., & Fegley, B., Jr. 2006, *ApJ*, 648, 1181
- Visser, R., van Dishoeck, E. F., & Black, J. H. 2009, *A&A*, 503, 323
- Voronov, G. S. 1997, *At. Data Nucl. Data Tables*, 65, 1
- Watson, A. J., Donahue, T. M., & Walker, J. C. G. 1981, *Icarus*, 48, 150
- Yamaleev, N. K., & Carpenter, M. H. 2009, *J. Comput. Phys.*, 228, 3025
- Yan, M., Sadeghpour, H. R., & Dalgarno, A. 1998, *ApJ*, 496, 1044
- Yan, P.-G., & Babb, J. F. 2023, *MNRAS*, 518, 6004
- Zhang, X., & Shu, C.-W. 2010a, *J. Comput. Phys.*, 229, 8918
- Zhang, X., & Shu, C.-W. 2010b, *J. Comput. Phys.*, 229, 3091